CS 476/676

Numerical Computation Methods for Financial Models

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LECTURE 1

Financial Markets, Derivatives, and Discounting

1.1 Financial assets and contingent claims

A financial model consists of an information structure, a collection of traded assets, and rules for constructing self-financing portfolios. Throughout, S_t denotes the price of one unit of a risky asset at time t, while B_t denotes the value of a cash account. The fundamental problem of derivative pricing is to determine the time-t value of a contingent cash flow whose magnitude depends on future market outcomes. The valuation depends on what risks can be traded and hedged, not merely on an investor's subjective prediction of future prices.

Definition 1.1 (European options)**.** A European call with strike $K > 0$ and maturity T grants the holder the right, but not the obligation, to buy the underlying at time T for K. Its terminal payoff is $(S_T - K)^+ = \max\{S_T - K, 0\}$. A European put grants the right to sell for K and has terminal payoff $(K - S_T)^+$. These payoffs depend only on the terminal value and hence are examples of European-style contingent claims.

Example 1.2 (A motivating call)**.** If $S_0 = 100$, $K = 105$, and $T = 1$ year, the call pays zero when $S_T \leq 105$ and pays $S_T - 105$ when $S_T > 105$. The initial call premium is distinct from the eventual payoff. For instance, a realized payoff of 20 does not mean an initial premium of 20 was fair; financing, trading opportunities, and uncertainty must be considered.

The following basic identities expose the piecewise-linear geometry of call and put payoffs:

$$\begin{aligned} (S_T - K)^+ &= (S_T - K)\mathbf{1}_{\{S_T > K\}}, \\ (K - S_T)^+ &= (K - S_T)\mathbf{1}_{\{S_T < K\}}, \\ (S_T - K)^+ - (K - S_T)^+ &= S_T - K, \\ (S_T - K)^+ + (K - S_T)^+ &= |S_T - K|\,. \end{aligned}$$

A portfolio of options adds the corresponding payoffs. For instance, simultaneously holding a call and a put at the same strike creates a *straddle*, paying $|S_T - K|$, whereas buying the call and selling the put produces the forward-like payoff $S_T - K$.

1.2 Path dependence and the computational problem

Not every payoff depends solely on S_T. For observation dates $0 < t_1 < \cdots < t_m = T$, an arithmetic-average Asian call pays

$$H(S_{t_1}, \ldots, S_{t_m}) = \left(\frac{1}{m} \sum_{j=1}^{m} S_{t_j} - K \right)^+ .$$

A lookback put may pay $\max_{0\le j\le m} S_{t_j} - S_T$. Such payoffs depend on a functional of an entire trajectory. In many models a simple closed-form valuation is unavailable, motivating tree-based recursion, simulation, and PDE methods. A *model price* must also be distinguished from a market quote: the model is useful only to the extent its assumptions and numerical approximations are appropriate.

1.3 Risk-free investment and time value of money

Definition 1.3 (Money-market account)**.** With a constant continuously compounded risk-free rate r, one unit invested at time zero grows to $B_t = e^{rt}$. The time-t value of one unit due at T is $e^{-r(T-t)}$.

For a quoted periodic rate r_m compounded m times per year, the account value at integer year horizons is $(1 + r_m/m)^{mt}$. Since $\log(1+x) = x + O(x^2)$ as $x \to 0$,

$$\begin{aligned}\log\left(1+\frac{r}{m}\right)^{mt} &= mt\log\left(1+\frac{r}{m}\right)\\ &= mt\left(\frac{r}{m} - \frac{r^2}{2m^2} + O(m^{-3})\right)\\ &= rt - \frac{r^2 t}{2m} + O(m^{-2}) \longrightarrow rt.\end{aligned}$$

Exponentiating yields continuous compounding: $\lim_{m\to\infty}(1 + r/m)^{mt} = e^{rt}$. The value at time t of a deterministic amount X paid at T is $Xe^{-r(T-t)}$. A risky cash flow, by contrast, cannot generally be valued by replacing X with its expectation under the real-world probability law.

1.4 Arbitrage, replication, and the modeling ingredients

Definition 1.4 (Arbitrage)**.** An arbitrage is a self-financing trading strategy with zero or negative initial cost, nonnegative terminal wealth in every state, and strictly positive terminal wealth in at least one state of positive probability (with the usual admissibility assumptions in continuous time).

Proposition 1.5 (Law of one price)**.** *If two attainable portfolios have identical terminal payoffs in every state, then absence of arbitrage requires their initial values to coincide.*

Proof. If two identical-payoff portfolios cost $v_1 < v_2$, buy the cheaper and short the more expensive. The initial gain $v_2 - v_1 > 0$ can be deposited in the cash account, while the two terminal portfolio payoffs cancel exactly. This produces an arbitrage, a contradiction. □

Thus, the basic task is to find either a replicating strategy or an equivalent risk-neutral probability distribution under which discounted asset prices are martingales. These two approaches will coincide in a complete binomial market.

1.5 Option portfolios and nonlinear exposure

The payoff diagram of a derivative is informative because it reveals which scenarios create gains and losses. A long call $C_T = (S_T - K)^+$ has nonnegative terminal payout but costs a positive

premium at inception; the total investment profit at maturity, if the premium is financed at rate r, is $(S_T - K)^+ - C_0 e^{rT}$. Thus a positive payoff does not automatically imply a positive net investment return. A short call receives the initial premium but has a potentially unbounded loss as $S_T \to \infty$.

Example 1.6 (Spreads and digital limits)**.** Consider a call spread consisting of a long call at strike K_1 and a short call at $K_2 > K_1$. Its terminal payoff is

$$H(s) = (s - K_1)^+ - (s - K_2)^+ = \begin{cases} 0, & s \le K_1, \\ s - K_1, & K_1 < s < K_2, \\ K_2 - K_1, & s \ge K_2. \end{cases}$$

The payoff is bounded above by $K_2 - K_1$. For $K_2 = K_1 + \varepsilon$, the normalized call spread $H(s)/\varepsilon$ approaches $\mathbf{1}_{\{s>K_1\}}$ for $s \neq K_1$, connecting ordinary calls with digital (binary) option payoffs. Such limiting arguments help explain why the strike derivatives of call prices convey information about risk-neutral distributions.

Remark 1.7 (Price versus risk). The curvature of an option value as a function of its underlying is a measure of nonlinear exposure. The kink at K in the terminal call payoff introduces a nonsmooth derivative at maturity. Even if a smooth pricing formula exists for positive time to maturity, finite-difference approximations of its derivatives may become unstable or require finer meshes close to expiry and strike.

1.6 Derivatives and distributional expectations

For a payoff $H(S_T)$, an elementary but incorrect pricing suggestion is to discount $\mathbb{E}^{\mathbb{P}}[H(S_T)]$ using the stock's historical drift μ. This is not generally a replication cost. In a complete arbitrage-free model, one instead constructs a probability measure $\mathbb{Q}$ such that the discounted prices of traded, properly dividend-adjusted assets are martingales. Under that measure, the fair price is a discounted expectation. The reason is *no arbitrage*, not a hypothesis that agents actually believe the stock's expected return equals r.

For a European claim with payoff H, a risk-neutral valuation functional $\pi(H) = e^{-rT}\mathbb{E}^{\mathbb{Q}}[H]$ is linear. Hence

$$\begin{aligned} \pi(aH_1 + bH_2) &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[aH_1 + bH_2] \\ &= a\pi(H_1) + b\pi(H_2). \end{aligned}$$

Linearity is the mathematical basis for combining derivatives into portfolios and valuing the components separately. It relies on frictionless trading and fixed linear contract cash flows; transaction costs, credit constraints, and nonlinear funding rules can destroy this simple structure.

1.7 Discounting, numéraires, and the precise object being valued

At time t an amount x in the money-market account corresponds to $xe^{r(T-t)}$ at T. To compare prices of payoffs at different dates we discount them to the same date. If $B_t = e^{rt}$, define a discounted wealth process by $\tilde{V}_t = V_t/B_t = e^{-rt}V_t$. For a self-financing portfolio holding Δ_t stocks and a cash position β_t, $V_t = \Delta_t S_t + \beta_t B_t$; if positions are rebalanced, the resulting change in wealth is produced by the assets themselves, not by an external deposit. This convention is essential to replication.

Proposition 1.8 (A deterministic stream of cash flows)**.** *For dates $0 < t_1 < \cdots < t_m$ and deterministic amounts $c_1, \ldots, c_m$, the unique no-arbitrage value at time zero of the stream is $\sum_{j=1}^{m} c_j e^{-rt_j}$.*

Proof. Deposit $c_j e^{-rt_j}$ at time zero in a distinct bank-account position, for every j. At time t_j it grows to exactly c_j, reproducing the liability. Summing initial costs gives the displayed price. A different price would contradict the law of one price by buying the cheaper and shorting the more expensive portfolio. □

A common error is to treat the exercise payoff as the total investment profit. If a European call costs C_0, its financed terminal profit is

$$\begin{aligned}\Pi_T &= (S_T - K)^+ - C_0 e^{rT} \\ &= \begin{cases} -C_0 e^{rT}, & S_T \le K, \\ S_T - K - C_0 e^{rT}, & S_T > K. \end{cases}\end{aligned}$$

Consequently, the call breaks even at $S_T = K + C_0 e^{rT}$, not at $S_T = K$. The terminal payoff and terminal profit are different functions.

1.8 Static replication of payoff identities

The elementary positive-part identity $(x)^+ - (-x)^+ = x$ is itself a replication statement. Indeed, separately considering $x \ge 0$ and $x < 0$ yields

$$(x)^+ - (-x)^+ = \begin{cases} x - 0, & x \ge 0, \\ 0 - (-x), & x < 0 \end{cases} = x.$$

Substituting $x = S_T - K$ establishes the equality of terminal cash flows between a long call combined with a short put and a forward position paying $S_T - K$. If the stock carries no dividends, a long stock plus a borrowing of Ke^{-rT} has exactly this terminal value. By the law of one price,

$$\begin{aligned} C_0 - P_0 &= S_0 - Ke^{-rT}, \\ C_0 + Ke^{-rT} &= P_0 + S_0. \end{aligned}$$

This argument works independently of the distribution of S_T, the volatility model, and the particular numerical approximation. Later a failure of a tree or simulation to approximately satisfy put–call parity will signal implementation error.

1.9 A precise probability-space interpretation

A stochastic model may be written $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\ge 0}, \mathbb{P})$, with $\mathcal{F}_t$ representing information available by time t. The price S_t is assumed $\mathcal{F}_t$-measurable; the claim H is $\mathcal{F}_T$-measurable. In a complete arbitrage-free market, the valuation measure $\mathbb{Q}$ makes appropriately discounted traded gains into martingales. The expression

$$\begin{aligned} V_t &= B_t \mathbb{E}^{\mathbb{Q}} \left[\frac{H}{B_T} \,\middle|\, \mathcal{F}_t \right] \\ &= e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t] \end{aligned}$$

is a *conditional expectation* rather than merely a conditional average of historical outcomes. In discrete binomial models the measure $\mathbb{Q}$ will be constructed from exact replication. This distinction explains why the physical up probability and the risk-neutral up probability need not agree.

1.10 Worked bounds from payoff inequalities

A model-free upper and lower price bound follows from a pointwise payoff inequality: if $0 \le H_1 \le H_2$ in every future state, monotonicity of arbitrage-free pricing requires $0 \le \pi(H_1) \le \pi(H_2)$. To prove this formally, if $\pi(H_1) > \pi(H_2)$, short the more expensive H_1-contract and buy the cheaper H_2-contract. The initial surplus is positive, and the terminal liability is $H_2 - H_1 \ge 0$. This is an arbitrage. For a European call,

$$0 \le (S_T - K)^+ \le S_T,$$
$$0 \le C_0 \le S_0$$

when the stock pays no dividends. A second bound comes from $(S_T - K)^+ \ge S_T - K$:

$$C_0 \ge S_0 - Ke^{-rT},$$
$$C_0 \ge \max\{S_0 - Ke^{-rT}, 0\}.$$

Similarly, $0 \le P_0 \le Ke^{-rT}$ and $P_0 \ge \max\{Ke^{-rT} - S_0, 0\}$. These bounds are useful unit tests for all subsequent numerical routines: a simulated or PDE European call violating them signals a modeling or computational problem.

1.11 A fully developed forward-delivery contract

A forward contract is entered with zero cost and pays $S_T - F$ at maturity. Since the claim can be replicated with one stock and a borrowing of Fe^{-rT}, its current replication cost is

$$0 = S_0 - Fe^{-rT},$$
$$Fe^{-rT} = S_0,$$
$$F = S_0 e^{rT}.$$

If the market forward price were $F^{\text{mkt}} > S_0 e^{rT}$, sell the overpriced forward, buy stock financed by borrowing S_0, and use delivered stock to settle the forward at T. The resulting terminal cash is $F^{\text{mkt}} - S_0 e^{rT} > 0$. If the market forward were cheaper than the fair price, reverse the positions. The argument makes the financing side of no-arbitrage pricing explicit.

1.12 Complete classification of basic option positions

A long call $C_T = (S_T - K)^+$, long put $P_T = (K - S_T)^+$, long straddle $C_T + P_T$, and synthetic forward $C_T - P_T$ can all be described by partitioning the state space into $S_T < K$, $S_T = K$, and $S_T > K$. On $S_T > K$,

$$C_T = S_T - K, \qquad P_T = 0,$$
$$C_T + P_T = S_T - K, \qquad C_T - P_T = S_T - K.$$

On $S_T < K$,

$$C_T = 0, \qquad P_T = K - S_T,$$
$$C_T + P_T = K - S_T, \qquad C_T - P_T = S_T - K.$$

Thus the straddle payoff is $|S_T - K|$, while the synthetic forward payoff is the linear function $S_T - K$ across the entire range of possible terminal prices. The strike kink is crucial: it causes option values to be nonlinear functions of the stock price and makes second-derivative numerical methods sensitive to maturity resolution.

1.13 A static superhedging argument

For any payoff H satisfying $0 \le H \le aS_T + b$ with $a, b \ge 0$, a portfolio long a shares and a deposit of be^{-rT} dominates the payoff at maturity. Its initial cost is $aS_0 + be^{-rT}$. If the claim sold for strictly more than this cost, one could sell it, purchase the dominating replicating superhedge, and retain the surplus without risking a negative terminal wealth. Therefore the no-arbitrage price must obey

$$\pi(H) \le aS_0 + be^{-rT}.$$

Similarly, any attainable subhedge gives a lower bound. Numerical derivatives should respect such static inequalities because they follow from dominated payoff comparisons and trading opportunities, independent of the particular stochastic specification.

1.14 A hierarchy of pricing tasks

The numerical pricing problem can be organized into distinct steps. First specify the terminal reward H and the exercise rights. Then specify the traded assets and the probability model. Next establish the appropriate pricing principle (replication, risk-neutral expectation, or optimal stopping), before selecting an approximation method. Finally, evaluate its statistical, discretization, and modeling errors. This order matters: simulating a stock process before identifying the payoff, discounting convention, or exercise rights can give a precise answer to the wrong mathematical question.

Example 1.9 (Three distinct valuations of a terminal call)**.** Fix the same payoff $H = (S_T - K)^+$. A sample-path realized payoff is $H(\omega)$; a physical expected payoff is $\mathbb{E}^{\mathbb{P}}[H]$; and a complete-market time-zero arbitrage-free price is $e^{-rT}\mathbb{E}^{\mathbb{Q}}[H]$. The first is a random outcome, the second a statistic of the historical return law, and the third a valuation consistent with dynamic hedging. They need not coincide. For example, in a one-step binomial model the physical up probability p may be 0.1 while q determined by replication is 0.7; then $pH_u + (1-p)H_d$ differs considerably from $qH_u + (1-q)H_d$ unless $H_u = H_d$.

LECTURE 2

One-Period Binomial Pricing and Put–Call Parity

2.1 A one-period stock model

Fix a time interval of length $h > 0$. At time zero the stock price is $S_0 > 0$. At time h, the stock takes one of two values:

$$S_h = \begin{cases} uS_0, & \text{up state,} \\ dS_0, & \text{down state,} \end{cases} \qquad 0 < d < u.$$

The cash account grows by e^{rh}. If the physical probability of an upward move is p, then $\mathbb{P}(S_h = uS_0) = p$. No-arbitrage valuations below will not require knowing p.

Proposition 2.1 (No-arbitrage condition)**.** *For the one-step binomial model with unrestricted trading and borrowing, absence of arbitrage is equivalent to*

$$d < e^{rh} < u.$$

Proof. If $d \geq e^{rh}$, buying one stock funded by borrowing S_0 in cash gives a nonnegative profit in both states and a strictly positive profit in at least one state, unless the model is degenerate. If $u \leq e^{rh}$, shorting stock and lending the proceeds yields the reverse arbitrage. Conversely, when $d < e^{rh} < u$, the risk-neutral probability defined below lies strictly between zero and one and every contingent payoff can be replicated, excluding arbitrage. □

2.2 Replication by stock and cash

Suppose a European payoff equals H_u in the up state and H_d in the down state. Seek a portfolio with Δ shares and β units of the money-market account, normalized so that $B_0 = 1$. The replication equations are

$$\begin{aligned} \Delta u S_0 + \beta e^{rh} &= H_u, \\ \Delta d S_0 + \beta e^{rh} &= H_d. \end{aligned}$$

Subtracting and substituting gives

$$\begin{aligned} \Delta &= \frac{H_u - H_d}{S_0(u-d)}, \\ \beta &= e^{-rh}\frac{uH_d - dH_u}{u-d}, \\ V_0 &= \Delta S_0 + \beta. \end{aligned}$$

Remark 2.2. The hedging ratio Δ measures how many shares the replicating portfolio holds. Its sign and magnitude are determined by the *difference* between the payoffs in the two states. This is the discrete ancestor of continuous-time delta hedging.

2.3 Risk-neutral probabilities

Rearranging the replication cost produces

$$\begin{aligned} V_0 &= \frac{H_u - H_d}{u-d} + e^{-rh}\frac{uH_d - dH_u}{u-d} \\ &= e^{-rh}\left[\frac{e^{rh}-d}{u-d}H_u + \frac{u-e^{rh}}{u-d}H_d\right]. \end{aligned}$$

Define the risk-neutral up probability by $q = (e^{rh}-d)/(u-d)$ and the down probability by $1-q = (u-e^{rh})/(u-d)$. Both are positive under the no-arbitrage condition and $q + (1-q) = 1$. Thus

$$V_0 = e^{-rh}\mathbb{E}^{\mathbb{Q}}[H(S_h)].$$

Proposition 2.3 (Discounted stock martingale)**.** *Under* $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}[e^{-rh}S_h] = S_0$.

Proof.

$$\begin{aligned} e^{-rh}\mathbb{E}^{\mathbb{Q}}[S_h] &= e^{-rh}S_0\left[qu + (1-q)d\right] \\ &= e^{-rh}S_0\left[d + \frac{e^{rh}-d}{u-d}(u-d)\right] = S_0. \end{aligned}$$

□

It is crucial that q is a *pricing* probability, not a claim about the empirical frequency of up moves. It is obtained from tradable replication constraints, not statistical estimation of p.

2.4 Put–call parity

Theorem 2.4 (European put–call parity)**.** *For European call and put contracts on the same non-dividend-paying underlying, with matching strike K and maturity T, the initial values satisfy*

$$C_0 - P_0 = S_0 - Ke^{-rT}.$$

Proof. A long call and short put has terminal payoff

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K.$$

A portfolio of one share of stock and a short position in Ke^{-rT} units of today's cash also pays $S_T - K$ at maturity. By the law of one price, $C_0 - P_0 = S_0 - Ke^{-rT}$. □

A parity violation would allow a static arbitrage. Under a deterministic continuous dividend yield a, stock replaced by prepaid forward exposure changes the right-hand side to $S_0e^{-aT} - Ke^{-rT}$.

2.5 A fully worked one-period replication

Let $S_0 = 50$, $u = 1.3$, $d = 0.8$, $e^{rh} = 1.05$, and consider a call with strike $K = 52$. The possible stock outcomes are 65 and 40, and the option payouts are 13 and 0. We first calculate

$$q = \frac{1.05 - 0.8}{1.3 - 0.8} = \frac{1}{2},$$
$$C_0 = e^{-rh}\left(\frac{1}{2}\cdot 13 + \frac{1}{2}\cdot 0\right) = \frac{6.5}{1.05}.$$

To derive the corresponding hedge without invoking expected values, solve the two-equation replication system:

$$\begin{aligned} 65\Delta + 1.05\beta &= 13, \\ 40\Delta + 1.05\beta &= 0, \\ 25\Delta &= 13, \\ \Delta &= \frac{13}{25} = 0.52, \\ \beta &= -\frac{40(0.52)}{1.05} = -\frac{20.8}{1.05}. \end{aligned}$$

The initial portfolio cost is

$$50\Delta + \beta = 26 - \frac{20.8}{1.05} = \frac{6.5}{1.05},$$

exactly equal to the risk-neutral expected discounted payoff. If the market sold the same call for a smaller amount, we could buy the call and short this replicating portfolio; if it sold the call for more, we could short the call and buy the replicating portfolio. Either discrepancy creates a state-independent net cash inflow at time zero without future obligations.

2.6 Uniqueness and completeness in a two-state market

The two replication equations form a linear system

$$\begin{pmatrix} uS_0 & e^{rh} \\ dS_0 & e^{rh} \end{pmatrix}\begin{pmatrix} \Delta \\ \beta \end{pmatrix} = \begin{pmatrix} H_u \\ H_d \end{pmatrix}.$$

Its determinant is $S_0 e^{rh}(u - d) \neq 0$ when $u > d$. Therefore every vector of state-contingent cash flows has a unique representation as a stock-and-bond portfolio. This is exactly what *market completeness* means in the one-period model. Completeness explains why the no-arbitrage price is unique: not only can prices be bounded by arbitrage arguments, but every payoff is attainable by trading existing securities.

The corresponding risk-neutral probability vector can be derived by imposing the two pricing conditions for the basis assets. If $\mathbb{Q}(u) = q$ and $\mathbb{Q}(d) = 1 - q$, then discounting the cash account must reproduce its current value and discounting the stock must reproduce S_0. The cash constraint is $q + (1 - q) = 1$, and the stock constraint is $e^{-rh}(qu + (1 - q)d) = 1$. The latter uniquely determines q. This algebra shows that replication and the equivalent martingale measure are two descriptions of the same linear system.

2.7 An end-to-end numerical one-step call calculation

Consider $S_0 = 100$, $u = 1.10$, $d = 0.90$, $K = 105$, $T = 1$ and a continuously compounded rate $r = 0.05$. The up and down stock prices are 110 and 90, and the call payoffs are $H_u = 5$ and $H_d = 0$. The replicating portfolio solves

$$\begin{aligned} 110\Delta + e^{0.05}\beta &= 5, \\ 90\Delta + e^{0.05}\beta &= 0, \\ (110-90)\Delta &= 5-0, \\ \Delta &= \frac{5}{20} = \frac{1}{4}, \\ e^{0.05}\beta &= -90\cdot\frac{1}{4} = -22.5, \\ \beta &= -22.5e^{-0.05}, \\ C_0 &= 100\Delta + \beta = 25 - 22.5e^{-0.05}. \end{aligned}$$

We verify this value using the risk-neutral probability:

$$\begin{aligned} q &= \frac{e^{0.05}-0.9}{1.1-0.9}, \\ 1-q &= \frac{1.1-e^{0.05}}{1.1-0.9}, \\ e^{-0.05}[qH_u + (1-q)H_d] &= 5e^{-0.05}\frac{e^{0.05}-0.9}{0.2} \\ &= 25 - 22.5e^{-0.05} = C_0. \end{aligned}$$

In the up state the hedge produces $\frac{1}{4}(110) - 22.5 = 5$, while in the down state it produces $\frac{1}{4}(90) - 22.5 = 0$. These are *pathwise* identities, not statements about average outcomes.

2.8 Why the stock's physical probability disappears

Let the physical up probability be $p \in (0,1)$. The expected stock price under $\mathbb{P}$ is $S_0[pu + (1-p)d]$, which generally differs from S_0e^{rT}. Nevertheless, the two replication equations determine the derivative price without any reference to p. One may derive q by requiring discounted stock prices to be martingales:

$$\begin{aligned} S_0 &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T] \\ &= e^{-rT}[q(uS_0) + (1-q)(dS_0)] \\ &= S_0e^{-rT}[d + q(u-d)], \\ e^{rT} &= d + q(u-d), \\ q &= \frac{e^{rT}-d}{u-d}. \end{aligned}$$

If $d < e^{rT} < u$, then $0 < q < 1$. At either boundary one risk-neutral state probability is zero; with both physical states possible, the corresponding redundant-price inequalities must be handled separately. The strict condition is the standard nondegenerate arbitrage-free binomial assumption.

2.9 State prices and a complete-market linear system

The same problem can be expressed without first introducing probabilities. A contract paying 1 in the up state and 0 in the down state has state price ψ_u, while its down-state counterpart has price ψ_d. We impose replication of the stock and of a unit risk-free maturity payment:

$$\begin{aligned}
\psi_u + \psi_d &= e^{-rT}, \\
uS_0\psi_u + dS_0\psi_d &= S_0, \\
u\psi_u + d(e^{-rT} - \psi_u) &= 1, \\
(u-d)\psi_u &= 1 - de^{-rT}, \\
\psi_u &= e^{-rT}\frac{e^{rT}-d}{u-d} = e^{-rT}q, \\
\psi_d &= e^{-rT}\frac{u-e^{rT}}{u-d} = e^{-rT}(1-q).
\end{aligned}$$

Any pair (H_u, H_d) has value $H_u\psi_u + H_d\psi_d$. The two traded assets span the two-dimensional vector space of state-contingent payoffs, explaining why this one-step market is complete. If there are more possible future stock states than linearly independent traded assets, exact replication may fail.

2.10 The digital option as an indicator payoff

Consider a one-step cash-or-nothing digital paying 1 if the stock goes up and 0 if it goes down. It satisfies $H_u = 1$ and $H_d = 0$. Its hedge and price are

$$\begin{aligned}
\Delta &= \frac{1-0}{S_0(u-d)} = \frac{1}{S_0(u-d)}, \\
\beta &= e^{-rT}\frac{u\cdot 0 - d\cdot 1}{u-d} = -e^{-rT}\frac{d}{u-d}, \\
D_0 &= \Delta S_0 + \beta = \frac{1-de^{-rT}}{u-d} \\
&= e^{-rT}\frac{e^{rT}-d}{u-d} = e^{-rT}q.
\end{aligned}$$

Its risk-neutral price is the discounted probability of the up state. For the complementary down digital the price is $e^{-rT}(1-q)$, so the prices add to e^{-rT}, the price of a bond delivering 1 in every state. This is a discrete prototype of state-price densities and Arrow–Debreu securities.

2.11 The exact boundary for no-arbitrage

A position long the stock and short S_0 in the bank has zero initial value and terminal wealth $S_0u - S_0e^{rT}$ in the up state and $S_0d - S_0e^{rT}$ in the down state. If $d \geq e^{rT}$ both values are nonnegative, so this position is an arbitrage when at least one is strictly positive. A position short the stock and long S_0 in the bank has terminal wealth $S_0(e^{rT} - u)$ and $S_0(e^{rT} - d)$, which is an arbitrage when $u \leq e^{rT}$ with a strict inequality in one state. Hence the order $d < e^{rT} < u$ follows from explicitly checking gains, rather than from treating a mathematical probability formula as a pricing axiom.

2.12 Matrix inversion and the uniqueness of the replicating hedge

Represent the state-dependent prices of the two traded assets as columns of a matrix,

$$A = \begin{pmatrix} uS_0 & e^{rT} \\ dS_0 & e^{rT} \end{pmatrix}, \qquad \vartheta = \begin{pmatrix} \Delta \\ \beta \end{pmatrix}, \qquad H = \begin{pmatrix} H_u \\ H_d \end{pmatrix}.$$

Replication is the linear system $A\vartheta = H$. Its determinant is $\det A = S_0 e^{rT}(u-d) > 0$, so it has exactly one solution. The inverse is

$$A^{-1} = \frac{1}{S_0 e^{rT}(u-d)} \begin{pmatrix} e^{rT} & -e^{rT} \\ -dS_0 & uS_0 \end{pmatrix}.$$

Thus

$$\begin{aligned} \vartheta &= A^{-1}H \\ &= \begin{pmatrix} \frac{H_u - H_d}{S_0(u-d)} \\ e^{-rT}\frac{uH_d - dH_u}{u-d} \end{pmatrix}, \end{aligned}$$

recovering the explicit hedge. The determinant also shows that when $u = d$, the stock does not distinguish future states, so an arbitrary state-dependent claim cannot be replicated using stock and cash alone. Completeness is a rank condition on attainable cash flows.

2.13 State prices as a positive linear pricing functional

The vector of discounted state prices can be derived from the matrix transpose:

$$\begin{pmatrix} S_0 \\ 1 \end{pmatrix} = \begin{pmatrix} uS_0 & dS_0 \\ e^{rT} & e^{rT} \end{pmatrix} \begin{pmatrix} \psi_u \\ \psi_d \end{pmatrix}.$$

The resulting $\psi_u = e^{-rT}q$ and $\psi_d = e^{-rT}(1-q)$ are both strictly positive exactly when $d < e^{rT} < u$. The price of any nonnegative nonzero payoff is then positive, since $\psi_u H_u + \psi_d H_d > 0$ when at least one payoff is strictly positive. A price functional with a negative state weight would make some strictly positive contingent cash flow have a negative cost, an arbitrage inconsistency.

2.14 An implementable one-period pricing procedure

Algorithm 2.5 (One-period binomial hedge and price)**.** Given S_0, u, d, r, T with $0 < d < e^{rT} < u$ and a payoff function g:

1. Compute the two terminal underlying prices $S_u = uS_0$ and $S_d = dS_0$.
2. Evaluate the terminal payoffs $H_u = g(S_u)$ and $H_d = g(S_d)$, including all contract-specific conditions.
3. Solve $\Delta = (H_u - H_d)/(S_u - S_d)$ and $\beta = e^{-rT}(uH_d - dH_u)/(u-d)$.
4. Compute the initial price $V_0 = \Delta S_0 + \beta$.
5. Independently calculate $q = (e^{rT} - d)/(u-d)$ and verify $V_0 = e^{-rT}[qH_u + (1-q)H_d]$.

6. Verify the two hedge residuals $\Delta S_u + \beta e^{rT} - H_u$ and $\Delta S_d + \beta e^{rT} - H_d$ are zero up to floating-point error.

All steps take constant arithmetic work. A practical implementation should reject $u = d$, $S_0 \leq 0$, and an invalid risk-neutral probability before attempting pricing. Using a hedging residual is particularly valuable because it tests the replication mechanism itself, not merely agreement between two algebraically equivalent price formulas.

LECTURE 3

Two-Period Trees and Dynamic Risk-Neutral Pricing

3.1 Dynamic replication

Extend the one-period model to two dates $0, h, 2h$ with independent binomial up/down innovations under $\mathbb{Q}$. If $S_h = s$, let the remaining payoff in the next step equal $H(us)$ or $H(ds)$. The conditional value at h must then be

$$V_h(s) = e^{-rh}\big[qH(us) + (1-q)H(ds)\big].$$

At time zero, applying the same replication principle once again yields

$$\begin{aligned} V_0 &= e^{-rh}\left[qV_h(uS_0) + (1-q)V_h(dS_0)\right] \\ &= e^{-2rh}\left[q^2H(u^2S_0) + 2q(1-q)H(udS_0) + (1-q)^2H(d^2S_0)\right], \end{aligned}$$

when the payoff depends only on the terminal stock price. The factor two counts the distinct up–down and down–up paths that reach the same node in a recombining tree.

3.2 Conditional expectations and the tower property

Let $\mathcal{F}_n$ be the information available by date $t_n = nh$. For a claim H payable at $T = Nh$, risk-neutral pricing at intermediate dates takes the form

$$V_{t_n} = \mathbb{E}^{\mathbb{Q}}\big[e^{-r(T-t_n)}H \mid \mathcal{F}_n\big].$$

For $n < N$ the tower property implies

$$\begin{aligned} V_{t_n} &= e^{-r(T-t_n)}\mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\right] \\ &= e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{t_{n+1}} \mid \mathcal{F}_n]. \end{aligned}$$

This is the mathematical basis for backward induction. It says that the *discounted value process* is a martingale under $\mathbb{Q}$, while the undiscounted option value need not be a martingale.

3.3 Node-by-node hedge ratios

A fixed hedge placed at inception is generally not sufficient after future information is revealed. At node (n, s), let the two next option values be V^u and V^d. The locally replicating positions are

$$\Delta_n(s) = \frac{V^u - V^d}{s(u-d)},$$

$$\beta_n(s) = e^{-rh}\frac{uV^d - dV^u}{u-d}.$$

The agent rebalances these positions as the path evolves. The strategy is self-financing when trading costs and cash flows satisfy the rebalancing identity: stock purchases must be financed by reductions in the cash position. Completeness of the binary tree follows because two traded assets span the two possible next-step payoffs at every node.

Example 3.1 (A European call in two periods)**.** Take $S_0 = 100$, $u = 1.2$, $d = 0.8$, $e^{rh} = 1.02$, and $K = 100$. The pricing probability is $q = (1.02 - 0.8)/(1.2 - 0.8) = 0.55$. Terminal stock values are $144, 96, 64$, corresponding to call payoffs $44, 0, 0$. Hence

$$C_0 = e^{-2rh}\left(0.55^2 \cdot 44\right) = \frac{13.31}{1.02^2}.$$

A physical up probability different from 0.55 changes the distribution of realized returns but not this complete-market replication price.

3.4 Induction proof of multi-period pricing

Theorem 3.2 (Backward valuation in a complete finite tree)**.** *Suppose each one-step stock tree obeys $d < e^{rh} < u$, trading is frictionless and self-financing, and the claim pays an integrable $\mathcal{F}_N$-measurable amount H at maturity. Then its arbitrage-free value at time n is*

$$V_n = e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n], \qquad V_N = H,$$

and hence $V_n = e^{-r(N-n)h}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_n]$.

Proof. At time $N-1$, the claim has only two possible payoffs conditional on the current state. The one-step replication argument establishes the discounted conditional expectation formula. Suppose the value V_{n+1} is attainable by trading over dates $n+1$ through N. At time n, replicate the two next-step values V^u_{n+1}, V^d_{n+1} using a stock/cash portfolio over one period, then switch to the already known continuation replicating strategies after the next state is observed. Concatenating these strategies yields a self-financing replication of H from time n. Its time-n cost is the one-step discounted conditional expectation. The tower property then establishes the full discounted expectation. Backward induction completes the proof. □

3.5 Nonrecombining state variables and path counts

A recombining stock tree with N time steps has only $N+1$ possible terminal *stock values*, but it still has 2^N distinct sequences of up and down moves. A European claim $H(S_N)$ assigns the same payoff to all paths ending at the same stock price; aggregation by binomial multiplicities is legitimate. By contrast, for a lookback option $H = (M_N - S_N)$ with $M_N = \max_{0\le k\le N} S_k$, two paths with the same terminal stock price may have distinct past maxima. The state of a valid dynamic program must distinguish them.

Example 3.3 (A four-step path distinction)**.** Take $u = 2$ and $d = 1/2$, starting from $S_0 = 1$. The paths $UUDD$ and $UDUD$ both finish at $S_4 = 1$, but their running maxima are 4 and 2. Their lookback-put payouts are therefore 3 and 1. A recursion using S_n alone would incorrectly merge these claims and misprice the option. Enlarging the state to (S_n, M_n) yields the correct transition

$$(s, m) \longmapsto (us, \max\{m, us\}) \quad \text{or} \quad (ds, \max\{m, ds\}).$$

3.6 A complete two-period calculation with the tower property

Let $\Delta t = T/2$, $R = e^{r\Delta t}$ and $q = (R-d)/(u-d)$. Assume the tree recombines and let the maturity payoffs be H_{uu}, $H_{ud} = H_{du}$ and H_{dd}. Starting from the last time step,

$$\begin{aligned} V_{1,u} &= R^{-1}\{qH_{uu} + (1-q)H_{ud}\}, \\ V_{1,d} &= R^{-1}\{qH_{ud} + (1-q)H_{dd}\}, \\ V_0 &= R^{-1}\{qV_{1,u} + (1-q)V_{1,d}\} \\ &= R^{-2}\{q[qH_{uu} + (1-q)H_{ud}] \\ &\qquad + (1-q)[qH_{ud} + (1-q)H_{dd}]\} \\ &= R^{-2}\{q^2H_{uu} + 2q(1-q)H_{ud} + (1-q)^2H_{dd}\}. \end{aligned}$$

The coefficient $2q(1-q)$ arises because both ud and du paths reach the same terminal stock price. The full two-period filtration still distinguishes their histories; they can only be merged when the payoff does not depend on their order.

3.7 The self-financing hedge at every interior node

At each time-one node $x \in \{u, d\}$ the option can be replicated by $\Delta_{1,x}$ shares and a cash position $\beta_{1,x}B_1$. Solving the two successor-state equations gives

$$\begin{aligned} \Delta_{1,u} &= \frac{H_{uu} - H_{ud}}{S_0 u(u-d)}, \\ \Delta_{1,d} &= \frac{H_{ud} - H_{dd}}{S_0 d(u-d)}, \\ \Delta_0 &= \frac{V_{1,u} - V_{1,d}}{S_0(u-d)}. \end{aligned}$$

These deltas are generally different: a delta hedge is dynamic. For a portfolio $\Pi_n = \Delta_n S_n + \beta_n B_n$, self-financing rebalancing requires

$$\begin{aligned} \Pi_{n+1} - \Pi_n &= \Delta_n(S_{n+1} - S_n) + \beta_n(B_{n+1} - B_n), \\ (\Delta_{n+1} - \Delta_n)S_{n+1} + (\beta_{n+1} - \beta_n)B_{n+1} &= 0. \end{aligned}$$

The second equality means buying extra shares at a rebalance is financed by reducing the cash holding. Without this accounting constraint, a purported hedge may implicitly inject money and the replication argument becomes invalid.

3.8 Induction through conditional expectation

Suppose $V_{n+1} = \mathbb{E}^{\mathbb{Q}}[R^{-(N-n-1)}H \mid \mathcal{F}_{n+1}]$. The one-period replication principle gives

$$\begin{aligned} V_n &= R^{-1}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n] \\ &= R^{-1}\mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}[R^{-(N-n-1)}H \mid \mathcal{F}_{n+1}] \,\middle|\, \mathcal{F}_n\right] \\ &= R^{-(N-n)}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_n]. \end{aligned}$$

The final step is the tower property because $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$. In particular $V_0 = R^{-N}\mathbb{E}^{\mathbb{Q}}[H]$. This proves backward induction is the conditional-expectation algorithm, not a heuristic averaging rule.

3.9 A two-period call computed node by node

Take $S_0 = 100$, $u = 1.1$, $d = 0.9$, $K = 105$, $r = 0.05$ per year, and two half-year intervals. Let $R = e^{0.025}$ and $q = (R - 0.9)/0.2$. The terminal stock nodes and call payoffs are

$$\begin{aligned} S_{2,uu} &= 121, & H_{uu} &= (121 - 105)^+ = 16, \\ S_{2,ud} = S_{2,du} &= 99, & H_{ud} &= (99 - 105)^+ = 0, \\ S_{2,dd} &= 81, & H_{dd} &= (81 - 105)^+ = 0. \end{aligned}$$

Backward induction gives

$$\begin{aligned} V_{1,u} &= R^{-1}[q \cdot 16 + (1-q) \cdot 0] = 16R^{-1}q, \\ V_{1,d} &= R^{-1}[q \cdot 0 + (1-q) \cdot 0] = 0, \\ V_0 &= R^{-1}[q(16R^{-1}q) + (1-q) \cdot 0] \\ &= 16R^{-2}q^2 = 16e^{-0.05}\left(\frac{e^{0.025} - 0.9}{0.2}\right)^2. \end{aligned}$$

The time-zero replicating stock position is

$$\begin{aligned} \Delta_0 &= \frac{V_{1,u} - V_{1,d}}{110 - 90} \\ &= \frac{16R^{-1}q}{20} = 0.8R^{-1}q. \end{aligned}$$

In the up branch, the new hedge becomes $\Delta_{1,u} = 16/(121 - 99) = 8/11$, whereas in the down branch it becomes $\Delta_{1,d} = 0$. The dynamic hedge reflects the change in option sensitivity as the underlying moves.

3.10 Discounted martingales in a numerical tree

At any node with stock price s,

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[S_{n+1} \mid S_n = s] &= qsu + (1-q)sd \\ &= s[d + q(u-d)] = sR, \\ \mathbb{E}^{\mathbb{Q}}[R^{-(n+1)}S_{n+1} \mid \mathcal{F}_n] &= R^{-(n+1)}RS_n = R^{-n}S_n. \end{aligned}$$

Thus the discounted stock is a discrete-time martingale. The same conditional-step identity, applied to V, yields $\mathbb{E}[R^{-(n+1)}V_{n+1} \mid \mathcal{F}_n] = R^{-n}V_n$ for a European replicable claim. These are exact model identities and are more informative than only checking the time-zero price.

3.11 A fully general recursion with node-dependent transition factors

A multi-period model need not have constant u, d, r. At node ω_n, let the next up and down multipliers be $u_n(\omega_n)$ and $d_n(\omega_n)$, with bank growth $R_n(\omega_n) > 0$. The local risk-neutral probability is

$$q_n(\omega_n) = \frac{R_n(\omega_n) - d_n(\omega_n)}{u_n(\omega_n) - d_n(\omega_n)}.$$

Assuming $d_n < R_n < u_n$ at every reachable node, the European backward step is

$$V_n(\omega_n) = \frac{1}{R_n(\omega_n)} \big[q_n(\omega_n) V_{n+1}(\omega_n, U) + (1 - q_n(\omega_n)) V_{n+1}(\omega_n, D) \big].$$

Inductively, the time-zero value equals the expectation of the terminal payoff discounted by the realized product $\prod_{k=0}^{N-1} R_k^{-1}$ under the measure determined by the state-dependent transition probabilities. One should not replace the product with e^{-rT} unless the interest rate is indeed constant (or the deterministic time integral of the rate is known).

3.12 Backward hedging and the dimension of the state

In the general local model, denote successor stock prices by $S_{n+1,u}$ and $S_{n+1,d}$. The unique local stock holding is

$$\Delta_n = \frac{V_{n+1,u} - V_{n+1,d}}{S_{n+1,u} - S_{n+1,d}}.$$

For a vanilla European payoff, it is often enough to label nodes by time and current stock value. For an arithmetic-average Asian claim the current running sum $A_n = \sum_{j=1}^{n} S_j$ is also needed. The relevant state is (n, S_n, A_n), and its update is $(S_{n+1}, A_{n+1}) = (S_{n+1}, A_n + S_{n+1})$. A backward algorithm that forgets A_n at an intermediate node merges histories whose future claims differ and cannot yield the correct path-dependent value.

3.13 Backward valuation and hedge extraction in a recombining tree

Algorithm 3.4 (European lattice pricing)**.** Choose N steps, step length $h = T/N$, multipliers u, d, and $q = (e^{rh} - d)/(u - d)$.

1. At maturity assign $S_{N,j} = S_0 u^j d^{N-j}$ and $V_{N,j} = g(S_{N,j})$ for $j = 0, \ldots, N$.
2. For $n = N-1, N-2, \ldots, 0$ and $j = 0, \ldots, n$, compute

$$V_{n,j} = e^{-rh}\{q V_{n+1,j+1} + (1-q) V_{n+1,j}\},$$
$$\Delta_{n,j} = \frac{V_{n+1,j+1} - V_{n+1,j}}{S_{n,j}(u - d)}.$$

3. Return $V_{0,0}$ and the lattice of hedge ratios $\Delta_{n,j}$.

This uses $\sum_{n=0}^{N}(n+1) = O(N^2)$ lattice entries if all hedges are retained. If only the price is needed, in-place overwriting can reduce memory to $O(N)$, provided the next-layer values are not overwritten before both successor values are used. For path-dependent rewards, one must enlarge the state and the recombining memory bounds need not apply.

LECTURE 4

Multi-Period Lattices, American Options, and Random Walks

4.1 Multi-period stock dynamics

Let $\xi_1, \dots, \xi_N$ be up/down multipliers equal to u or d. The price process is

$$S_n = S_0 \prod_{j=1}^{n} \xi_j, \quad S_{n+1} \in \{uS_n, dS_n\}.$$

If $d = u^{-1}$, the possible terminal prices are $S_N = S_0 u^{2k-N}$ for $k = 0, \dots, N$. If the multipliers are constant, the tree recombines because $ud = du$. A European terminal claim $H(S_N)$ has value

$$V_0 = e^{-rNh} \sum_{k=0}^{N} \binom{N}{k} q^k (1-q)^{N-k} H(S_0 u^k d^{N-k}).$$

This formula is useful for path-independent payoffs. For path-dependent claims, distinct paths reaching the same stock price can have different payoff-relevant histories, so they cannot always be merged into a single state.

Algorithm 4.1 (Backward induction for a European claim)**.** Initialize the terminal node values $V_{N,k} = H(S_0 u^k d^{N-k})$. For $n = N-1, \dots, 0$ and $k = 0, \dots, n$, compute

$$V_{n,k} = e^{-rh} \left[q V_{n+1,k+1} + (1-q) V_{n+1,k} \right].$$

The time-zero price is $V_{0,0}$. Reconstruct hedge ratios at each node using the two successor values.

4.2 Path-dependent and early-exercise claims

For an arithmetic Asian claim, a natural enlarged state variable is (S_n, A_n), where $A_n = \sum_{j=1}^{n} S_j$. The transition satisfies $A_{n+1} = A_n + S_{n+1}$. For a lookback option, the running maximum $M_n = \max_{0 \le j \le n} S_j$ serves as an additional state. The backward pricing function is therefore $V_n(s, m)$ or $V_n(s, a)$ rather than $V_n(s)$ alone.

Definition 4.2 (American-style early exercise)**.** An American derivative with intrinsic exercise value $g(s)$ may be exercised at any permitted date. On a discrete exercise grid it satisfies

$$V_{n,k} = \max \left\{ g(S_{n,k}),\ e^{-rh} \left[q V_{n+1,k+1} + (1-q) V_{n+1,k} \right] \right\}.$$

The second entry is the continuation value, whereas the first is the gain from exercising now. The recursion optimizes over stopping decisions that use only available information, anticipating the optimal stopping and complementary problems developed later.

4.3 Log returns as a random walk

Let $Y_n = \log(S_n/S_0)$. Then

$$Y_n = \sum_{j=1}^{n} Z_j, \qquad Z_j = \begin{cases} \log u, & \xi_j = u, \\ \log d, & \xi_j = d. \end{cases}$$

If $d = u^{-1}$, write $a = \log u$. Then $Z_j \in \{-a, a\}$ and

$$\begin{aligned} \mathbb{E}[Z_j] &= a(2p-1), \\ \mathrm{Var}(Z_j) &= \mathbb{E}[Z_j^2] - \mathbb{E}[Z_j]^2 = a^2 - a^2(2p-1)^2 = 4a^2p(1-p). \end{aligned}$$

By independence, $\mathbb{E}[Y_n] = na(2p-1)$ and $\mathrm{Var}(Y_n) = 4na^2p(1-p)$. The law under $\mathbb{P}$ uses p; under $\mathbb{Q}$ the same identities use q. A diffusive scaling of these log returns leads toward Brownian motion and geometric Brownian motion.

4.4 Optimal stopping on a lattice: a proof

Proposition 4.3 (Bermudan dynamic programming)**.** *Suppose a claim offers reward $G_n = g(S_n)$ upon exercise at date n. Set $V_N = G_N$ and, for $n < N$, define*

$$V_n = \max\{G_n, e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]\}.$$

Then V_n is the essential supremum of conditional discounted exercise payoffs over all stopping times $\tau \in \{n, \dots, N\}$, and a stopping time that first enters the exercise set is optimal.

Proof. At maturity there is no choice, so $V_N = G_N$. Suppose the assertion holds starting from $n+1$. At date n, an admissible strategy either exercises immediately for G_n, or waits one period. In the latter case, the best attainable continuation value is, by the induction hypothesis, $e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]$. Taking the larger of the two values captures all admissible decisions, and the rule choosing the maximizing action is adapted. Backward induction proves optimality. □

4.5 Binomial convergence to a diffusion

In the symmetric-multiplier scheme $u = e^{\sigma\sqrt{h}}$, $d = e^{-\sigma\sqrt{h}}$, write the log increment as $\Delta \log S = \sigma\sqrt{h}\,\varepsilon$ where $\varepsilon \in \{-1, 1\}$. Its risk-neutral mean and variance can be expanded using $q_h = 1/2 + (r - \sigma^2/2)\sqrt{h}/(2\sigma) + O(h)$:

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[\Delta \log S] &= \sigma\sqrt{h}(2q_h - 1) \\ &= (r - \sigma^2/2)h + O(h^{3/2}), \\ \mathrm{Var}^{\mathbb{Q}}(\Delta \log S) &= \sigma^2 h - \left(\mathbb{E}^{\mathbb{Q}}[\Delta \log S]\right)^2 \\ &= \sigma^2 h + O(h^2). \end{aligned}$$

Thus the aggregate log return over $N = T/h$ steps has limiting mean $(r - \sigma^2/2)T$ and variance $\sigma^2 T$. A functional central limit theorem can strengthen this marginal statement to convergence of the entire rescaled price process. The resulting stock process is geometric Brownian motion, whose exponential form guarantees positive prices.

4.6 The closed binomial formula from counting paths

Suppose N steps have multipliers u and d, with an upward move having risk-neutral probability q. The stock value after k up moves equals $S_0 u^k d^{N-k}$, regardless of their order. There are $\binom{N}{k}$ distinct paths with k up moves. Thus the binomial probability is

$$\begin{aligned}
\mathbb{Q}(N_u = k) &= \binom{N}{k} q^k (1-q)^{N-k}, \\
\binom{N}{k} &= \frac{N!}{k!(N-k)!}, \\
C_0 &= e^{-rT} \sum_{k=0}^{N} \binom{N}{k} q^k (1-q)^{N-k} \left(S_0 u^k d^{N-k} - K \right)^+ .
\end{aligned}$$

An implementation can evaluate this with $O(N)$ terminal states and $O(N^2)$ backward-node updates, instead of enumerating 2^N paths. If one uses a Markov state Y_n recording the running maximum, the enlarged state (S_n, Y_n) may contain substantially more nodes; collapsing solely by S_n then incorrectly discards path information.

4.7 Deriving the American exercise recursion rather than guessing it

Let $g_n(S_n)$ denote the reward from exercising at date t_n. Denote by $\mathcal{T}_n$ the set of stopping times taking values in $\{n, \ldots, N\}$. The holder's value is

$$V_n = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_n} \mathbb{E}^{\mathbb{Q}}[R^{-(\tau - n)} g_\tau(S_\tau) \mid \mathcal{F}_n].$$

Every permissible stopping time either exercises now (payoff $g_n(S_n)$) or continues at least one step. Conditional on continuing, the best attainable value at the next node is V_{n+1}. Therefore

$$\begin{aligned}
V_N &= g_N(S_N), \\
V_n &= \max \left\{ g_n(S_n), R^{-1} \mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n] \right\} \\
&= \max \left\{ g_n(S_n), R^{-1}[q V_{n+1,u} + (1-q) V_{n+1,d}] \right\}.
\end{aligned}$$

The exercise rule is $\tau^* = \inf\{n : g_n(S_n) \geq R^{-1} \mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]\}$. It is an *adapted* decision; the holder cannot use future returns when deciding whether to exercise at a current node.

4.8 Random-walk representation and moment matching

Write $X_n = \log S_n$ and $\xi_j = \log u$ with probability q or $\log d$ with probability $1-q$. Then

$$\begin{aligned} X_n &= \log S_0 + \sum_{j=1}^{n} \xi_j, \\ \mathbb{E}^{\mathbb{Q}}[\xi_j] &= q \log u + (1-q)\log d, \\ \mathrm{Var}^{\mathbb{Q}}(\xi_j) &= q(1-q)(\log u - \log d)^2, \\ \mathbb{E}^{\mathbb{Q}}[X_n] &= \log S_0 + n[q \log u + (1-q)\log d], \\ \mathrm{Var}^{\mathbb{Q}}(X_n) &= nq(1-q)(\log u - \log d)^2. \end{aligned}$$

The variance adds because the increments are independent. The continuous limit will choose u and d as functions of Δt so the mean is of order Δt and variance of order Δt, permitting a nontrivial diffusion limit.

4.9 Complete three-step lookback example from Practice Set 1

Consider three one-year binomial steps with $S_0 = 4$, $u = 2$, $d = 1/2$ and $r = 1/4$. The terminal payoff is the difference between the maximum observed stock price and its terminal value,

$$H = \max_{0 \le n \le 3} S_n - S_3.$$

Write $M_n = \max\{S_0, \ldots, S_n\}$. The path-by-path stock prices, running maximum, and terminal reward are

Path	S_3	M_3	$M_3 - S_3$
UUU	32	32	0
UUD	8	16	8
UDU	8	8	0
UDD	2	8	6
DUU	8	8	0
DUD	2	4	2
DDU	2	4	2
DDD	1/2	4	7/2

The risk-neutral upward probability is $q = (e^{1/4} - 1/2)/(2 - 1/2)$; write $p = 1 - q$. Summing all eight discounted terminal payoffs gives

$$\begin{aligned} V_0 &= e^{-3/4}\{0 \cdot q^3 + 8q^2 p + 0 \cdot qpq + 6qp^2 \\ &\quad + 0 \cdot pq^2 + 2pqp + 2p^2 q + \tfrac{7}{2}p^3\} \\ &= e^{-3/4}\left[8q^2 p + 10qp^2 + \frac{7}{2}p^3\right]. \end{aligned}$$

With $q \approx 0.522684$, the result is $V_0 \approx 1.23508$ without intermediate rounding. Notice that UUD and UDU have the same terminal stock price but different payoffs. Thus the reward cannot be priced as a function of S_3 alone.

4.10 All stages of the lookback backward recursion

A time-two node is indexed by the entire history or by the Markov state (S_2, M_2). Starting from the terminal rewards,

$$\begin{aligned}
V_{2,UU} &= e^{-r}[q \cdot 0 + p \cdot 8] = 8e^{-r}p, \\
V_{2,UD} &= e^{-r}[q \cdot 0 + p \cdot 6] = 6e^{-r}p, \\
V_{2,DU} &= e^{-r}[q \cdot 0 + p \cdot 2] = 2e^{-r}p, \\
V_{2,DD} &= e^{-r}[q \cdot 2 + p \cdot \tfrac{7}{2}] = e^{-r}(2q + \tfrac{7}{2}p).
\end{aligned}$$

Here UD and DU both have stock price 4 at time two but differ in running maximum, so their option values are not equal. Continue to time one:

$$\begin{aligned}
V_{1,U} &= e^{-r}[qV_{2,UU} + pV_{2,UD}] \\
&= e^{-2r}[8qp + 6p^2], \\
V_{1,D} &= e^{-r}[qV_{2,DU} + pV_{2,DD}] \\
&= e^{-2r}[2qp + 2qp + \tfrac{7}{2}p^2] \\
&= e^{-2r}[4qp + \tfrac{7}{2}p^2].
\end{aligned}$$

Finally,

$$\begin{aligned}
V_0 &= e^{-r}[qV_{1,U} + pV_{1,D}] \\
&= e^{-3r}\{q[8qp + 6p^2] + p[4qp + \tfrac{7}{2}p^2]\} \\
&= e^{-3r}[8q^2p + 10qp^2 + \tfrac{7}{2}p^3],
\end{aligned}$$

exactly agreeing with the full-path expectation. The time-zero stock hedge and bank position are

$$\begin{aligned}
\Delta_0 &= \frac{V_{1,U} - V_{1,D}}{4u - 4d} = \frac{V_{1,U} - V_{1,D}}{6}, \\
\beta_0 &= V_0 - 4\Delta_0.
\end{aligned}$$

This is the precise way to compute a dynamic replicating portfolio for a path-dependent option.

4.11 An American put: backward induction and an exercise comparison

Let $S_0 = 100$, $u = 1.2$, $d = 0.8$, $K = 100$, a one-year maturity split into two half-year steps, and $r = 0.05$. The risk-neutral up probability is $q = (e^{0.025} - 0.8)/(1.2 - 0.8)$. Terminal put payoffs at stock prices $144, 96, 64$ are

$$\begin{aligned}
P_{2,uu} &= (100 - 144)^+ = 0, \\
P_{2,ud} &= (100 - 96)^+ = 4, \\
P_{2,dd} &= (100 - 64)^+ = 36.
\end{aligned}$$

At the up node $S_1 = 120$, immediate exercise gives zero and continuation gives $e^{-0.025}[4(1 - q)]$. Hence

$$P_{1,u} = \max\{0, 4e^{-0.025}(1 - q)\} = 4e^{-0.025}(1 - q).$$

At the down node $S_1 = 80$, immediate exercise gives 20 and continuation gives

$$\begin{aligned} C_{1,d} &= e^{-0.025}[q \cdot 4 + (1-q) \cdot 36] \\ &= e^{-0.025}(36 - 32q). \end{aligned}$$

Since $q \approx 0.56329$, continuation is approximately 17.54, below the immediate payoff 20, and exercise is optimal:

$$P_{1,d} = \max\{20, e^{-0.025}(36 - 32q)\} = 20.$$

At time zero the immediate payoff is zero, so

$$\begin{aligned} P_0 &= \max\{0, e^{-0.025}[qP_{1,u} + (1-q)P_{1,d}]\} \\ &= e^{-0.025}\left[4qe^{-0.025}(1-q) + 20(1-q)\right]. \end{aligned}$$

This worked tree demonstrates that early exercise can occur at intermediate nodes even when the stock remains random and future payoffs can be high. The holder compares immediate payment with the *conditional expected* continuation value, not with the single largest future payoff.

4.12 An implementable American lattice routine

Algorithm 4.4 (Backward American exercise)**.** For a nonpath-dependent reward $g(s,t)$ on a recombining tree:

1. Set $V_{N,j} = g(S_{N,j}, T)$ and mark the maturity nodes as exercise nodes.
2. For each earlier time $n = N-1, \ldots, 0$, calculate $C_{n,j} = e^{-rh}[qV_{n+1,j+1} + (1-q)V_{n+1,j}]$.
3. Compare $E_{n,j} = g(S_{n,j}, t_n)$ with continuation: set $V_{n,j} = \max\{E_{n,j}, C_{n,j}\}$.
4. Record the exercise decision when $E_{n,j} \geq C_{n,j}$; a tie convention may be selected for numerical reproducibility.
5. Return the price and exercise boundary implied by the marked exercise nodes.

The exercise comparison is made *after* the later values have been computed. A version that inserts the terminal exercise payoff into an otherwise European discounting calculation but does not recheck every interior node is not an American pricing algorithm. The routine's fundamental validation inequalities are $V_{n,j} \geq g(S_{n,j}, t_n)$ and $V_{n,j} \geq C_{n,j}$.

LECTURE 5

From Binomial Models to Itô Processes

5.1 The diffusion limit

Consider an interval of duration h with symmetric log-price increments of order $\sqrt{h}$. The Cox–Ross–Rubinstein multipliers are

$$u = e^{\sigma\sqrt{h}}, \qquad d = e^{-\sigma\sqrt{h}}, \qquad q_h = \frac{e^{rh} - d}{u - d}.$$

Expand around $h = 0$ to see the risk-neutral drift:

$$\begin{aligned} u &= 1 + \sigma\sqrt{h} + \tfrac{1}{2}\sigma^2 h + O(h^{3/2}), \\ d &= 1 - \sigma\sqrt{h} + \tfrac{1}{2}\sigma^2 h + O(h^{3/2}), \\ e^{rh} &= 1 + rh + O(h^2), \\ q_h &= \frac{2\sigma\sqrt{h} + (r - \sigma^2/2)h + O(h^{3/2})}{2\sigma\sqrt{h} + O(h^{3/2})} \\ &= \tfrac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}\sqrt{h} + O(h). \end{aligned}$$

As $h \downarrow 0$ with $Nh = T$ fixed, the log-price converges in distribution toward a process with drift $r - \sigma^2/2$ and diffusion coefficient σ.

5.2 Brownian motion and stochastic calculus

Definition 5.1 (Standard Brownian motion)**.** A process $(W_t)_{t \geq 0}$ is standard Brownian motion if $W_0 = 0$ almost surely, its sample paths are continuous, it has independent increments, and $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$.

An *Itô process* has the differential representation

$$\mathrm{d}X_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}W_t,$$

interpreted through $X_t = X_0 + \int_0^t a_s \,\mathrm{d}s + \int_0^t b_s \,\mathrm{d}W_s$. For a smooth function $f(t, x)$, Itô's formula is

$$\mathrm{d}f(t, X_t) = \left(f_t + a_t f_x + \tfrac{1}{2} b_t^2 f_{xx}\right) \mathrm{d}t + b_t f_x \,\mathrm{d}W_t.$$

The extra second-order derivative is forced by the fact that Brownian increments are of size $\sqrt{\mathrm{d}t}$, so their squares contribute at order $\mathrm{d}t$. Formally, $(\mathrm{d}W_t)^2 = \mathrm{d}t$, $\mathrm{d}W_t \,\mathrm{d}t = 0$, and $(\mathrm{d}t)^2 = 0$.

5.3 The Black–Scholes stock model

Under a physical probability measure, the geometric Brownian motion model is

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{P}}.$$

Applying Itô's formula to $f(s) = \log s$ yields

$$\begin{aligned}\mathrm{d}\log S_t &= \frac{1}{S_t}\,\mathrm{d}S_t - \frac{1}{2S_t^2}(\,\mathrm{d}S_t)^2 \\ &= \left(\mu - \frac{1}{2}\sigma^2\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}.\end{aligned}$$

Integration and exponentiation give the exact solution

$$S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{P}}\right\}.$$

Under the risk-neutral measure, the non-dividend-paying stock has $\mu = r$. The associated pricing identity for an integrable claim H is

$$V_t = \mathbb{E}^{\mathbb{Q}}\left[e^{-r(T-t)}H \mid \mathcal{F}_t\right].$$

5.4 Brownian scaling and martingale structure

Brownian motion possesses the scaling property $(W_{ct})_{t\ge 0} \overset{d}{=} (\sqrt{c}\,W_t)_{t\ge 0}$ for every $c > 0$. To verify its finite-dimensional distributions, note that both processes have zero means and covariance

$$\operatorname{Cov}(W_{cs}, W_{ct}) = c\min(s,t) = \operatorname{Cov}(\sqrt{c}\,W_s, \sqrt{c}\,W_t).$$

Since both are centered Gaussian processes with continuous paths, their laws agree. Scaling explains why increments are of order $\sqrt{h}$ rather than order h: $W_{t+h} - W_t \sim \sqrt{h}\,Z$. As $h \to 0$, the ratio $(W_{t+h} - W_t)/h$ has variance $1/h \to \infty$, indicating why classical pathwise differentiation is impossible.

Brownian motion is a martingale because for $s < t$,

$$\begin{aligned}\mathbb{E}[W_t \mid \mathcal{F}_s] &= \mathbb{E}[W_s + (W_t - W_s) \mid \mathcal{F}_s] \\ &= W_s + \mathbb{E}[W_t - W_s] = W_s.\end{aligned}$$

The process $W_t^2 - t$ is likewise a martingale, but this second fact requires knowing that the conditional second moment of a Brownian increment is $t - s$:

$$\begin{aligned}\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] &= W_s^2 + 2W_s\mathbb{E}[W_t - W_s] + \mathbb{E}[(W_t - W_s)^2] - t \\ &= W_s^2 + (t-s) - t = W_s^2 - s.\end{aligned}$$

These are prototypes for the stochastic integral identities proved later.

5.5 Dynamics of a nonlinear stock transform

If S_t follows GBM and $Y_t = S_t^\alpha$ for a real exponent α, then $f(s) = s^\alpha$ has first two derivatives $\alpha s^{\alpha-1}$ and $\alpha(\alpha-1)s^{\alpha-2}$. Itô's formula gives

$$\begin{aligned} \mathrm{d}Y_t &= \alpha S_t^{\alpha-1}(\mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t) \\ &\quad + \tfrac{1}{2}\alpha(\alpha-1)S_t^{\alpha-2}\sigma^2 S_t^2\,\mathrm{d}t \\ &= Y_t\left[\left(\alpha\mu + \tfrac{1}{2}\alpha(\alpha-1)\sigma^2\right)\,\mathrm{d}t + \alpha\sigma\,\mathrm{d}W_t\right]. \end{aligned}$$

The variance-driven drift correction is present whenever f is nonlinear. For $\alpha = 1$ the correction disappears, but for $\alpha = 2$ it contributes $\sigma^2 S_t^2\,\mathrm{d}t$.

5.6 Scaling a lattice to obtain the risk-neutral diffusion

For $h = T/N$ choose the Cox–Ross–Rubinstein multipliers $u_h = e^{\sigma\sqrt{h}}$, $d_h = e^{-\sigma\sqrt{h}}$. The risk-neutral up probability is

$$q_h = \frac{e^{rh} - e^{-\sigma\sqrt{h}}}{e^{\sigma\sqrt{h}} - e^{-\sigma\sqrt{h}}}.$$

Using Taylor expansion in powers of $\sqrt{h}$ gives

$$\begin{aligned} e^{rh} &= 1 + rh + O(h^2), \\ e^{\sigma\sqrt{h}} &= 1 + \sigma\sqrt{h} + \frac{\sigma^2 h}{2} + \frac{\sigma^3 h^{3/2}}{6} + O(h^2), \\ e^{-\sigma\sqrt{h}} &= 1 - \sigma\sqrt{h} + \frac{\sigma^2 h}{2} - \frac{\sigma^3 h^{3/2}}{6} + O(h^2), \\ e^{rh} - e^{-\sigma\sqrt{h}} &= \sigma\sqrt{h} + \left(r - \frac{\sigma^2}{2}\right)h + \frac{\sigma^3 h^{3/2}}{6} + O(h^2), \\ e^{\sigma\sqrt{h}} - e^{-\sigma\sqrt{h}} &= 2\sigma\sqrt{h} + \frac{\sigma^3 h^{3/2}}{3} + O(h^{5/2}), \\ q_h &= \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}\sqrt{h} + O(h^{3/2}). \end{aligned}$$

Let $\eta_j = +1$ on an up step and -1 on a down step. The log increment is $\sigma\sqrt{h}\,\eta_j$ and hence

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[\Delta \log S_j] &= \sigma\sqrt{h}(2q_h - 1) \\ &= \left(r - \frac{\sigma^2}{2}\right)h + O(h^2), \\ \mathrm{Var}^{\mathbb{Q}}(\Delta \log S_j) &= \sigma^2 h\{1 - (2q_h - 1)^2\} \\ &= \sigma^2 h + O(h^2). \end{aligned}$$

After $N = T/h$ steps, the log-price therefore has mean tending to $\log S_0 + (r - \sigma^2/2)T$ and variance tending to $\sigma^2 T$. A functional central limit theorem provides convergence of the interpolated processes to Brownian-driven log-price dynamics under appropriate conditions; convergence of individual terminal distributions alone is not the complete pathwise result.

5.7 The multiplicative solution and the Itô correction

Assume under $\mathbb{Q}$ that $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ and $S_t > 0$. Let $f(s) = \log s$. Itô's formula, using $f'(s) = 1/s$ and $f''(s) = -1/s^2$, gives

$$\begin{aligned}
\mathrm{d}\log S_t &= f'(S_t)\,\mathrm{d}S_t + \frac{1}{2}f''(S_t)(\sigma S_t)^2\,\mathrm{d}t \\
&= \frac{1}{S_t}(rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t) - \frac{1}{2}\frac{1}{S_t^2}\sigma^2 S_t^2\,\mathrm{d}t \\
&= \left(r - \frac{\sigma^2}{2}\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \\
\log S_T - \log S_0 &= \left(r - \frac{\sigma^2}{2}\right)T + \sigma(W_T - W_0), \\
S_T &= S_0 \exp\left\{\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T\right\}.
\end{aligned}$$

The $-\sigma^2/2$ correction is essential. Omitting it would imply $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{(r+\sigma^2/2)T}$, inconsistent with the risk-neutral martingale restriction.

5.8 A direct check by the moment-generating function

Because $W_T \sim \mathcal{N}(0, T)$, for every $a \in \mathbb{R}$,

$$\begin{aligned}
\mathbb{E}[e^{aW_T}] &= \int_{-\infty}^{\infty} e^{aw}\frac{e^{-w^2/(2T)}}{\sqrt{2\pi T}}\,\mathrm{d}w \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi T}}\exp\left\{-\frac{(w-aT)^2}{2T} + \frac{a^2T}{2}\right\}\,\mathrm{d}w \\
&= e^{a^2T/2}.
\end{aligned}$$

Therefore $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{(r-\sigma^2/2)T}e^{\sigma^2 T/2} = S_0 e^{rT}$. More generally,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[S_T^k] &= S_0^k e^{k(r-\sigma^2/2)T}\mathbb{E}[e^{k\sigma W_T}] \\
&= S_0^k \exp\left\{krT + \frac{k(k-1)}{2}\sigma^2 T\right\}.
\end{aligned}$$

For $k = 2$, $\mathrm{Var}(S_T) = S_0^2 e^{2rT}(e^{\sigma^2 T} - 1)$, providing an exact benchmark for testing a numerical simulation.

5.9 Simple random walk convergence: finite-dimensional normality

Let $\eta_1, \ldots, \eta_n$ be independent Rademacher random variables with equal probabilities for $+1$ and -1. Define a diffusively rescaled random walk by $W_t^{(n)} = n^{-1/2} \sum_{j=1}^{\lfloor nt \rfloor} \eta_j$. For any fixed $t > 0$,

$$\mathbb{E}[W_t^{(n)}] = \frac{1}{\sqrt{n}} \sum_{j=1}^{\lfloor nt \rfloor} \mathbb{E}[\eta_j] = 0,$$

$$\mathrm{Var}(W_t^{(n)}) = \frac{1}{n} \sum_{j=1}^{\lfloor nt \rfloor} \mathrm{Var}(\eta_j) = \frac{\lfloor nt \rfloor}{n} \longrightarrow t.$$

By the central limit theorem, $W_t^{(n)} \Rightarrow \mathcal{N}(0, t)$. More generally, for disjoint time intervals the corresponding random-walk increments depend on disjoint sets of the independent η_j, so they are independent. Their limiting variances equal their interval lengths. These finite-dimensional facts match Brownian motion. To obtain process-level convergence in a suitable path space, one needs a tightness argument in addition to finite-dimensional convergence; this is the content of a functional invariance principle.

5.10 Brownian covariance derived from independent increments

For $s \leq t$, write $W_t = W_s + (W_t - W_s)$. Because Brownian increments are independent and centered,

$$\begin{aligned} \mathrm{Cov}(W_s, W_t) &= \mathrm{Cov}(W_s, W_s) + \mathrm{Cov}(W_s, W_t - W_s) \\ &= \mathrm{Var}(W_s) + 0 = s. \end{aligned}$$

By symmetry the covariance for any $s, t \geq 0$ is $\mathrm{Cov}(W_s, W_t) = \min(s, t)$. This covariance formula is used repeatedly in simulation, variance reduction, geometric-average Asian options, and Gaussian process interpolation. It follows from the increment axioms and does not need to be memorized as an independent postulate.

5.11 Why a diffusion requires the square-root time scale

Suppose a random-walk increment has magnitude a_h and is repeated T/h times. The accumulated variance behaves like

$$\mathrm{Var}\left(\sum_{j=1}^{T/h} a_h \eta_j \right) = \frac{T}{h} a_h^2.$$

For a finite nonzero diffusion variance as $h \to 0$, one requires $a_h^2/h \to c \in (0, \infty)$, hence a_h of order $\sqrt{h}$. An increment of order h instead yields total variance $O(h) \to 0$, producing a deterministic limit; an increment of order 1 yields variance of order $h^{-1} \to \infty$. This scaling is the key to the $\sigma\sqrt{\Delta t} Z$ term used in every Brownian Monte Carlo and SDE numerical scheme.

5.12 A characteristic-function derivation of the Brownian limit

For a symmetric random-walk step $\eta \in \{-1, +1\}$, the characteristic function is $\mathbb{E}[e^{i\theta\eta}] = \frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \cos\theta$. For the scaled sum $Y_n = \sqrt{T/n}\sum_{j=1}^{n}\eta_j$, independence gives

$$\begin{aligned}\mathbb{E}[e^{i\theta Y_n}] &= \prod_{j=1}^{n}\mathbb{E}[e^{i\theta\sqrt{T/n}\eta_j}] \\ &= \left[\cos\left(\theta\sqrt{\frac{T}{n}}\right)\right]^n.\end{aligned}$$

Since $\log\cos x = -x^2/2 + O(x^4)$ near zero,

$$\begin{aligned}\log\mathbb{E}[e^{i\theta Y_n}] &= n\log\cos\left(\theta\sqrt{\frac{T}{n}}\right) \\ &= n\left[-\frac{\theta^2 T}{2n} + O(n^{-2})\right] \\ &= -\frac{1}{2}\theta^2 T + O(n^{-1}).\end{aligned}$$

Exponentiation gives the limiting characteristic function $e^{-\theta^2 T/2}$ of $\mathcal{N}(0, T)$. This proves convergence in distribution at one time by the continuity theorem. The functional Brownian limit adds joint-time consistency and path-space tightness, as discussed earlier.

5.13 Conditional lognormal transitions

If $0 \le s < t$, then $W_t - W_s \sim \mathcal{N}(0, t-s)$ independently of $\mathcal{F}_s$. Therefore under GBM,

$$\begin{aligned}S_t &= S_s\exp\{(r - \tfrac{1}{2}\sigma^2)(t-s) + \sigma(W_t - W_s)\}, \\ \log(S_t/S_s) \mid \mathcal{F}_s &\sim \mathcal{N}\left((r - \tfrac{1}{2}\sigma^2)(t-s), \sigma^2(t-s)\right).\end{aligned}$$

The conditional mean and variance follow from lognormal moments:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[S_t \mid \mathcal{F}_s] &= S_s e^{r(t-s)}, \\ \mathrm{Var}^{\mathbb{Q}}(S_t \mid \mathcal{F}_s) &= S_s^2 e^{2r(t-s)}\{e^{\sigma^2(t-s)} - 1\}.\end{aligned}$$

These exact conditional formulas justify iterating GBM transition samples across unequal time steps, and they provide conditional moment tests at every date of a simulated trajectory.

5.14 A diffusion-limit verification protocol

For a sequence of CRR lattices with $N \to \infty$, keep T, r, σ, S_0 fixed and set $h = T/N$, $u = e^{\sigma\sqrt{h}}$, $d = e^{-\sigma\sqrt{h}}$ and $q = (e^{rh} - d)/(u - d)$. For each N, compute the exact binomial mean and second moment of the terminal stock:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[S_N] &= S_0\{qu + (1-q)d\}^N = S_0 e^{rT}, \\ \mathbb{E}^{\mathbb{Q}}[S_N^2] &= S_0^2\{qu^2 + (1-q)d^2\}^N.\end{aligned}$$

As N increases, the second moment should approach $S_0^2 e^{2rT+\sigma^2 T}$. Option prices for smooth and nonsmooth payoffs may converge at different rates and may oscillate with lattice parity. To assess convergence, evaluate a sequence of N values rather than only one coarsely discretized tree, compare with a known Black–Scholes benchmark where available, and distinguish model approximation from floating-point instability.

LECTURE 6

Probability Review, Black–Scholes, and Monte Carlo

6.1 Normal and lognormal random variables

If $Z \sim \mathcal{N}(0,1)$, its density is $\varphi(z) = (2\pi)^{-1/2} e^{-z^2/2}$. The moment-generating function follows by completing the square:

$$\begin{aligned}\mathbb{E}[e^{tZ}] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{tz - z^2/2}\, \mathrm{d}z \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(z-t)^2/2}\, \mathrm{d}z = e^{t^2/2}.\end{aligned}$$

Thus, for $X \sim \mathcal{N}(m, v)$, $\mathbb{E}[e^{tX}] = e^{tm + t^2 v/2}$. Consequently, if $S_T = S_0 e^{(r - \sigma^2/2)T + \sigma\sqrt{T}Z}$,

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{(r-\sigma^2/2)T} \mathbb{E}[e^{\sigma\sqrt{T}Z}] = S_0 e^{rT}.$$

This is precisely the risk-neutral stock martingale requirement after discounting.

6.2 Exact simulation of Brownian motion and geometric Brownian motion

On a grid $0 = t_0 < \cdots < t_N = T$, sample independent $Z_j \sim \mathcal{N}(0,1)$. Then

$$\begin{aligned}W_{t_{j+1}} &= W_{t_j} + \sqrt{t_{j+1} - t_j}\, Z_{j+1}, \\ S_{t_{j+1}} &= S_{t_j} \exp\left\{ (r - \tfrac{1}{2}\sigma^2)(t_{j+1} - t_j) + \sigma \sqrt{t_{j+1} - t_j}\, Z_{j+1} \right\}.\end{aligned}$$

These updates are exact at the chosen observation dates in the GBM model; the randomness in between grid points is not needed for discretely sampled Asian payoffs, but it can matter for continuously monitored barriers and extrema.

Example 6.1 (Asian call Monte Carlo)**.** For m equally spaced fixing dates t_j, simulate n independent GBM paths. For each path i, compute

$$Y_i = e^{-rT} \left(\frac{1}{m} \sum_{j=1}^{m} S_{t_j}^{(i)} - K \right)^+.$$

Then $\widehat{V}_n = n^{-1} \sum_{i=1}^{n} Y_i$ estimates the risk-neutral option value. Since the averaging and positive-part operations are nonlinear, one must compute the payoff *on each path before averaging across paths*.

6.3 Statistical foundations of crude Monte Carlo

Let $Y_1, \dots, Y_n$ be independent, identically distributed, and integrable, with target $\theta = \mathbb{E}[Y_1]$. Define $\widehat{\theta}_n = n^{-1} \sum_{i=1}^n Y_i$. By linearity, $\mathbb{E}[\widehat{\theta}_n] = \theta$. If $\mathrm{Var}(Y_1) = v < \infty$, independence gives

$$\mathrm{Var}(\widehat{\theta}_n) = \frac{1}{n^2} \sum_{i=1}^n \mathrm{Var}(Y_i) = \frac{v}{n}.$$

The strong law of large numbers gives $\widehat{\theta}_n \to \theta$ almost surely. The central limit theorem gives

$$\sqrt{n} \frac{\widehat{\theta}_n - \theta}{\sqrt{v}} \Rightarrow \mathcal{N}(0,1).$$

Replacing v with $s_n^2 = (n-1)^{-1} \sum_{i=1}^n (Y_i - \widehat{\theta}_n)^2$ gives an approximate two-sided confidence interval

$$\left[\widehat{\theta}_n - z_{1-\alpha/2} \frac{s_n}{\sqrt{n}},\ \widehat{\theta}_n + z_{1-\alpha/2} \frac{s_n}{\sqrt{n}} \right].$$

To halve the standard error, the number of independent samples must be multiplied by four. This slow $O(n^{-1/2})$ rate motivates variance reduction.

6.4 Risk-neutral European option valuation: a normal integral

Under the no-dividend Black–Scholes model,

$$S_T = s \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau} Z \right), \qquad Z \sim \mathcal{N}(0,1), \quad \tau = T - t.$$

Define $a = (\log(K/s) - (r - \sigma^2/2)\tau)/(\sigma\sqrt{\tau}) = -d_2$. Since $S_T > K$ exactly when $Z > a$, the call price is

$$\begin{aligned} C(t,s) &= e^{-r\tau} \mathbb{E}[(S_T - K)^+] \\ &= e^{-r\tau} \left(\mathbb{E}[S_T \mathbf{1}_{\{Z > a\}}] - K \mathbb{P}(Z > a) \right). \end{aligned}$$

The truncated lognormal expectation is

$$\begin{aligned} \mathbb{E}[S_T \mathbf{1}_{\{Z>a\}}] &= s e^{(r-\sigma^2/2)\tau} \int_a^\infty e^{\sigma\sqrt{\tau} z} \varphi(z)\, \mathrm{d}z \\ &= s e^{(r-\sigma^2/2)\tau} e^{\sigma^2\tau/2} \int_{a-\sigma\sqrt{\tau}}^\infty \varphi(y)\, \mathrm{d}y \\ &= s e^{r\tau} \Phi(\sigma\sqrt{\tau} - a) = s e^{r\tau} \Phi(d_1). \end{aligned}$$

Also, $\mathbb{P}(Z > a) = \Phi(d_2)$, so the resulting call value is $s\Phi(d_1) - K e^{-r\tau} \Phi(d_2)$. This explicit derivation explains why pricing a European call requires a *truncated* lognormal moment, not merely $\mathbb{E}[S_T]$.

6.5 Estimator design and confidence intervals for options

A Monte Carlo estimate should report more than a point value. Suppose the simulated discounted payoffs are $Y_1, \dots, Y_n$, where Y_i may be heavily right-skewed for a deep-out-of-the-money call. The

sample standard error is

$$\widehat{\mathrm{SE}}(\widehat{V}_n) = \frac{s_Y}{\sqrt{n}}, \qquad s_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \widehat{V}_n)^2.$$

If the payoff has finite variance and n is sufficiently large for the normal approximation to be reliable, the interval $\widehat{V}_n \pm 1.96 s_Y/\sqrt{n}$ has approximately 95% coverage. For rare-event options, most Y_i may be zero. In this situation the CLT can converge slowly in practice and an apparently tight interval from a small sample may be misleading. A simulation diagnostic should include nonzero-payoff frequency, independent-run replicates, and stability as n increases.

6.6 A derivation of the Black–Scholes call formula from an integral

At current time t, let $\tau = T - t$. Under the risk-neutral Black–Scholes model, conditional on $S_t = s$,

$$S_T = s\exp\left\{(r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z\right\}, \qquad Z \sim \mathcal{N}(0,1).$$

The exercise event $S_T > K$ is equivalent, step by step, to

$$\begin{aligned} s\exp\left\{(r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z\right\} &> K, \\ (r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z &> \log(K/s), \\ Z &> \frac{\log(K/s) - (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} = -d_2, \end{aligned}$$

where

$$\begin{aligned} d_2 &= \frac{\log(s/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \\ d_1 &= d_2 + \sigma\sqrt{\tau} = \frac{\log(s/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}. \end{aligned}$$

Using $(S_T - K)^+ = S_T\mathbf{1}_{\{S_T>K\}} - K\mathbf{1}_{\{S_T>K\}}$, the call price separates into two expectations:

$$\begin{aligned} C(s,t) &= e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ \mid S_t = s] \\ &= e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{Z>-d_2\}} \mid S_t = s] - Ke^{-r\tau}\mathbb{Q}(Z > -d_2). \end{aligned}$$

By symmetry of the standard normal cdf, $\mathbb{Q}(Z > -d_2) = \Phi(d_2)$. For the stock term we compute the Gaussian integral rather than quoting a known result:

$$\begin{aligned} &\mathbb{E}[e^{\sigma\sqrt{\tau}Z}\mathbf{1}_{\{Z>-d_2\}}] \\ &= \frac{1}{\sqrt{2\pi}}\int_{-d_2}^{\infty}\exp\left\{\sigma\sqrt{\tau}z - \frac{z^2}{2}\right\}\mathrm{d}z \\ &= \frac{1}{\sqrt{2\pi}}\int_{-d_2}^{\infty}\exp\left\{-\frac{1}{2}(z - \sigma\sqrt{\tau})^2 + \frac{1}{2}\sigma^2\tau\right\}\mathrm{d}z \\ &= e^{\sigma^2\tau/2}\frac{1}{\sqrt{2\pi}}\int_{-d_2-\sigma\sqrt{\tau}}^{\infty}e^{-y^2/2}\,\mathrm{d}y \\ &= e^{\sigma^2\tau/2}\Phi(d_2 + \sigma\sqrt{\tau}) = e^{\sigma^2\tau/2}\Phi(d_1). \end{aligned}$$

Substitution now produces the closed form

$$\begin{aligned} C(s,t) &= e^{-r\tau}\left[se^{(r-\sigma^2/2)\tau}e^{\sigma^2\tau/2}\Phi(d_1) - K\Phi(d_2)\right] \\ &= s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2). \end{aligned}$$

The derivation exhibits which normal probabilities occur and exactly why the strike and stock terms involve different d-values. It also yields $P(s,t) = Ke^{-r\tau}\Phi(-d_2) - s\Phi(-d_1)$ via put–call parity.

6.7 The mean, variance, and confidence interval of crude Monte Carlo

Generate independent standard normals $Z_1, \ldots, Z_M$, and let

$$\begin{aligned} Y_i &= e^{-rT}\left(S_0 \exp\{(r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i\} - K\right)^+, \\ \widehat{C}_M &= \frac{1}{M}\sum_{i=1}^{M} Y_i. \end{aligned}$$

Since Y_i are identically distributed with finite variance,

$$\begin{aligned} \mathbb{E}[\widehat{C}_M] &= \frac{1}{M}\sum_{i=1}^{M}\mathbb{E}[Y_i] = C_0, \\ \mathrm{Var}(\widehat{C}_M) &= \frac{1}{M^2}\sum_{i=1}^{M}\mathrm{Var}(Y_i) = \frac{\mathrm{Var}(Y_1)}{M}, \\ \mathrm{SE}(\widehat{C}_M) &= \frac{\sqrt{\mathrm{Var}(Y_1)}}{\sqrt{M}}. \end{aligned}$$

Estimate the unknown variance by

$$\begin{aligned} s_Y^2 &= \frac{1}{M-1}\sum_{i=1}^{M}(Y_i - \widehat{C}_M)^2, \\ \widehat{\mathrm{SE}}(\widehat{C}_M) &= \frac{s_Y}{\sqrt{M}}. \end{aligned}$$

For large M, an approximate 95% interval is $[\widehat{C}_M - 1.96 s_Y/\sqrt{M}, \widehat{C}_M + 1.96 s_Y/\sqrt{M}]$. The central limit theorem justifies the normal approximation as $M \to \infty$, but it does not guarantee finite-sample coverage for a very skewed option payoff distribution. Independent replication batches or a pilot study may be required to assess the approximation.

6.8 From terminal simulation to an Asian payoff path

For $0 = t_0 < t_1 < \cdots < t_n = T$, set $\Delta t_j = t_j - t_{j-1}$. Draw independent $Z_j \sim \mathcal{N}(0,1)$ and update

$$\begin{aligned} S_{t_j} &= S_{t_{j-1}} \exp\{(r - \tfrac{1}{2}\sigma^2)\Delta t_j + \sigma\sqrt{\Delta t_j}Z_j\}, \\ \log S_{t_j} &= \log S_0 + \sum_{k=1}^{j}\{(r - \tfrac{1}{2}\sigma^2)\Delta t_k + \sigma\sqrt{\Delta t_k}Z_k\}. \end{aligned}$$

The arithmetic-average Asian call payoff is $(n^{-1}\sum_{j=1}^{n} S_{t_j} - K)^+$. The different S_{t_j} are *not* independent: Brownian increments are independent, whereas price levels share the increments from all earlier dates. Drawing independent terminal lognormals at every observation date gives the wrong joint path distribution and therefore the wrong Asian price.

6.9 Conditional Gaussian laws at multiple observation times

For Brownian times $0 < t_1 < \cdots < t_n$, the vector $(W_{t_1}, \ldots, W_{t_n})$ is jointly normal with mean zero and covariance matrix

$$\Sigma_{ij} = \mathrm{Cov}(W_{t_i}, W_{t_j}) = \min(t_i, t_j).$$

One may sample the same vector recursively from independent normals:

$$\begin{aligned} W_{t_1} &= \sqrt{t_1} Z_1, \\ W_{t_2} &= W_{t_1} + \sqrt{t_2 - t_1} Z_2, \\ &\vdots \\ W_{t_n} &= W_{t_{n-1}} + \sqrt{t_n - t_{n-1}} Z_n. \end{aligned}$$

To check the covariance, if $i \le j$, W_{t_i} is the sum of increments through i, while W_{t_j} contains those same increments plus independent future increments. Thus

$$\begin{aligned} \mathrm{Cov}(W_{t_i}, W_{t_j}) &= \sum_{k=1}^{i} \mathrm{Var}(\sqrt{t_k - t_{k-1}} Z_k) \\ &= \sum_{k=1}^{i} (t_k - t_{k-1}) = t_i. \end{aligned}$$

This construction is a Cholesky factorization of the Brownian covariance matrix in disguise.

6.10 A lognormal moment computation with integration details

If $Y = e^{\mu + \sigma Z}$ with $Z \sim \mathcal{N}(0,1)$,

$$\begin{aligned} \mathbb{E}[Y^k] &= \int_{-\infty}^{\infty} e^{k(\mu+\sigma z)} \frac{e^{-z^2/2}}{\sqrt{2\pi}} \, \mathrm{d}z \\ &= e^{k\mu} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left\{ -\frac{1}{2}[z^2 - 2k\sigma z] \right\} \mathrm{d}z \\ &= e^{k\mu} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left\{ -\frac{1}{2}(z - k\sigma)^2 + \frac{1}{2}k^2\sigma^2 \right\} \mathrm{d}z \\ &= e^{k\mu + k^2\sigma^2/2}. \end{aligned}$$

Taking $k = 1$ and $k = 2$, and subtracting the squared mean,

$$\begin{aligned} \mathrm{Var}(Y) &= e^{2\mu + 2\sigma^2} - (e^{\mu + \sigma^2/2})^2 \\ &= e^{2\mu + \sigma^2}(e^{\sigma^2} - 1). \end{aligned}$$

These exact moments let us check whether a simulated lognormal sample behaves sensibly before using it inside a more complicated option price estimator.

6.11 A risk-neutral Gaussian integral for the European put

Let $\tau = T - t$, $S_T = s\exp\{(r - \sigma^2/2)\tau + \sigma\sqrt{\tau}Z\}$, and d_1, d_2 as previously defined. The put payoff can be written

$$(K - S_T)^+ = (K - S_T)\mathbf{1}_{\{S_T < K\}} = (K - S_T)\mathbf{1}_{\{Z < -d_2\}}.$$

The put value separates into

$$\begin{aligned} P(s,t) &= e^{-r\tau}\mathbb{E}[(K - S_T)\mathbf{1}_{\{Z<-d_2\}}] \\ &= Ke^{-r\tau}\Phi(-d_2) - se^{-\sigma^2\tau/2}\mathbb{E}[e^{\sigma\sqrt{\tau}Z}\mathbf{1}_{\{Z<-d_2\}}]. \end{aligned}$$

Complete the square in the truncated Gaussian integral:

$$\begin{aligned} \mathbb{E}[e^{\sigma\sqrt{\tau}Z}\mathbf{1}_{\{Z<-d_2\}}] &= \int_{-\infty}^{-d_2} \frac{1}{\sqrt{2\pi}} e^{\sigma\sqrt{\tau}z - z^2/2}\,\mathrm{d}z \\ &= e^{\sigma^2\tau/2}\int_{-\infty}^{-d_2-\sigma\sqrt{\tau}} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\,\mathrm{d}y \\ &= e^{\sigma^2\tau/2}\Phi(-d_1). \end{aligned}$$

Therefore

$$P(s,t) = Ke^{-r\tau}\Phi(-d_2) - s\Phi(-d_1).$$

Subtracting yields

$$\begin{aligned} C - P &= s[\Phi(d_1) + \Phi(-d_1)] - Ke^{-r\tau}[\Phi(d_2) + \Phi(-d_2)] \\ &= s - Ke^{-r\tau}, \end{aligned}$$

which independently verifies put–call parity.

6.12 A complete Monte Carlo pipeline for an arithmetic Asian call

Algorithm 6.2 (Path simulation, discounted rewards, and confidence interval)**.** Given S_0, K, r, σ, T, monitoring times $0 < t_1 < \cdots < t_n = T$, and $M \geq 2$ independent replications:

1. For each path $i = 1, \ldots, M$, initialize $S \leftarrow S_0$ and a running sum $A \leftarrow 0$.
2. For $j = 1, \ldots, n$, independently generate $Z_{ij} \sim \mathcal{N}(0,1)$, set $h_j = t_j - t_{j-1}$, update $S \leftarrow S\exp\{(r - \sigma^2/2)h_j + \sigma\sqrt{h_j}Z_{ij}\}$ and $A \leftarrow A + S$.
3. Set the discounted path payoff $Y_i = e^{-rT}(A/n - K)^+$.
4. Compute $\widehat{C} = M^{-1}\sum_i Y_i$ and $s_Y^2 = (M-1)^{-1}\sum_i (Y_i - \widehat{C})^2$.
5. Report $\widehat{C}$ and the asymptotic 95% interval $\widehat{C} \pm 1.96 s_Y/\sqrt{M}$.

The procedure costs $O(Mn)$ updates and $O(M)$ storage if path payoffs are retained, or $O(1)$ extra storage if running mean/variance updates are used. Its time-grid simulation is exact for GBM at the monitoring dates; therefore, for this discretely monitored Asian payoff there is no extra Euler discretization bias. A continuously monitored Asian payoff would require additional analysis between monitoring dates.

LECTURE 7

Random Number Generation and Monte Carlo Inversion

7.1 Pseudo-random numbers

A pseudo-random number generator is a deterministic algorithm designed to produce numbers behaving approximately like independent $U(0,1)$ random variables. For example, a linear congruential generator uses

$$X_{k+1} = (aX_k + c) \bmod m, \qquad U_k = \frac{X_k}{m}.$$

Its sequence repeats after at most m values; poor choices of (a, c, m) can lead to short periods and severe lattice artifacts. Famous defective generators, including RANDU, illustrate why marginal uniformity does not suffice: triples of successive outputs can lie on a small number of planes. Financial simulation is sensitive to the *joint* properties of the generated numbers, especially for high-dimensional path-dependent claims.

7.2 Inverse transform sampling

Theorem 7.1 (Probability integral/inverse transform)**.** *Let F be a cumulative distribution function and let $U \sim U(0,1)$. Define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. Then $X = F^{-1}(U)$ has distribution function F.*

Proof. For every continuity point x of F, the generalized inverse has the event equivalence

$$\{F^{-1}(U) \leq x\} = \{U \leq F(x)\}.$$

Since U is uniform, $\mathbb{P}(X \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$. □

Example 7.2 (Exponential sampling)**.** If $X \sim \text{Exp}(\lambda)$, its CDF is $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. Set $U = F(X)$ and solve:

$$\begin{aligned} U &= 1 - e^{-\lambda X}, \\ e^{-\lambda X} &= 1 - U, \\ X &= -\lambda^{-1} \log(1 - U). \end{aligned}$$

Because $1 - U$ is also uniform, one can equivalently simulate $X = -\log U / \lambda$, provided $U > 0$.

For distributions lacking analytically invertible CDFs, one can numerically invert F or use alternative techniques such as rejection sampling. Numerical stability in the tails matters for rare-event payoffs.

7.3 Integration as an expectation

A Lebesgue-integrable function $g : (0,1)^d \to \mathbb{R}$ defines the integral $\theta = \int_{(0,1)^d} g(u)\,\mathrm{d}u$. If $U \sim U((0,1)^d)$, then $\theta = \mathbb{E}[g(U)]$. This converts a deterministic high-dimensional integration task to a sampling problem. The Monte Carlo estimate $n^{-1}\sum_i g(U_i)$ is consistent under integrability and admits a classical CLT under finite variance. Inverse CDFs, conditional distributions, and normal transforms allow more general pricing expectations to be expressed in this form.

7.4 A formal proof of the Monte Carlo central limit scaling

Suppose Y_i are i.i.d. with mean θ and finite positive variance v. Set $Z_i = (Y_i - \theta)/\sqrt{v}$. Then $\mathbb{E}[Z_i] = 0$, $\mathrm{Var}(Z_i) = 1$ and

$$\frac{\sqrt{n}(\widehat{\theta}_n - \theta)}{\sqrt{v}} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i.$$

The classical Lindeberg–Lévy central limit theorem yields convergence in distribution to $\mathcal{N}(0,1)$. The sample variance s_n^2 converges in probability to v under finite second moment, so Slutsky's theorem permits replacing the unknown $\sqrt{v}$ by s_n when constructing confidence intervals. This is an *asymptotic* statement and does not guarantee accurate nominal coverage for small samples or unusually heavy-tailed payoffs.

7.5 Computing expectations with one uniform variate

For a measurable g and a random variable X with CDF F, the inverse transform representation gives $X \overset{d}{=} F^{-1}(U)$ and

$$\mathbb{E}[g(X)] = \int_0^1 g(F^{-1}(u))\,\mathrm{d}u,$$

provided $g(X)$ is integrable. This formula unifies numerical quadrature and Monte Carlo. A deterministic quadrature rule selects a structured collection of u values; ordinary Monte Carlo selects independent uniforms; quasi-Monte Carlo selects a low-discrepancy set. In multidimensional simulation, one needs a mapping from uniform vectors into the joint law of the entire stochastic path, not just the correct marginals.

Example 7.3 (Exponential transform and a check)**.** Let $U \sim U(0,1)$ and $X = -\log U/\lambda$. Compute its first moment directly:

$$\begin{aligned}\mathbb{E}[X] &= -\frac{1}{\lambda}\int_0^1 \log u\,\mathrm{d}u \\ &= -\frac{1}{\lambda}\left[u\log u - u\right]_{u=0}^{u=1} = \frac{1}{\lambda}.\end{aligned}$$

A corresponding second-moment calculation gives $\mathbb{E}[X^2] = 2/\lambda^2$, hence $\mathrm{Var}(X) = 1/\lambda^2$. These moments provide useful numerical unit tests for an exponential random variate generator.

7.6 Inverse-transform sampling as a theorem

Let U be uniform on $(0,1)$, and let F be a continuous strictly increasing cdf with inverse F^{-1}. For $X = F^{-1}(U)$,

$$\begin{aligned}\mathbb{P}(X \le x) &= \mathbb{P}(F^{-1}(U) \le x)\\ &= \mathbb{P}(U \le F(x))\\ &= F(x).\end{aligned}$$

For a general cdf with jumps one replaces the ordinary inverse by $F^{-1}(u) = \inf\{x : F(x) \ge u\}$; then $\{F^{-1}(U) \le x\} = \{U \le F(x)\}$, so the same conclusion holds. The generalized inverse is important when simulation involves discrete or mixed distributions.

7.7 Exponential waiting times from a uniform variable

For $X \sim \mathrm{Exp}(\lambda)$, $F(x) = 1 - e^{-\lambda x}$ for $x \ge 0$. Taking $U = F(X)$, solve explicitly:

$$\begin{aligned}U &= 1 - e^{-\lambda X},\\ e^{-\lambda X} &= 1 - U,\\ -\lambda X &= \log(1-U),\\ X &= -\frac{1}{\lambda}\log(1-U).\end{aligned}$$

Because $1 - U \overset{d}{=} U$, $X = -\lambda^{-1}\log U$ is an equivalent expression. Its survival probability can be verified directly:

$$\begin{aligned}\mathbb{P}(X > x) &= \mathbb{P}(-\lambda^{-1}\log U > x)\\ &= \mathbb{P}(\log U < -\lambda x)\\ &= \mathbb{P}(U < e^{-\lambda x}) = e^{-\lambda x}.\end{aligned}$$

No discretization of time is necessary to simulate the event time exactly.

7.8 A derivation of the Box–Muller normal transform

Let U_1, U_2 be independent uniforms on $(0,1)$ and define $R = \sqrt{-2\log U_1}$, $\Theta = 2\pi U_2$. Set $Z_1 = R\cos\Theta$ and $Z_2 = R\sin\Theta$. The radius obeys

$$\begin{aligned}\mathbb{P}(R \le r) &= \mathbb{P}(-2\log U_1 \le r^2)\\ &= \mathbb{P}(U_1 \ge e^{-r^2/2})\\ &= 1 - e^{-r^2/2}, \qquad r \ge 0.\end{aligned}$$

Thus $f_R(r) = re^{-r^2/2}$ for $r > 0$, and Θ is independent and uniform on $[0, 2\pi)$. With polar-coordinate Jacobian $|\partial(x,y)/\partial(r,\theta)| = r$, the joint density is

$$\begin{aligned} f_{Z_1,Z_2}(x,y) &= \frac{f_R(\sqrt{x^2+y^2}) f_\Theta(\operatorname{atan2}(y,x))}{\sqrt{x^2+y^2}} \\ &= \frac{1}{2\pi} \exp\left\{-\frac{x^2+y^2}{2}\right\} \\ &= \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}. \end{aligned}$$

The factorization proves Z_1 and Z_2 are independent $\mathcal{N}(0,1)$ variables. Although numerical software commonly uses other algorithms, the transformation is a transparent mathematical link between uniform and Gaussian simulation.

7.9 How sample size changes simulation precision

If $\widehat{\mu}_M$ is an average of M independent samples with variance σ_Y^2, then

$$\begin{aligned} \mathrm{RMSE}(\widehat{\mu}_M) &= \sqrt{\mathbb{E}[(\widehat{\mu}_M - \mu)^2]} \\ &= \sqrt{\mathrm{Var}(\widehat{\mu}_M) + (\mathbb{E}[\widehat{\mu}_M] - \mu)^2} \\ &= \frac{\sigma_Y}{\sqrt{M}}. \end{aligned}$$

To halve the standard error we need $M_{\text{new}} = 4M_{\text{old}}$. To reduce it by a factor 10, we need 100 times as many paths. Monte Carlo has a rate essentially independent of integration dimension, but a large number of dimensions can still inflate the integrand variance and cause computational cost from constructing complete paths.

7.10 The classical Monte Carlo estimator for π

Let (U_i, V_i) be independent points uniformly distributed on $[0,1]^2$. Define $I_i = \mathbf{1}_{\{U_i^2+V_i^2\le 1\}}$. The area of a quarter unit disk is $\pi/4$, so I_i is Bernoulli with success probability $p = \pi/4$. Therefore $\widehat{\pi}_M = 4M^{-1}\sum_{i=1}^M I_i$ satisfies

$$\begin{aligned} \mathbb{E}[\widehat{\pi}_M] &= 4\frac{1}{M}\sum_i \mathbb{E}[I_i] = 4p = \pi, \\ \mathrm{Var}(\widehat{\pi}_M) &= 16\frac{1}{M^2}\sum_i \mathrm{Var}(I_i) \\ &= \frac{16p(1-p)}{M}. \end{aligned}$$

A consistent variance estimate is $16\widehat{p}(1-\widehat{p})/M$, where $\widehat{p} = M^{-1}\sum_i I_i$. The approximate large-sample interval for π is

$$\widehat{\pi}_M \pm z_{1-\alpha/2}\, 4\sqrt{\frac{\widehat{p}(1-\widehat{p})}{M}}.$$

For small samples, the Bernoulli-normal approximation can be inaccurate, and one may first construct an appropriate binomial proportion interval for p and then multiply both endpoints by 4.

7.11 Why estimating a tail probability is a Bernoulli Monte Carlo problem

For any simulated random output X, the target $\mu = \mathbb{P}(X > K)$ equals $\mathbb{E}[\mathbf{1}_{\{X>K\}}]$. Independent simulation paths X_i yield indicators $I_i \in \{0, 1\}$ with variance $\mu(1-\mu)$. Then

$$\begin{aligned}
\widehat{\mu}_M &= \frac{1}{M}\sum_{i=1}^{M} I_i, \\
\mathrm{Var}(\widehat{\mu}_M) &= \frac{\mu(1-\mu)}{M}, \\
\widehat{\mathrm{SE}}(\widehat{\mu}_M) &= \sqrt{\frac{\widehat{\mu}_M(1-\widehat{\mu}_M)}{M}}.
\end{aligned}$$

For rare events with $\mu \ll 1$, crude Monte Carlo often observes zero successes unless M is extremely large; the apparent estimated standard error of zero is then misleading. Importance sampling and conditional Monte Carlo are natural extensions, though they require new change-of-measure or conditioning arguments.

7.12 Proof of the strong law statement used in Monte Carlo

For integrable i.i.d. random variables $Y_1, Y_2, \ldots$ with mean μ, Kolmogorov's strong law states $M^{-1}\sum_{i=1}^{M} Y_i \to \mu$ almost surely. A simple second-moment argument proves a weaker but still informative result under finite variance: by independence,

$$\begin{aligned}
\mathbb{E}\left[\left(\frac{1}{M}\sum_{i=1}^{M} Y_i - \mu\right)^2\right] &= \frac{1}{M^2}\sum_{i=1}^{M}\mathrm{Var}(Y_i) \\
&= \frac{\mathrm{Var}(Y_1)}{M} \longrightarrow 0.
\end{aligned}$$

Thus the estimator converges in L^2, and by Markov/Chebyshev,

$$\mathbb{P}(|\widehat{\mu}_M - \mu| > \varepsilon) \le \frac{\mathrm{Var}(Y_1)}{M\varepsilon^2} \longrightarrow 0.$$

This establishes consistency in probability, but not almost-sure consistency by this inequality alone, because $\sum_M 1/M$ diverges. The actual strong law requires a stronger theorem or a refined subsequence/blocking proof. Distinguishing these statements prevents an invalid jump from variance convergence to pathwise convergence.

7.13 An asymptotic confidence interval through the central limit theorem

With i.i.d. observations of variance $0 < \sigma_Y^2 < \infty$, the standardized estimator obeys

$$\sqrt{M}\frac{\widehat{\mu}_M - \mu}{\sigma_Y} \Rightarrow \mathcal{N}(0,1).$$

If s_M^2 is the usual sample variance, consistency of $s_M \to \sigma_Y$ in probability (under the stated moment conditions) and Slutsky's theorem yield

$$\sqrt{M}\frac{\widehat{\mu}_M - \mu}{s_M} \Rightarrow \mathcal{N}(0,1).$$

Rearranging the approximate probability event

$$\mathbb{P}\left(-z_{1-\alpha/2} \le \frac{\sqrt{M}(\widehat{\mu}_M - \mu)}{s_M} \le z_{1-\alpha/2}\right) \approx 1-\alpha$$

produces the two-sided interval $\widehat{\mu}_M \pm z_{1-\alpha/2} s_M/\sqrt{M}$. The width is random because it uses estimated variance. Reporting a standard error without stating the number of *independent replications* can be seriously misleading under antithetic pairing, quasi-Monte Carlo dependence, or reused paths.

7.14 Pseudorandom generators are deterministic algorithms

A linear congruential generator (LCG) produces integers by

$$\begin{aligned} X_{n+1} &= (aX_n + c) \bmod m, \\ U_n &= X_n/m. \end{aligned}$$

Once the seed X_0 and constants a, c, m are fixed, the entire sequence is deterministic. Its statistical usefulness depends on period length and high-dimensional distribution quality, not on true randomness. If two simulations use the same seed and generator, they can reproduce the same samples; this is valuable for debugging and common-random-number comparisons. But using closely related seeds, or a historically flawed generator such as RANDU, can create strong geometric structure across successive coordinates and corrupt high-dimensional integration.

7.15 Inversion sampling under numerical limitations

For a strictly increasing continuous cdf F, one may compute $X = F^{-1}(U)$. In software, U should be chosen strictly inside $(0,1)$ when the quantile diverges at the endpoints, as in the normal or exponential distribution. For distributions whose quantile is not available analytically, solve $F(x) - U = 0$ by a bracketed root finder. Monotonicity of F ensures one-sided ordering of the quantile, but flat or discontinuous cdf regions require the generalized inverse definition. Numerical inversion tolerance contributes a separate simulation error distinct from sampling variability.

LECTURE 8

Variance Reduction: Antithetic and Control Variates

8.1 Antithetic sampling

To estimate $\theta = \mathbb{E}[g(U)]$ for $U \sim U(0,1)$, sample independent $U_1, \dots, U_m$ and construct paired averages

$$A_i = \frac{g(U_i) + g(1-U_i)}{2}, \qquad \widehat{\theta}_{\mathrm{AV}} = \frac{1}{m}\sum_{i=1}^{m} A_i.$$

Because $1 - U \sim U(0,1)$, $\mathbb{E}[A_i] = \theta$. At equal computational budgets, compare this $2m$-evaluation estimator with $2m$ independent crude samples:

$$\begin{aligned}
\mathrm{Var}(\widehat{\theta}_{\mathrm{AV}}) &= \frac{1}{4m}\left[\mathrm{Var}(g(U)) + \mathrm{Var}(g(1-U)) + 2\,\mathrm{Cov}(g(U), g(1-U))\right] \\
&= \frac{v + \mathrm{Cov}(g(U), g(1-U))}{2m}, \\
\mathrm{Var}(\widehat{\theta}_{\mathrm{MC},2m}) &= \frac{v}{2m}.
\end{aligned}$$

Antithetic sampling improves precision only when the covariance of the paired payoffs is negative; this often holds for monotone g, since $g(U)$ and $g(1-U)$ then move in opposite directions. For nonmonotone g the covariance can be positive and the estimator can perform worse.

Example 8.1 (Integrating a quadratic)**.** For $g(u) = u^2$, $\theta = 1/3$ and

$$\begin{aligned}
\mathrm{Var}(U^2) &= \mathbb{E}[U^4] - (\mathbb{E}[U^2])^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}, \\
\mathrm{Cov}(U^2, (1-U)^2) &= \mathbb{E}[U^2(1-U)^2] - \frac{1}{9} \\
&= \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right) - \frac{1}{9} = \frac{1}{30} - \frac{1}{9} = -\frac{7}{90}.
\end{aligned}$$

Consequently, $\mathrm{Var}(A_i) = \frac{1}{2}(4/45 - 7/90) = 1/180$, a substantial reduction from the variance $4/45$ of a single evaluation.

8.2 Control variates

Suppose Y is the target payoff and C is a correlated random variable with a *known* expectation $c = \mathbb{E}[C]$. For any fixed coefficient $b \in \mathbb{R}$, define

$$Z_b = Y - b(C - c), \qquad \mathbb{E}[Z_b] = \mathbb{E}[Y] = \theta.$$

Its variance is

$$\mathrm{Var}(Z_b) = \mathrm{Var}(Y) + b^2 \mathrm{Var}(C) - 2b \mathrm{Cov}(Y, C).$$

Differentiating this quadratic in b gives

$$b^* = \frac{\mathrm{Cov}(Y, C)}{\mathrm{Var}(C)}, \qquad \mathrm{Var}(Z_{b^*}) = \mathrm{Var}(Y)\left(1 - \mathrm{Corr}(Y, C)^2\right),$$

when $\mathrm{Var}(C) > 0$. Thus any nonzero correlation provides theoretical variance reduction with the optimal coefficient. In practice b^* is unknown; it can be estimated from an independent pilot sample to preserve exact unbiasedness of the main estimator conditional on that pilot.

Example 8.2 (Asian option control)**.** For an arithmetic-average Asian payoff Y, a geometric-average Asian payoff can be a useful control because it is driven by the same path and often has a known analytic model price. Draw Y_i and C_i using identical stock paths, estimate b^*, and form

$$\widehat{V}_{\mathrm{CV}} = \overline{Y} - b(\overline{C} - c).$$

The expected discounted geometric payoff c must correspond exactly to the same observation schedule, model, and discounting conventions.

8.3 Several controls and practical confidence intervals

For a control vector $C \in \mathbb{R}^k$ with known mean c, minimize $\mathrm{Var}(Y - b^\top(C - c))$ over b. Under invertibility of $\Sigma_{CC} = \mathrm{Cov}(C, C)$,

$$b^* = \Sigma_{CC}^{-1} \mathrm{Cov}(C, Y).$$

A multivariate regression of Y on C estimates these coefficients. When calculating error bars for an antithetic estimator, treat the *pair means* A_i as independent sampling units; treating all $2m$ evaluations as independent ignores within-pair correlation.

8.4 Why antithetic covariances can have either sign

The efficacy of antithetic sampling depends on the shape of g. If g is increasing, then $g(U)$ increases with U while $g(1 - U)$ decreases. A general rearrangement/covariance inequality then implies $\mathrm{Cov}(g(U), g(1 - U)) \leq 0$ when the required moments exist. If g is nonmonotone, the two transforms can move together. For example, set $g(u) = (u - 1/2)^2$. Then $g(1 - u) = g(u)$ for every u, so

$$\mathrm{Cov}(g(U), g(1 - U)) = \mathrm{Var}(g(U)) > 0.$$

At equal total payoff evaluations, pairing identical evaluations uses half as many independent inputs and doubles the estimator variance relative to independent Monte Carlo. This counterexample shows that the word *antithetic* does not itself guarantee variance reduction.

8.5 Optimal controls derived from a projection

Let $\widetilde{Y} = Y - \mathbb{E}[Y]$ and $\widetilde{C} = C - \mathbb{E}[C]$ be centered square-integrable variables. Think of L^2 as an inner-product space with $\langle X, Z\rangle = \mathbb{E}[XZ]$. The optimal scalar control coefficient minimizes

$$\left\|\widetilde{Y} - b\widetilde{C}\right\|_{L^2}^2.$$

Orthogonality of the residual to the control demands

$$\begin{aligned} 0 &= \langle \widetilde{Y} - b^*\widetilde{C}, \widetilde{C}\rangle \\ &= \mathrm{Cov}(Y, C) - b^* \,\mathrm{Var}(C), \\ b^* &= \frac{\mathrm{Cov}(Y, C)}{\mathrm{Var}(C)}. \end{aligned}$$

This is the population least-squares regression coefficient. For k controls, orthogonality to every centered control produces the normal equations $\Sigma_{CC} b = \mathrm{Cov}(C, Y)$. Strong correlation means that the payoff's random fluctuations can be predicted from control fluctuations with known expectation and subsequently subtracted.

8.6 Cost-aware variance reduction

Suppose method A has per-sample cost c_A and variance v_A for an independent sample unit, whereas method B has cost c_B and variance v_B. At a fixed total computational budget B, their approximate squared errors are proportional to $v_A c_A / B$ and $v_B c_B / B$. Consequently, the appropriate efficiency comparison is

$$\text{Efficiency ratio} = \frac{v_A c_A}{v_B c_B},$$

not merely the two variances at unequal computational effort. Antithetic sampling can yield a dramatic variance reduction while using paired evaluations; controls may require substantial pilot-estimation or analytic computation. Performance claims should always include cost as well as precision.

9.7 Antithetic estimators: a complete covariance calculation

Let $Z \sim \mathcal{N}(0, 1)$ and write $Y = f(Z)$, $Y^- = f(-Z)$. Both have the same mean μ because $Z \overset{d}{=} -Z$. The paired sample is $A = (Y + Y^-)/2$, so

$$\begin{aligned} \mathbb{E}[A] &= \frac{1}{2}\{\mathbb{E}[Y] + \mathbb{E}[Y^-]\} = \mu, \\ \mathrm{Var}(A) &= \frac{1}{4}\,\mathrm{Var}(Y + Y^-) \\ &= \frac{1}{4}\{\mathrm{Var}(Y) + \mathrm{Var}(Y^-) + 2\,\mathrm{Cov}(Y, Y^-)\} \\ &= \frac{1}{2}\{\mathrm{Var}(Y) + \mathrm{Cov}(Y, Y^-)\}. \end{aligned}$$

For a comparison with the same number of payoff evaluations, two *independent* values Y_1, Y_2 have average variance $\mathrm{Var}(Y)/2$. Therefore antithetic pairing improves variance exactly when $\mathrm{Cov}(Y, Y^-) < 0$. Monotonicity is a sufficient condition for this nonpositive covariance: if f is increasing, $f(-z)$ is decreasing in z. An increasing function and a decreasing function of the same variable have nonpositive covariance when the necessary moments exist. However, nonmonotone path-dependent payoffs need not satisfy this property.

9.8 The optimal scalar control-variate coefficient

Let Y be the discounted payoff with unknown mean μ and X a cheap correlated variable with known mean m_X. For any constant b, define $Y_b = Y - b(X - m_X)$. Then

$$\begin{aligned}
\mathbb{E}[Y_b] &= \mathbb{E}[Y] - b(\mathbb{E}[X] - m_X) = \mu, \\
\mathrm{Var}(Y_b) &= \mathrm{Var}\{Y - b(X - m_X)\} \\
&= \mathrm{Var}(Y) + b^2 \,\mathrm{Var}(X) - 2b \,\mathrm{Cov}(X, Y).
\end{aligned}$$

Differentiating the quadratic with respect to b gives

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}b} \mathrm{Var}(Y_b) &= 2b \,\mathrm{Var}(X) - 2\,\mathrm{Cov}(X, Y), \\
0 &= 2b^* \,\mathrm{Var}(X) - 2\,\mathrm{Cov}(X, Y), \\
b^* &= \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}.
\end{aligned}$$

Substituting the minimizing coefficient yields

$$\begin{aligned}
\mathrm{Var}(Y_{b^*}) &= \mathrm{Var}(Y) - \frac{\mathrm{Cov}(X, Y)^2}{\mathrm{Var}(X)} \\
&= \mathrm{Var}(Y)(1 - \rho_{XY}^2),
\end{aligned}$$

where $\rho_{XY} = \mathrm{Cov}(X, Y)/[\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}]$. The squared correlation is the *maximum proportional variance reduction* for a single linear control when population moments are known. A regression coefficient estimated on the very same sample introduces additional finite-sample subtleties; independent pilot regression or asymptotic arguments should be used when claiming exact unbiasedness.

9.9 Multiple controls and orthogonal projections

Let $X = (X_1, \ldots, X_k)^\top$ be a vector of controls with known expectation m and invertible covariance matrix Σ_{XX}. The estimator $Y_b = Y - b^\top (X - m)$ has variance

$$\begin{aligned}
\mathrm{Var}(Y_b) &= \mathrm{Var}(Y) + b^\top \Sigma_{XX} b - 2b^\top \Sigma_{XY}, \\
\nabla_b \,\mathrm{Var}(Y_b) &= 2\Sigma_{XX} b - 2\Sigma_{XY}, \\
b^* &= \Sigma_{XX}^{-1} \Sigma_{XY}, \\
\mathrm{Var}(Y_{b^*}) &= \mathrm{Var}(Y) - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}.
\end{aligned}$$

The normal equations $\mathbb{E}[(Y - b^{*\top}(X - m) - \mu)(X - m)] = 0$ say the residual is uncorrelated with every chosen control. Thus the optimal control is the orthogonal projection of the payoff onto the span of the centered controls in L^2.

9.10 A geometric-average Asian control and its exact moments

For equally spaced observations at times $t_j = jT/n$, define the geometric average $G = (\prod_{j=1}^{n} S_{t_j})^{1/n}$. Under geometric Brownian motion,

$$\begin{aligned}\log G &= \frac{1}{n}\sum_{j=1}^{n} \log S_{t_j} \\ &= \log S_0 + \left(r - \frac{1}{2}\sigma^2\right)\frac{1}{n}\sum_{j=1}^{n} t_j + \frac{\sigma}{n}\sum_{j=1}^{n} W_{t_j}.\end{aligned}$$

Since $\mathrm{Cov}(W_s, W_t) = \min(s,t)$,

$$\begin{aligned}\mathbb{E}[\log G] &= \log S_0 + \left(r - \frac{1}{2}\sigma^2\right)\frac{T(n+1)}{2n}, \\ \mathrm{Var}(\log G) &= \frac{\sigma^2}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\min(t_i, t_j) \\ &= \frac{\sigma^2 T}{n^3}\sum_{i=1}^{n}\sum_{j=1}^{n}\min(i,j) \\ &= \frac{\sigma^2 T(n+1)(2n+1)}{6n^2}.\end{aligned}$$

The last equality uses $\sum_{i,j=1}^{n}\min(i,j) = n(n+1)(2n+1)/6$. Therefore G is lognormal with known moments, and geometric Asian payoffs can be analytically priced under this model. They are useful controls for the usually more expensive arithmetic Asian payoff because the two averages are strongly correlated.

9.11 A complete antithetic example with zero variance

Let $U \sim \mathrm{Uniform}(0,1)$ and $g(u) = au + b$ for constants a, b. The target integral is

$$\begin{aligned}\mu &= \int_0^1 (au + b)\,\mathrm{d}u \\ &= \left[\frac{au^2}{2} + bu\right]_0^1 = \frac{a}{2} + b.\end{aligned}$$

For an antithetic pair,

$$\begin{aligned}A(U) &= \frac{g(U) + g(1-U)}{2} \\ &= \frac{aU + b + a(1-U) + b}{2} \\ &= \frac{a + 2b}{2} = \mu.\end{aligned}$$

Every paired observation equals the exact integral, so $\mathrm{Var}(A(U)) = 0$ for any sample size. More generally, for $g(u) = a^\top u + b$ on $[0,1]^d$, $[g(U) + g(\mathbf{1} - U)]/2 = \frac{1}{2}a^\top \mathbf{1} + b$ identically. This is an exactness property of reflection symmetry, not a numerical coincidence.

9.12 A complete example in which antithetic sampling doubles variance

Let $g(U) = (U - 1/2)^2$ for $U \sim \mathrm{Uniform}(0,1)$. Reflection about $1/2$ leaves this function unchanged:

$$\begin{aligned} g(1-U) &= (1 - U - 1/2)^2 \\ &= (1/2 - U)^2 = (U - 1/2)^2 = g(U). \end{aligned}$$

Thus the paired output is

$$A(U) = \frac{g(U) + g(1-U)}{2} = g(U).$$

Fix a total budget of M evaluations with M even. Crude Monte Carlo averages M independent values and has variance $\mathrm{Var}(g(U))/M$. The antithetic estimator has only $M/2$ independent pairs and variance

$$\begin{aligned} \mathrm{Var}(\widehat{\mu}_{\mathrm{AV}}) &= \frac{\mathrm{Var}(A(U))}{M/2} \\ &= \frac{2\,\mathrm{Var}(g(U))}{M} \\ &= 2\,\mathrm{Var}(\widehat{\mu}_{\mathrm{crude}}). \end{aligned}$$

Antithetic pairing is beneficial when the reflection creates negative covariance, not when it merely duplicates the payoff under the same computational budget.

9.13 The analytical variance of the square function

For the example $g(U) = U^2$,

$$\begin{aligned} \mathbb{E}[U^2] &= \int_0^1 u^2 \,\mathrm{d}u = \frac{1}{3}, \\ \mathbb{E}[U^4] &= \int_0^1 u^4 \,\mathrm{d}u = \frac{1}{5}, \\ \mathrm{Var}(U^2) &= \frac{1}{5} - \frac{1}{9} = \frac{4}{45}. \end{aligned}$$

Its antithetic pair has

$$\begin{aligned} A(U) &= \frac{U^2 + (1-U)^2}{2} = U^2 - U + \frac{1}{2}, \\ \mathbb{E}[A(U)] &= \frac{1}{3} - \frac{1}{2} + \frac{1}{2} = \frac{1}{3}. \end{aligned}$$

Center the polynomial: $A(U) - 1/3 = U^2 - U + 1/6$. Integrate its square,

$$\begin{aligned}\mathrm{Var}(A(U)) &= \int_0^1 \left(u^2 - u + \frac{1}{6}\right)^2 \mathrm{d}u \\ &= \int_0^1 \left(u^4 - 2u^3 + \frac{4}{3}u^2 - \frac{1}{3}u + \frac{1}{36}\right) \mathrm{d}u \\ &= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}.\end{aligned}$$

Under a budget of M calls to g, the antithetic variance is $2(1/180)/M = 1/(90M)$, versus $4/(45M) = 8/(90M)$ for crude Monte Carlo: an eightfold variance reduction with the same number of g evaluations.

9.14 An exact integral and its control-variate optimization

Let $U \sim \mathrm{Uniform}(0,1)$, $Y = U^2$, and take $X = U$ as a control, with known mean $\mathbb{E}[U] = 1/2$. The target is $\mu = \mathbb{E}[U^2] = 1/3$. Compute the moments:

$$\begin{aligned}\mathrm{Var}(U) &= \frac{1}{12}, \\ \mathrm{Cov}(U^2, U) &= \mathbb{E}[U^3] - \mathbb{E}[U^2]\mathbb{E}[U] \\ &= \frac{1}{4} - \frac{1}{3}\cdot\frac{1}{2} = \frac{1}{12}, \\ \mathrm{Var}(U^2) &= \frac{1}{5} - \frac{1}{9} = \frac{4}{45}.\end{aligned}$$

Therefore the optimal coefficient is $b^* = \mathrm{Cov}(Y, X)/\,\mathrm{Var}(X) = 1$. The adjusted sample is

$$Y_{b^*} = U^2 - (U - \tfrac{1}{2}) = U^2 - U + \frac{1}{2}.$$

Its variance is

$$\begin{aligned}\mathrm{Var}(Y_{b^*}) &= \mathrm{Var}(U^2) - \frac{\mathrm{Cov}(U^2, U)^2}{\mathrm{Var}(U)} \\ &= \frac{4}{45} - \frac{(1/12)^2}{1/12} \\ &= \frac{4}{45} - \frac{1}{12} = \frac{1}{180}.\end{aligned}$$

Thus the control reduces variance from $4/45$ to $1/180$, a sixteenfold improvement. Compare with antithetic pairing of U^2, which has pair variance $1/180$ but requires two function evaluations per pair: the control can be more efficient when evaluating U is essentially free.

9.15 Two-layer variance for an estimated control coefficient

Suppose a pilot sample produces a random estimate $\widehat{b}$, independent of the M subsequent production pairs (Y_i, X_i). Conditional on the pilot,

$$\begin{aligned}\mathbb{E}[\widehat{\mu}_{\mathrm{CV}} \mid \widehat{b}] &= \mu,\\ \mathrm{Var}(\widehat{\mu}_{\mathrm{CV}} \mid \widehat{b}) &= \frac{1}{M}\,\mathrm{Var}(Y - \widehat{b}(X - m_X) \mid \widehat{b}).\end{aligned}$$

By the law of total variance,

$$\begin{aligned}\mathrm{Var}(\widehat{\mu}_{\mathrm{CV}}) &= \mathbb{E}[\mathrm{Var}(\widehat{\mu}_{\mathrm{CV}} \mid \widehat{b})] + \mathrm{Var}(\mathbb{E}[\widehat{\mu}_{\mathrm{CV}} \mid \widehat{b}])\\ &= \frac{1}{M}\mathbb{E}[\mathrm{Var}(Y - \widehat{b}(X - m_X) \mid \widehat{b})].\end{aligned}$$

The second term is zero precisely because independent pilot estimation preserves unbiasedness conditional on the pilot. It does not say the pilot-estimated coefficient achieves the optimal population variance: finite pilot noise can raise the residual variance. This is the bias–variance–cost tradeoff in practical variance reduction.

9.16 An antithetic-control Monte Carlo routine

Algorithm 9.3 (Paired path payoffs with an independent pilot control)**.** Suppose $Y = f(Z)$ is the discounted target payoff, $X = h(Z)$ is a control with known expectation m_X, and Z is a standard normal vector.

1. Generate an independent pilot sample of n_{pilot} pairs (Y_i, X_i) and estimate b by sample covariance divided by sample control variance.
2. For production pair i, draw one vector Z_i, evaluate both Z_i and $-Z_i$, and form

$$\begin{aligned}A_i &= \frac{1}{2}\{f(Z_i) + f(-Z_i)\},\\ B_i &= \frac{1}{2}\{h(Z_i) + h(-Z_i)\}.\end{aligned}$$

3. Compute the adjusted pair $R_i = A_i - b(B_i - m_X)$.
4. Average the independent pair-level residuals $\widehat{\mu} = M^{-1}\sum_{i=1}^{M} R_i$ and estimate its standard error from the M residuals, not from the $2M$ dependent path evaluations.

Conditional on the independent pilot estimate b, $\mathbb{E}[R_i \mid b] = \mu$. If the target payoff is highly correlated with a geometric Asian control, and the antithetic pairing reduces variability in both target and control, the combination can outperform either variance-reduction technique alone. The combined optimal control coefficient is generally the coefficient for the *pair-level* variables (A, B), not automatically the coefficient for individual paths (Y, X).

LECTURE 10

Correlated Simulation, Dynamic Hedging, and Risk Measures

10.1 Multivariate Gaussian simulation

Let $Z = (Z_1, \dots, Z_d)^\top$ have independent $\mathcal{N}(0,1)$ coordinates. If $\mu \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d\times d}$, then

$$X = \mu + AZ, \qquad \mathbb{E}[X] = \mu, \qquad \operatorname{Cov}(X) = AA^\top.$$

Given a positive-definite covariance matrix Σ, the Cholesky decomposition supplies a lower-triangular L such that $\Sigma = LL^\top$. Hence $X = \mu + LZ \sim \mathcal{N}_d(\mu, \Sigma)$. If Σ is merely positive semidefinite, an eigenvalue or singular-value factorization provides a suitable rectangular square root.

Example 10.1 (Two correlated standard normals)**.** For target correlation $\rho \in [-1, 1]$, take independent $Z_1, Z_2 \sim \mathcal{N}(0,1)$ and set

$$X_1 = Z_1, \qquad X_2 = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2.$$

Then $\operatorname{Var}(X_1) = \operatorname{Var}(X_2) = 1$ and

$$\begin{aligned}\operatorname{Cov}(X_1, X_2) &= \mathbb{E}\left[Z_1\left(\rho Z_1 + \sqrt{1-\rho^2}Z_2\right)\right] \\ &= \rho\mathbb{E}[Z_1^2] + \sqrt{1-\rho^2}\mathbb{E}[Z_1 Z_2] = \rho.\end{aligned}$$

For a two-asset risk-neutral diffusion driven by correlated Brownian motions,

$$\frac{\mathrm{d}S_t^{(i)}}{S_t^{(i)}} = r\,\mathrm{d}t + \sigma_i\,\mathrm{d}W_t^{(i)}, \qquad \mathrm{d}[W^{(1)}, W^{(2)}]_t = \rho\,\mathrm{d}t,$$

apply the normal construction independently at each time step; independent draws across steps preserve the Brownian independent-increment property.

10.2 Dynamic trading and discrete delta hedging

A European derivative price $V(t,S)$ can be hedged by holding $\Delta_t = V_S(t, S_t)$ shares. If a trader sells the option, the hedge portfolio (stock plus cash minus option) is

$$\Pi_t = -V(t, S_t) + \Delta_t S_t + B_t.$$

With initial cash $B_0 = V(0,S_0) - \Delta_0 S_0$, the initial net value is zero. At rebalancing times, choose $\Delta_{t_{j+1}}$ using current information and move cash to keep the strategy self-financing. Under a discretized strategy, residual profit and loss is generally nonzero because the hedge is not continuously adjusted.

Proposition 10.2 (Leading-order unhedged option change)**.** *Over a short interval with stock change ΔS, a twice-differentiable option value satisfies*

$$\Delta V \approx V_t \Delta t + \Delta \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2, \qquad \Gamma = V_{SS}.$$

For a delta hedge fixed over the interval, the stochastic linear term is canceled, but the realized quadratic move contributes $-\frac{1}{2}\Gamma(\Delta S)^2$ to a short-option hedged portfolio, before financing and time-decay effects.

If the hedge model uses volatility $\widehat{\sigma}$ while realized instantaneous quadratic variation follows $\sigma_t^2 S_t^2\,\mathrm{d}t$, the continuous delta-hedged profit of a short European option is, under suitable assumptions,

$$\mathrm{d}\Pi_t = \tfrac{1}{2}\Gamma_t S_t^2(\widehat{\sigma}^2 - \sigma_t^2)\,\mathrm{d}t.$$

The sign reverses for a long-option position. This identity explains why realized variance relative to the model's implied variance influences the performance of delta hedging.

10.3 Gamma-neutral portfolios

Suppose, in addition to stock, one can trade another derivative $I(t, S)$ with nonzero I_{SS}. A portfolio short one unit of V, long δ_S shares, and long δ_I units of I has local delta and gamma

$$\begin{aligned}\Pi_S &= -V_S + \delta_S + \delta_I I_S,\\ \Pi_{SS} &= -V_{SS} + \delta_I I_{SS}.\end{aligned}$$

Imposing both equal to zero yields

$$\delta_I = \frac{V_{SS}}{I_{SS}}, \qquad \delta_S = V_S - \frac{V_{SS}}{I_{SS}} I_S.$$

The remaining cash holding makes the overall strategy self-financing. This hedge is local and depends on both derivatives' sensitivities, which can change over time.

10.4 Value-at-Risk and Expected Shortfall

Let L represent monetary *loss* over a fixed horizon, so that large positive values are undesirable. For a confidence level $\alpha \in (0, 1)$, define

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell \in \mathbb{R} : \mathbb{P}(L \le \ell) \ge \alpha\}.$$

For an integrable loss with a continuous distribution at the α-quantile,

$$\mathrm{ES}_\alpha(L) = \mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha(L)].$$

A more general tail-average definition, suitable in the presence of atoms, is

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,\mathrm{d}u.$$

For $L \sim \mathcal{N}(m, s^2)$, put $z_\alpha = \Phi^{-1}(\alpha)$ and note $L = m + sZ$. Then

$$\mathrm{VaR}_\alpha(L) = m + s z_\alpha,$$
$$\mathrm{ES}_\alpha(L) = m + s\frac{\varphi(z_\alpha)}{1-\alpha}.$$

Indeed, $\mathbb{E}[Z\mathbf{1}_{\{Z>z\}}] = \int_z^\infty x\varphi(x)\,\mathrm{d}x = \varphi(z)$. For simulated losses $L_1, \ldots, L_n$, estimate VaR by the empirical quantile and ES by the average of the worst tail observations (with appropriate treatment of the cutoff). Tail estimators can have considerable sampling uncertainty, particularly for large α.

10.5 Portfolio correlation and pathwise dependence

Correlation must be imposed at the level of the driving shocks, not by merely correlating terminal outputs after the simulation. Suppose $\mathrm{d}W_1\,\mathrm{d}W_2 = \rho\,\mathrm{d}t$ in the Itô multiplication notation, and let two stocks solve

$$\mathrm{d}S_t^{(1)} = \mu_1 S_t^{(1)}\,\mathrm{d}t + \sigma_1 S_t^{(1)}\,\mathrm{d}W_t^{(1)}, \qquad \mathrm{d}S_t^{(2)} = \mu_2 S_t^{(2)}\,\mathrm{d}t + \sigma_2 S_t^{(2)}\,\mathrm{d}W_t^{(2)}.$$

For log prices, $\mathrm{d}\log S_t^{(i)} = (\mu_i - \sigma_i^2/2)\,\mathrm{d}t + \sigma_i\,\mathrm{d}W_t^{(i)}$. Therefore

$$\begin{aligned}\mathrm{Cov}\left(\log\frac{S_T^{(1)}}{S_0^{(1)}}, \log\frac{S_T^{(2)}}{S_0^{(2)}}\right) &= \sigma_1\sigma_2\,\mathrm{Cov}(W_T^{(1)}, W_T^{(2)}) \\ &= \rho\sigma_1\sigma_2 T.\end{aligned}$$

Yet the *correlation of the stock returns* is not exactly ρ for finite T: exponentiation changes correlation. This matters when constructing basket option payoffs and estimating tail dependence. The driver correlation ρ belongs to the joint normal Brownian increments, while nonlinear transformations generate a different dependence measure for the resulting asset prices.

10.6 Risk estimation with a simulated loss sample

To estimate the loss distribution of a hedged option book, simulate market paths under an appropriate scenario law, update the derivatives and hedge positions at the required trading dates, and record the terminal change in value net of cash financing and any specified transaction costs. The distribution used for *risk* calculations is typically a physical or stress-scenario distribution, not automatically the same as the risk-neutral measure used for pricing. A properly implemented Monte Carlo workflow must distinguish the probability law under which it computes expected discounted payoffs from the law under which it estimates realized losses.

For example, if L_i are i.i.d. simulated portfolio losses and $L_{(1)} \le \cdots \le L_{(n)}$ are their order statistics, an empirical α-VaR estimate is $L_{(\lceil n\alpha \rceil)}$. If the upper tail has m observations and the distribution is sufficiently smooth, a simple ES estimator averages the m largest losses. In the presence of ties or a desired exact tail fraction $(1-\alpha)$, the cutoff observation may receive fractional weight. Both estimators depend strongly on the quality of the simulated tail, which is particularly challenging for portfolios short convex options.

10.7 Numerical Greeks by bump-and-revalue

If a derivative is priced by a numerical routine $V(s, \sigma, r, t)$, finite differences offer approximate sensitivities without requiring a closed formula. Holding all other inputs fixed, a central-difference delta is

$$\widehat{\Delta}_h = \frac{V(s+h, \sigma, r, t) - V(s-h, \sigma, r, t)}{2h},$$

and a central-difference gamma is

$$\widehat{\Gamma}_h = \frac{V(s+h, \sigma, r, t) - 2V(s, \sigma, r, t) + V(s-h, \sigma, r, t)}{h^2}.$$

Taylor expansion shows that if V is sufficiently smooth and the prices are exact, both formulas have truncation error $O(h^2)$. In a Monte Carlo pricer, however, each evaluated price has sampling noise. A small h reduces Taylor bias but amplifies uncorrelated simulation noise by factors of order h^{-1} for delta and h^{-2} for gamma. It is therefore often advantageous to use the same random input paths for the bumped prices, inducing positive covariance that partially cancels when differences are formed.

An alternative is the pathwise derivative method. Under an exact GBM simulation $S_T = sF(Z)$ with $F(Z) > 0$ independent of s, a smooth European payoff H permits, under suitable differentiation-under-the-expectation assumptions,

$$\begin{aligned}\Delta &= \frac{\partial}{\partial s}\left(e^{-rT}\mathbb{E}[H(sF(Z))]\right) \\ &= e^{-rT}\mathbb{E}[H'(sF(Z))F(Z)].\end{aligned}$$

For a call, $H'(x) = \mathbf{1}_{\{x>K\}}$ away from its strike, so a valid pathwise-delta estimator is

$$\widehat{\Delta}_{\mathrm{PW}} = \frac{e^{-rT}}{n}\sum_{i=1}^{n} \mathbf{1}_{\{S_T^{(i)}>K\}}\frac{S_T^{(i)}}{s}.$$

The kink does not prevent this first derivative under the continuous lognormal model, because the event $S_T = K$ has probability zero. The analogous pathwise second derivative is more delicate because the payoff derivative is discontinuous; blindly differentiating the indicator gives an incorrect zero gamma.

10.8 Constructing a bivariate Gaussian pair with exactly the desired covariance

Suppose Z_1, Z_2 are independent standard normals. For $-1 \le \rho \le 1$ set

$$\begin{aligned}X &= Z_1, \\ Y &= \rho Z_1 + \sqrt{1-\rho^2}Z_2.\end{aligned}$$

We verify the covariance structure term by term:

$$\begin{aligned}
\mathbb{E}[X] &= 0, \qquad \mathbb{E}[Y] = 0,\\
\operatorname{Var}(X) &= \operatorname{Var}(Z_1) = 1,\\
\operatorname{Var}(Y) &= \rho^2 \operatorname{Var}(Z_1) + (1-\rho^2)\operatorname{Var}(Z_2)\\
&\qquad + 2\rho\sqrt{1-\rho^2}\operatorname{Cov}(Z_1, Z_2)\\
&= \rho^2 + (1-\rho^2) = 1,\\
\operatorname{Cov}(X, Y) &= \operatorname{Cov}(Z_1, \rho Z_1 + \sqrt{1-\rho^2} Z_2)\\
&= \rho \operatorname{Var}(Z_1) + \sqrt{1-\rho^2}\operatorname{Cov}(Z_1, Z_2) = \rho.
\end{aligned}$$

The vector is jointly normal because it is a linear transform of the Gaussian vector (Z_1, Z_2). Thus its correlation is exactly ρ, not merely approximately ρ in population. The empirical correlation from a finite sample will fluctuate around ρ.

10.9 The general Cholesky algorithm and its limitations

For d asset returns with covariance matrix Σ symmetric and positive definite, factor $\Sigma = LL^\top$ with L lower triangular. If $Z \sim \mathcal{N}(0, I_d)$, set $X = \mu + LZ$. Then

$$\begin{aligned}
\mathbb{E}[X] &= \mu + L\mathbb{E}[Z] = \mu,\\
\operatorname{Cov}(X) &= \mathbb{E}[(X-\mu)(X-\mu)^\top]\\
&= L\mathbb{E}[ZZ^\top]L^\top\\
&= LI_dL^\top = \Sigma.
\end{aligned}$$

The factorization requires positive definiteness for the usual unpivoted algorithm. If Σ is only positive semidefinite, an eigenvalue or pivoted factorization may be used; if an estimated covariance matrix has small negative eigenvalues from numerical error, it should not simply be fed to a Cholesky routine without validation.

10.10 A self-financing hedging-error recursion

Let $V_n = V(S_{t_n}, t_n)$ be the theoretical option value. If one is short the option and holds δ_n shares plus cash A_n just after rebalancing at t_n, the portfolio's marked-to-model wealth is $\Pi_n = \delta_n S_{t_n} + A_n - V_n$. Over the next time step, before rebalancing, the trading gains are

$$\Pi_{n+1}^- = \delta_n S_{t_{n+1}} + A_n e^{r\Delta t_n} - V_{n+1}.$$

A self-financing rebalance changes δ_n to δ_{n+1} while adjusting cash by $-(\delta_{n+1} - \delta_n)S_{t_{n+1}}$, so the rebalance leaves Π_{n+1} unchanged. With a fair initial option premium and $A_0 = V_0 - \delta_0 S_0$, we get $\Pi_0 = 0$ and the accumulated hedging P&L equals the trading strategy's terminal wealth minus the option payoff.

10.11 Why delta hedging leaves a gamma term

Expand $V(S+\Delta S, t+\Delta t)$ around (S,t), assuming sufficiently many derivatives:

$$\begin{aligned}\Delta V = V_t\Delta t + V_S\Delta S + \frac{1}{2}V_{SS}(\Delta S)^2 \\ + V_{St}\Delta S\Delta t + \frac{1}{2}V_{tt}(\Delta t)^2 + \text{higher-order terms.}\end{aligned}$$

For a short option and $\delta = V_S$ long shares, the price-sensitive first-order term cancels:

$$\begin{aligned}\delta\Delta S - \Delta V &= (\delta - V_S)\Delta S - V_t\Delta t - \frac{1}{2}V_{SS}(\Delta S)^2 + \cdots \\ &= -V_t\Delta t - \frac{1}{2}\Gamma(\Delta S)^2 + \cdots .\end{aligned}$$

Under a diffusion $(\Delta S)^2$ has leading order $\sigma^2 S^2\Delta t$, so the gamma term remains at order Δt. The Black–Scholes PDE balances this curvature term against the drift and financing return; discrete trading still incurs nonzero realized hedging error, especially for coarse hedging grids or transaction costs.

10.12 Closed-form Gaussian Value-at-Risk and Expected Shortfall

Let portfolio loss $L \sim \mathcal{N}(m, s^2)$ with $s > 0$. Write $L = m + sZ$ and let $z_\alpha = \Phi^{-1}(\alpha)$. For a continuous loss distribution,

$$\begin{aligned}\mathrm{VaR}_\alpha(L) &= \inf\{\ell : \mathbb{P}(L \le \ell) \ge \alpha\} \\ &= m + sz_\alpha.\end{aligned}$$

Expected Shortfall is the conditional upper-tail mean:

$$\begin{aligned}\mathrm{ES}_\alpha(L) &= \mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha(L)] \\ &= m + s\mathbb{E}[Z \mid Z \ge z_\alpha] \\ &= m + s\frac{\int_{z_\alpha}^{\infty} z\varphi(z)\,\mathrm{d}z}{1-\Phi(z_\alpha)}.\end{aligned}$$

Since $\varphi'(z) = -z\varphi(z)$,

$$\begin{aligned}\int_{z_\alpha}^{\infty} z\varphi(z)\,\mathrm{d}z &= -\int_{z_\alpha}^{\infty}\varphi'(z)\,\mathrm{d}z \\ &= \varphi(z_\alpha), \\ \mathrm{ES}_\alpha(L) &= m + s\frac{\varphi(z_\alpha)}{1-\alpha}.\end{aligned}$$

The continuous-normal formula differs from the empirical treatment of atoms at the quantile; the optimization appendix will use an Expected Shortfall representation that is valid beyond normality.

10.13 A portfolio loss distribution under correlated Gaussian returns

Suppose dollar returns on two positions are R_1 and R_2, with means m_1, m_2, standard deviations s_1, s_2, and correlation ρ. Let weights be w_1, w_2. The total dollar profit is $P = w_1R_1 + w_2R_2$, and the loss is $L = -P$. By linearity,

$$\begin{aligned}\mathbb{E}[L] &= -w_1m_1 - w_2m_2,\\ \mathrm{Var}(L) &= \mathrm{Var}(w_1R_1 + w_2R_2)\\ &= w_1^2s_1^2 + w_2^2s_2^2 + 2w_1w_2\rho s_1s_2.\end{aligned}$$

With jointly normal returns, L is normal and the analytical VaR formula applies. The covariance term can make risk larger or smaller than the sum of isolated standard deviations. Setting $\rho = 0$ without justification discards diversification and concentration effects.

10.14 Deriving finite-difference delta, gamma, and vega estimators

For a smooth pricing function $V(s, \sigma, t)$ and stock bump $h_s > 0$,

$$\begin{aligned}V(s + h_s, \sigma, t) &= V + h_sV_s + \frac{h_s^2}{2}V_{ss} + \frac{h_s^3}{6}V_{sss} + O(h_s^4),\\ V(s - h_s, \sigma, t) &= V - h_sV_s + \frac{h_s^2}{2}V_{ss} - \frac{h_s^3}{6}V_{sss} + O(h_s^4).\end{aligned}$$

Thus

$$\begin{aligned}\widehat{\Delta} &= \frac{V(s + h_s) - V(s - h_s)}{2h_s} = V_s + O(h_s^2),\\ \widehat{\Gamma} &= \frac{V(s + h_s) - 2V(s) + V(s - h_s)}{h_s^2} = V_{ss} + O(h_s^2),\\ \widehat{\mathrm{Vega}} &= \frac{V(s, \sigma + h_\sigma) - V(s, \sigma - h_\sigma)}{2h_\sigma} = V_\sigma + O(h_\sigma^2).\end{aligned}$$

When V is itself estimated by Monte Carlo, using the *same random numbers* for the bumped evaluations can substantially reduce the variance of differences. For example,

$$\mathrm{Var}(\widehat{\Delta}) = \frac{1}{4h_s^2}\left[\mathrm{Var}(\widehat{V}_+) + \mathrm{Var}(\widehat{V}_-) - 2\,\mathrm{Cov}(\widehat{V}_+, \widehat{V}_-)\right].$$

If independent random numbers are used, the covariance term vanishes and the variance can blow up like h_s^{-2} as the bump becomes small. Common random numbers counteract this instability by increasing the covariance between paired estimates.

10.15 Correct empirical tail estimation

Let $L_{(1)} \le \cdots \le L_{(M)}$ be the sorted independent simulated losses. An empirical quantile is $\widehat{\mathrm{VaR}}_\alpha = L_{(\lceil M\alpha \rceil)}$ (one of several standard conventions). For the upper-tail average, it is tempting to

average all observations strictly above the quantile, but this can mishandle tied observations and the exact tail fraction $1-\alpha$. The sample optimization formula

$$\widehat{\mathrm{ES}}_\alpha = \min_{z\in\mathbb{R}} \left[z + \frac{1}{M(1-\alpha)} \sum_{i=1}^{M} (L_i - z)^+ \right]$$

provides a consistent coherent definition that also handles quantile ties via fractional mass at the boundary. At confidence $\alpha = 0.995$ with only $M = 50$ independent simulations, the expected number of observations in the upper half-percent tail is $M(1-\alpha) = 0.25$, so such a sample cannot support a reliable empirical ES estimate.

10.16 Brownian bridge conditional simulation

Let $0 < s < t < T$ and suppose Brownian endpoints $W_0 = 0$ and $W_T = w$ are known. The pair (W_t, W_T) is jointly normal with covariance $\mathrm{Cov}(W_t, W_T) = t$ and variances t and T. The conditional Gaussian formulas give

$$\begin{aligned} \mathbb{E}[W_t \mid W_T = w] &= \frac{\mathrm{Cov}(W_t, W_T)}{\mathrm{Var}(W_T)} w = \frac{t}{T} w, \\ \mathrm{Var}(W_t \mid W_T) &= \mathrm{Var}(W_t) - \frac{\mathrm{Cov}(W_t, W_T)^2}{\mathrm{Var}(W_T)} \\ &= t - \frac{t^2}{T} = \frac{t(T-t)}{T}. \end{aligned}$$

More generally, conditioning on two endpoints at $s < t < u$, one obtains

$$W_t \mid (W_s = x, W_u = y) \sim \mathcal{N}\left(x + \frac{t-s}{u-s}(y-x), \frac{(t-s)(u-t)}{u-s} \right).$$

To simulate a path on a fine grid, sample the final endpoint, then sample intermediate midpoints conditional on their surrounding endpoints, recursively refining. The resulting vector has the same Brownian joint law as forward-increment simulation, but the first coordinates of the underlying random vector can control large-scale path variation. This is helpful for quasi-Monte Carlo, where the order of coordinates can affect practical effective dimension.

10.17 Common random numbers as a covariance method

Suppose two scenarios or pricing models produce outputs $Y^{(1)}$ and $Y^{(2)}$ and we wish to estimate the difference $\Delta = \mathbb{E}[Y^{(1)} - Y^{(2)}]$. Under independent simulation of the models,

$$\mathrm{Var}(\widehat{\Delta}) = \frac{1}{M} \left[\mathrm{Var}(Y^{(1)}) + \mathrm{Var}(Y^{(2)}) \right].$$

If the scenarios are driven by a common random input, the paired difference instead satisfies

$$\begin{aligned} \mathrm{Var}(\widehat{\Delta}) &= \frac{1}{M} \mathrm{Var}(Y^{(1)} - Y^{(2)}) \\ &= \frac{1}{M} \left[\mathrm{Var}(Y^{(1)}) + \mathrm{Var}(Y^{(2)}) - 2\,\mathrm{Cov}(Y^{(1)}, Y^{(2)}) \right]. \end{aligned}$$

For a positive covariance this is smaller. The estimator remains unbiased because each scenario retains its own correct marginal distribution. Common random numbers are particularly useful for scenario differences, bump-and-revalue Greeks, and the coarse–fine coupling used in discretization extrapolation.

10.18 A dynamic portfolio's complete cash recursion

Suppose a short derivative is hedged using δ_n stock units and a bank balance A_n just after trading at time t_n. The value of the hedge assets is $H_n = \delta_n S_n + A_n$. During the next interval, holdings remain unchanged; immediately before rebalance,

$$H_{n+1}^- = \delta_n S_{n+1} + e^{r\Delta t} A_n.$$

Rebalance to δ_{n+1} and choose the cash holding

$$A_{n+1} = e^{r\Delta t} A_n - (\delta_{n+1} - \delta_n) S_{n+1}.$$

Then

$$\delta_{n+1} S_{n+1} + A_{n+1} = \delta_n S_{n+1} + e^{r\Delta t} A_n = H_{n+1}^-,$$

so rebalancing injects no external capital. If $H_0 = V_0$, terminal hedging error for selling the claim is $H_N - g(S_N)$. The recursion should be simulated pathwise, and the resulting distribution evaluated under the *physical* law when assessing real-world risk, even if hedge ratios and initial prices were computed under a risk-neutral model.

10.19 A complete simulation workflow for dynamic hedging risk

Algorithm 10.3 (Pathwise delta-hedging P&L)**.** Given an initial derivative premium V_0, a model for simulating S_{n+1} conditional on S_n, pricing delta function $\Delta(s,t)$, time grid $t_n = nh$, and risk-free rate r:

1. Initialize the hedge stock position $\delta_0 = \Delta(S_0, 0)$ and cash $A_0 = V_0 - \delta_0 S_0$.
2. At each step n, simulate the next stock value using the model, accrue cash $A_n \leftarrow e^{rh} A_n$, then calculate the new delta $\delta_{n+1} = \Delta(S_{n+1}, t_{n+1})$ if another rebalance is permitted.
3. Finance the stock rebalance through cash: $A_{n+1} = e^{rh} A_n - (\delta_{n+1} - \delta_n) S_{n+1}$.
4. At maturity compute the hedge P&L $\Pi_T = \delta_{N-1} S_N + e^{rh} A_{N-1} - g(S_N)$ (or use the equivalent recursion with the last cash/stock positions adjusted consistently).
5. Repeat on independent complete paths; estimate the sample distribution of losses $L = -\Pi_T$ and summarize VaR and ES.

It is essential to distinguish the measure used for derivative price and delta calculations (often $\mathbb{Q}$) from the measure used for realized risk scenarios (often $\mathbb{P}$). If a jump-diffusion or stochastic-volatility process is used to generate real-world outcomes, a Black–Scholes delta is generally an imperfect hedge; its P&L distribution reveals model risk as well as discrete-hedging error.

10.20 The impact of hedge-frequency and trading cost

In an ideal diffusion model, more frequent trading may reduce discretization error of delta hedging. With proportional transaction cost κ_{tc} per dollar traded, each rebalance incurs $\kappa_{\text{tc}}|\delta_{n+1} - \delta_n|S_{n+1}$. Total transaction cost is then

$$C_{\text{tc}} = \kappa_{\text{tc}} \sum_{n=0}^{N-1} |\delta_{n+1} - \delta_n| S_{n+1}.$$

Trading ever more frequently no longer guarantees better realized net P&L, because total turnover can increase as the grid is refined. Thus hedge frequency is an optimization problem balancing replication error and execution costs rather than a pure numerical-limit question.

LECTURE 11

Quadratic Variation and Construction of the Itô Integral

11.1 From ordinary to stochastic integration

For a continuously differentiable path $x : [0, T] \to \mathbb{R}$, ordinary calculus relates its increments to $\int_0^T x'(t)\,\mathrm{d}t$. Brownian paths, however, are almost surely nowhere differentiable, so $\int H_t\,\mathrm{d}W_t$ cannot be interpreted through the classical Riemann–Stieltjes integral for arbitrary adapted integrands. The correct construction begins with simple predictable processes and then passes to a suitable L^2 limit.

11.2 Quadratic variation of Brownian motion

Let $\pi_n = \{0 = t_0^{(n)} < \cdots < t_{m_n}^{(n)} = t\}$ be a sequence of partitions whose maximum mesh tends to zero. Its discrete quadratic variation is

$$Q_{\pi_n}(W) = \sum_{j=0}^{m_n-1} (W_{t_{j+1}^{(n)}} - W_{t_j^{(n)}})^2.$$

Theorem 11.1. *For Brownian motion, $Q_{\pi_n}(W) \to t$ in L^2 whenever the partition mesh tends to zero. In particular $[W]_t = t$.*

Proof. Write $\Delta_j W = W_{t_{j+1}} - W_{t_j}$ and $h_j = t_{j+1} - t_j$. Independence and $\Delta_j W \sim \mathcal{N}(0, h_j)$ give

$$\begin{aligned}
\mathbb{E}[Q_{\pi_n}(W)] &= \sum_j \mathbb{E}[(\Delta_j W)^2] = \sum_j h_j = t, \\
\operatorname{Var}(Q_{\pi_n}(W)) &= \sum_j \operatorname{Var}((\Delta_j W)^2) \\
&= \sum_j \big(\mathbb{E}[(\Delta_j W)^4] - h_j^2\big) \\
&= 2\sum_j h_j^2 \le 2\,|\pi_n| \sum_j h_j = 2t\,|\pi_n| \longrightarrow 0.
\end{aligned}$$

Thus $\mathbb{E}[(Q_{\pi_n}(W) - t)^2] \to 0$. □

A continuously differentiable path f has zero quadratic variation because

$$\sum_j (f(t_{j+1}) - f(t_j))^2 \le \Big(\max_j |f(t_{j+1}) - f(t_j)|\Big) \sum_j |f(t_{j+1}) - f(t_j)| \to 0.$$

Consequently, time has $[t]_T = 0$ but Brownian motion has positive quadratic variation. This distinction is what changes the chain rule.

11.3 Simple predictable integrands

Definition 11.2 (Itô integral of a simple process)**.** Let $H_t = \sum_{j=0}^{n-1} H_j \mathbf{1}_{(t_j, t_{j+1}]}(t)$ with H_j measurable with respect to $\mathcal{F}_{t_j}$ and $\mathbb{E}[H_j^2] < \infty$. Define

$$\int_0^T H_t \, \mathrm{d}W_t = \sum_{j=0}^{n-1} H_j (W_{t_{j+1}} - W_{t_j}).$$

The use of *left-endpoint* information matters: H_j must not depend on the future increment it multiplies. Conditioned on $\mathcal{F}_{t_j}$, the next Brownian increment has mean zero, so the integral has zero expectation.

Theorem 11.3 (Itô isometry)**.** *For every square-integrable predictable integrand H,*

$$\mathbb{E}\left[\left(\int_0^T H_t \, \mathrm{d}W_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right].$$

Proof. For simple H, expand the square of the sum. Cross terms vanish because earlier terms are measurable at the start of a later increment, which has conditional mean zero. The diagonal terms satisfy

$$\begin{aligned}\mathbb{E}\left[H_j^2 (\Delta_j W)^2\right] &= \mathbb{E}\left[H_j^2 \mathbb{E}[(\Delta_j W)^2 \mid \mathcal{F}_{t_j}]\right] \\ &= h_j \mathbb{E}[H_j^2].\end{aligned}$$

Summing proves the identity for simple processes. Approximate a general predictable H in $L^2(\Omega \times [0,T])$ by simple predictable processes; the isometry makes their stochastic integrals Cauchy in $L^2(\Omega)$ and defines the limiting integral. □

11.4 Riemann and Lebesgue–Stieltjes integration

For a smooth deterministic integrator F with derivative F', classical Riemann–Stieltjes integration reduces to

$$\int_a^b f(x) \, \mathrm{d}F(x) = \int_a^b f(x) F'(x) \, \mathrm{d}x.$$

When F is a cumulative distribution function, the Lebesgue–Stieltjes integral $\int g(x) \, \mathrm{d}F(x)$ is the expectation $\mathbb{E}[g(X)]$ if X has CDF F and $g(X)$ is integrable. This integral encompasses both continuous and discrete distributions. If X takes values x_j with probabilities p_j, then

$$\int_{\mathbb{R}} g(x) \, \mathrm{d}F(x) = \sum_j g(x_j) p_j.$$

If X has density f, the same expression becomes $\int_{\mathbb{R}} g(x)f(x)\,\mathrm{d}x$. In either case the CDF is a probability measure's cumulative mass, so expectations can be written without choosing a density or probability-mass representation.

Brownian sample paths are continuous but have infinite total variation almost surely on every nontrivial interval. Therefore, trying to treat W_t as a bounded-variation Stieltjes integrator fails. The Itô integral is defined probabilistically, with left-endpoint adapted integrands and an L^2 completion, rather than by naive pathwise Riemann–Stieltjes limits.

11.5 A Brownian quadratic-variation calculation with an explicit error bound

For an equidistant grid $t_j = jT/n$,

$$Q_n = \sum_{j=1}^{n} (W_{t_j} - W_{t_{j-1}})^2.$$

Each increment has variance T/n, so $\mathbb{E}[Q_n] = T$. Because a centered normal $Z \sim \mathcal{N}(0, v)$ satisfies $\mathbb{E}[Z^4] = 3v^2$,

$$\mathrm{Var}(Q_n) = n\left(3\frac{T^2}{n^2} - \frac{T^2}{n^2}\right) = \frac{2T^2}{n},$$
$$\mathbb{E}[(Q_n - T)^2] = \frac{2T^2}{n}.$$

By Chebyshev's inequality,

$$\mathbb{P}(|Q_n - T| > \varepsilon) \le \frac{2T^2}{n\varepsilon^2}.$$

This gives a quantitative probability bound for discretized quadratic variation. The result is useful both as a proof tool and as a practical simulation diagnostic: aggregating squared Brownian increments over increasingly fine partitions should approach elapsed time.

11.6 Brownian quadratic variation: a proof in L^2

Fix $T > 0$ and an equally spaced partition $t_j = jT/n$. Set $h = T/n$ and define

$$Q_n = \sum_{j=0}^{n-1} (W_{t_{j+1}} - W_{t_j})^2.$$

Brownian increments are independent, $\Delta W_j \sim \mathcal{N}(0,h)$, and for a centered normal of variance h one has $\mathbb{E}[(\Delta W_j)^2] = h$ and $\mathbb{E}[(\Delta W_j)^4] = 3h^2$. Therefore

$$\begin{aligned}
\mathbb{E}[Q_n] &= \sum_{j=0}^{n-1} \mathbb{E}[(\Delta W_j)^2] \\
&= nh = T, \\
\mathrm{Var}((\Delta W_j)^2) &= \mathbb{E}[(\Delta W_j)^4] - \{\mathbb{E}[(\Delta W_j)^2]\}^2 \\
&= 3h^2 - h^2 = 2h^2, \\
\mathrm{Var}(Q_n) &= \sum_{j=0}^{n-1} \mathrm{Var}((\Delta W_j)^2) \\
&= 2nh^2 = \frac{2T^2}{n}.
\end{aligned}$$

Thus

$$\begin{aligned}
\mathbb{E}[(Q_n - T)^2] &= \mathrm{Var}(Q_n) + \{\mathbb{E}[Q_n] - T\}^2 \\
&= \frac{2T^2}{n} \longrightarrow 0.
\end{aligned}$$

This proves $Q_n \to T$ in L^2 and hence in probability. It does not by itself prove almost-sure convergence along *every* possible sequence of partitions; such statements require additional hypotheses and separate arguments.

11.7 Why Brownian motion is not of finite variation

If a continuous real-valued function w has finite total variation $V(w;[0,T])$, then along any partition with mesh tending to zero,

$$\begin{aligned}
\sum_j (w(t_{j+1}) - w(t_j))^2 &\le \left(\max_j |w(t_{j+1}) - w(t_j)| \right) \sum_j |w(t_{j+1}) - w(t_j)| \\
&\le \left(\max_j |w(t_{j+1}) - w(t_j)| \right) V(w;[0,T]) \\
&\longrightarrow 0
\end{aligned}$$

by uniform continuity. Brownian quadratic variation, however, converges in probability to $T > 0$. Thus Brownian sample paths cannot have finite variation on $[0,T]$ almost surely. This is why a naive pathwise Riemann–Stieltjes construction is inadequate for generic $\int H_s \, \mathrm{d}W_s$.

11.8 Simple predictable processes and the Itô isometry

Take a predictable simple process $H_t = \sum_{j=0}^{n-1} H_j \mathbf{1}_{(t_j, t_{j+1}]}(t)$, where H_j is square-integrable and $\mathcal{F}_{t_j}$-measurable. Its Itô integral is defined by

$$I_T(H) = \sum_{j=0}^{n-1} H_j (W_{t_{j+1}} - W_{t_j}).$$

Because H_j is known at t_j and the subsequent Brownian increment has conditional mean zero,

$$\begin{aligned}\mathbb{E}[H_j \Delta W_j] &= \mathbb{E}[\mathbb{E}[H_j \Delta W_j \mid \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[H_j \mathbb{E}[\Delta W_j \mid \mathcal{F}_{t_j}]] = 0.\end{aligned}$$

If $j < k$, conditioning at time t_k gives $\mathbb{E}[H_j \Delta W_j H_k \Delta W_k] = 0$. Consequently all cross terms vanish in the squared integral:

$$\begin{aligned}\mathbb{E}[I_T(H)^2] &= \sum_j \mathbb{E}[H_j^2 (\Delta W_j)^2] \\ &\quad + 2 \sum_{j<k} \mathbb{E}[H_j \Delta W_j H_k \Delta W_k] \\ &= \sum_j \mathbb{E}[\mathbb{E}[H_j^2 (\Delta W_j)^2 \mid \mathcal{F}_{t_j}]] \\ &= \sum_j \mathbb{E}[H_j^2] \Delta t_j \\ &= \mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right].\end{aligned}$$

This is the Itô isometry. It permits extension by L^2 completion to general square-integrable predictable integrands, making stochastic integration rigorous rather than symbolic.

11.9 An explicit integral: $\int_0^T W_t \, \mathrm{d}W_t$

Begin with the algebraic identity $b^2 - a^2 = 2a(b-a) + (b-a)^2$. Taking $a = W_{t_j}$ and $b = W_{t_{j+1}}$ and summing gives

$$\begin{aligned}W_T^2 - W_0^2 &= \sum_j \{W_{t_{j+1}}^2 - W_{t_j}^2\} \\ &= 2 \sum_j W_{t_j} \Delta W_j + \sum_j (\Delta W_j)^2, \\ \sum_j W_{t_j} \Delta W_j &= \frac{1}{2}(W_T^2 - W_0^2) - \frac{1}{2} \sum_j (\Delta W_j)^2.\end{aligned}$$

Letting the mesh shrink, the left sums converge to the Itô integral and the last sums converge to T, so

$$\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2}(W_T^2 - T).$$

The ordinary antiderivative $W_T^2/2$ is missing the $-T/2$ correction. This elementary calculation foreshadows the second-order term in Itô's formula.

11.10 Why an Itô integral has zero mean but not zero variance

Let H be square-integrable and predictable on $[0, T]$. The Itô integral $I_T = \int_0^T H_t \, dW_t$ is defined by completion from simple predictable processes. The Itô isometry gives

$$\begin{aligned} \mathbb{E}[I_T] &= 0, \\ \mathrm{Var}(I_T) &= \mathbb{E}[I_T^2] - (\mathbb{E}[I_T])^2 \\ &= \mathbb{E}\left[\int_0^T H_t^2 \, dt\right]. \end{aligned}$$

Thus a stochastic integral can have mean zero while exhibiting substantial random fluctuation. The zero mean is a *martingale property*, not a claim that the stochastic term can be dropped from sample-path dynamics.

11.11 Adaptedness is not a cosmetic condition

For example, W_T is not $\mathcal{F}_t$-measurable when $t < T$. If one were to insert the anticipating integrand $H_t = W_T$ into the simple Itô-sum formula, then formally

$$\sum_{j=0}^{n-1} W_T \Delta W_j = W_T(W_T - W_0) = W_T^2,$$

which has expectation T rather than zero. This is not an Itô integral of a predictable process. Anticipating integrals require different theories and rules. A self-financing hedge uses an adapted trading position because the trader must select holdings before the subsequent price increment is known.

11.12 An Itô isometry computation for a deterministic kernel

Consider $I_T = \int_0^T e^{-\kappa(T-s)} \, dW_s$ for $\kappa > 0$. Because the integrand is deterministic, I_T is normal with mean zero. Its variance follows directly:

$$\begin{aligned} \mathrm{Var}(I_T) &= \mathbb{E}\left[\left(\int_0^T e^{-\kappa(T-s)} \, dW_s\right)^2\right] \\ &= \int_0^T e^{-2\kappa(T-s)} \, ds \\ &= \int_0^T e^{-2\kappa u} \, du \\ &= \left[-\frac{e^{-2\kappa u}}{2\kappa}\right]_0^T \\ &= \frac{1 - e^{-2\kappa T}}{2\kappa}. \end{aligned}$$

This calculation recurs in Ornstein–Uhlenbeck and Vasicek solutions, and is an example of the precise relationship between stochastic-integration theory and exact numerical simulation.

11.13 Left, right, and midpoint stochastic sums are not interchangeable

For a partition $0 = t_0 < \cdots < t_n = T$, consider the left-endpoint sum $I_n^L = \sum_j W_{t_j}(W_{t_{j+1}} - W_{t_j})$. From the squared-increment identity,

$$I_n^L = \frac{1}{2}\left(W_T^2 - W_0^2 - \sum_j (\Delta W_j)^2\right) \longrightarrow \frac{1}{2}(W_T^2 - T).$$

The right-endpoint sum is $I_n^R = \sum_j W_{t_{j+1}} \Delta W_j$. Since $W_{t_{j+1}} = W_{t_j} + \Delta W_j$,

$$I_n^R = I_n^L + \sum_j (\Delta W_j)^2 \longrightarrow \frac{1}{2}(W_T^2 + T).$$

The arithmetic-average endpoint sum is

$$\begin{aligned} I_n^M &= \sum_j \frac{W_{t_j} + W_{t_{j+1}}}{2} \Delta W_j \\ &= \frac{1}{2}(I_n^L + I_n^R) \longrightarrow \frac{1}{2} W_T^2. \end{aligned}$$

The left limit is the Itô integral $\int_0^T W_t \, \mathrm{d}W_t$. The midpoint limit corresponds to a Stratonovich integral $\int_0^T W_t \circ \mathrm{d}W_t$ in this setting. Their difference is $T/2$, exactly half of Brownian quadratic variation. This is why defining the stochastic integral requires specifying the evaluation convention, rather than simply writing a Riemann-like sum.

11.14 A general Itô-to-Stratonovich conversion

For sufficiently regular adapted coefficients $H_t = h(W_t)$, the Stratonovich integral satisfies

$$\int_0^T h(W_t) \circ \mathrm{d}W_t = \int_0^T h(W_t) \, \mathrm{d}W_t + \frac{1}{2}[h(W), W]_T.$$

By Itô's formula, the diffusion coefficient of $h(W_t)$ is $h'(W_t)$, so the quadratic covariation is

$$[h(W), W]_T = \int_0^T h'(W_t) \, \mathrm{d}t.$$

Consequently,

$$\int_0^T h(W_t) \circ \mathrm{d}W_t = \int_0^T h(W_t) \, \mathrm{d}W_t + \frac{1}{2} \int_0^T h'(W_t) \, \mathrm{d}t.$$

Taking $h(x) = x$ recovers $\frac{1}{2} W_T^2 = \frac{1}{2}(W_T^2 - T) + T/2$. The conversion explains the altered drift when translating an SDE between the Itô and Stratonovich conventions.

11.15 Constructing an Itô integral numerically

Let $H_t = h(t, W_t)$ be adapted and satisfy $\mathbb{E}\int_0^T H_t^2\,\mathrm{d}t < \infty$. On a grid $t_j = jT/N$, approximate

$$I_T^{(N)} = \sum_{j=0}^{N-1} h(t_j, W_{t_j})(W_{t_{j+1}} - W_{t_j}).$$

The integrand is evaluated at the *left* endpoint. Its mean should be near zero across independent sample paths, and its mean square should approach $\mathbb{E}\int_0^T H_t^2\,\mathrm{d}t$. For the test integrand $H_t = W_t$, the exact result is $I_T = (W_T^2 - T)/2$. Because the approximation admits the identity

$$I_T^{(N)} = \frac{1}{2}\left(W_T^2 - \sum_{j=0}^{N-1}(\Delta W_j)^2\right),$$

its approximation error is exactly

$$I_T^{(N)} - I_T = \frac{1}{2}\left(T - \sum_j (\Delta W_j)^2\right).$$

Thus

$$\begin{aligned}\mathbb{E}[(I_T^{(N)} - I_T)^2] &= \frac{1}{4}\,\mathrm{Var}\left(\sum_j (\Delta W_j)^2\right) \\ &= \frac{1}{4}\cdot\frac{2T^2}{N} = \frac{T^2}{2N}.\end{aligned}$$

The root mean square error is $T/\sqrt{2N}$, an explicit pathwise convergence rate for this elementary left-sum approximation. This gives a quantitative test of an Itô integral simulation routine.

LECTURE 12

Itô's Formula and Stochastic Integral Examples

12.1 Why the classical chain rule fails

Take a smooth function f and expand over a short Brownian increment:

$$f(W_{t+h}) - f(W_t) = f'(W_t)\Delta W_t + \tfrac{1}{2}f''(W_t)(\Delta W_t)^2 + R_h.$$

Summing across a fine partition, the first term converges to the Itô integral and the second to $\frac{1}{2}\int_0^T f''(W_t)\,\mathrm{d}t$ because Brownian quadratic variation accumulates. The higher-order remainder vanishes under suitable regularity. Therefore

$$f(W_T) - f(W_0) = \int_0^T f'(W_t)\,\mathrm{d}W_t + \tfrac{1}{2}\int_0^T f''(W_t)\,\mathrm{d}t.$$

Theorem 12.1 (One-dimensional Itô–Doeblin formula)**.** *Suppose* $\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t$*, and* $f \in C^{1,2}$*. Under usual local integrability assumptions,*

$$\begin{aligned}\mathrm{d}f(t, X_t) = {} & \left[f_t(t, X_t) + a(t, X_t)f_x(t, X_t) + \tfrac{1}{2}b(t, X_t)^2 f_{xx}(t, X_t)\right]\,\mathrm{d}t \\ & + b(t, X_t)f_x(t, X_t)\,\mathrm{d}W_t.\end{aligned}$$

Example 12.2 (A stochastic integral evaluated explicitly)**.** Apply the formula to $f(x) = x^2$. Since $f' = 2x$ and $f'' = 2$,

$$\begin{aligned}\mathrm{d}W_t^2 &= 2W_t\,\mathrm{d}W_t + \mathrm{d}t, \\ W_T^2 - W_0^2 &= 2\int_0^T W_t\,\mathrm{d}W_t + T, \\ \int_0^T W_t\,\mathrm{d}W_t &= \frac{W_T^2 - T}{2}.\end{aligned}$$

For $f(x) = \cos x$, the same argument gives

$$\int_0^T \sin(W_t)\,\mathrm{d}W_t = 1 - \cos(W_T) - \frac{1}{2}\int_0^T \cos(W_t)\,\mathrm{d}t.$$

The second result is useful for the supplied practice problem: the stochastic integral is replaced by an ordinary time integral.

12.2 Martingales generated by stochastic integrals

If H is square-integrable and predictable, $M_t = \int_0^t H_s\,\mathrm{d}W_s$ is a martingale with respect to the Brownian filtration. In particular, for $0 \le s \le t$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad \mathbb{E}[M_t] = 0.$$

The representation $W_t^2 - t = 2\int_0^t W_s \, \mathrm{d}W_s$ immediately proves that $W_t^2 - t$ is a martingale, whereas W_t^2 is not. Itô's formula and quadratic variation thus explain why stochastic calculus differs from classical calculus even for simple polynomial functions.

12.3 Product formula and covariance variation

For continuous Itô processes X and Y,

$$\mathrm{d}(X_t Y_t) = X_t \, \mathrm{d}Y_t + Y_t \, \mathrm{d}X_t + \mathrm{d}[X, Y]_t.$$

If $\mathrm{d}X_t = a_X \, \mathrm{d}t + b_X \, \mathrm{d}W_t$ and $\mathrm{d}Y_t = a_Y \, \mathrm{d}t + b_Y \, \mathrm{d}W_t$, then $\mathrm{d}[X, Y]_t = b_X b_Y \, \mathrm{d}t$. Driven by independent Brownian motions, the instantaneous cross variation is zero. The product rule is the two-variable case of multidimensional Itô's formula.

12.4 The stochastic integration-by-parts identity

For sufficiently regular deterministic functions f and g, integration by parts states that $\mathrm{d}(fg) = f \, \mathrm{d}g + g \, \mathrm{d}f$. In stochastic calculus an additional term is needed. Let $X_t = W_t$ and $Y_t = W_t$, then

$$\begin{aligned} \mathrm{d}(W_t^2) &= 2W_t \, \mathrm{d}W_t + \mathrm{d}[W]_t \\ &= 2W_t \, \mathrm{d}W_t + \mathrm{d}t. \end{aligned}$$

Omitting $\mathrm{d}t$ would imply $\mathbb{E}[W_t^2] = 0$, contradicting the normal variance t. This provides a direct consistency check on the stochastic product rule and underlines why the quadratic variation term is not optional.

For deterministic, continuously differentiable h, the product $h(t)W_t$ satisfies

$$\mathrm{d}(h(t)W_t) = h'(t)W_t \, \mathrm{d}t + h(t) \, \mathrm{d}W_t,$$

with no cross-variation correction because h has zero quadratic variation. Therefore

$$\int_0^T h(t) \, \mathrm{d}W_t = h(T)W_T - h(0)W_0 - \int_0^T h'(t)W_t \, \mathrm{d}t.$$

The right-hand side involves an ordinary time integral. This representation is sometimes useful for both theoretical calculations and numerical checks of deterministic-integrand stochastic integrals.

12.5 Conditional moments of Itô integrals

If h is deterministic and square-integrable, then $I = \int_0^T h(t) \, \mathrm{d}W_t$ is Gaussian with mean zero and variance $\int_0^T h(t)^2 \, \mathrm{d}t$. One proof approximates h by deterministic step functions. Each approximating integral is a linear combination of independent normal increments, hence is normal. The Itô isometry gives convergence of means and second moments; Gaussian characteristic functions identify the limit. For example,

$$\int_0^T e^{-\kappa(T-s)} \, \mathrm{d}W_s \sim \mathcal{N}\left(0, \frac{1 - e^{-2\kappa T}}{2\kappa}\right).$$

This simple normal-integral identity provides the exact innovation distribution in the Vasicek model and is a foundational tool for Gaussian affine processes.

12.6 A second-order Taylor derivation of Itô's formula

For a smooth $f(t,x)$ and a diffusion $\mathrm{d}X_t = a(t,X_t)\,\mathrm{d}t + b(t,X_t)\,\mathrm{d}W_t$, consider one short increment h. Taylor's theorem gives formally

$$\begin{aligned}\Delta f = {} & f_t h + f_x \Delta X + \frac{1}{2} f_{xx} (\Delta X)^2 \\ & + f_{tx} h \Delta X + \frac{1}{2} f_{tt} h^2 + \text{higher-order terms.}\end{aligned}$$

The stochastic increment has the size $\Delta W = O_{L^2}(h^{1/2})$, while $\Delta X \approx ah + b\Delta W$. Expanding its square,

$$\begin{aligned}(\Delta X)^2 &\approx (ah + b\Delta W)^2 \\ &= a^2 h^2 + 2abh\Delta W + b^2 (\Delta W)^2.\end{aligned}$$

Summed over a refining partition, h^2 and $h\Delta W$ vanish in the appropriate limit, but $(\Delta W)^2$ accumulates to h in quadratic variation. Retaining precisely the terms of aggregate order h yields

$$\begin{aligned}\mathrm{d}f(t,X_t) &= f_t\,\mathrm{d}t + f_x(a\,\mathrm{d}t + b\,\mathrm{d}W_t) + \frac{1}{2} f_{xx} b^2\,\mathrm{d}t \\ &= \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right)\,\mathrm{d}t + b f_x\,\mathrm{d}W_t.\end{aligned}$$

This identifies the infinitesimal generator $Lf = a f_x + (b^2/2) f_{xx}$, and also clarifies why a classical first-order chain rule cannot be used for a Brownian-driven path.

12.7 Deriving a nontrivial polynomial Brownian martingale

For $f(t,x) = x^3 - 3tx$, calculate

$$f_t = -3x, \qquad f_x = 3x^2 - 3t, \qquad f_{xx} = 6x.$$

Apply Itô's formula to $X_t = W_t$ (so $a = 0$, $b = 1$):

$$\begin{aligned}\mathrm{d}f(t,W_t) &= \left[-3W_t + \frac{1}{2}(6W_t) \right]\,\mathrm{d}t + (3W_t^2 - 3t)\,\mathrm{d}W_t \\ &= 3(W_t^2 - t)\,\mathrm{d}W_t.\end{aligned}$$

Thus $W_t^3 - 3tW_t$ is a local martingale and is a true martingale on each finite time horizon by standard moment bounds. Its stochastic-integral representation immediately implies $\mathbb{E}[W_t^3 - 3tW_t] = 0$, consistent with the vanishing odd moments of a normal random variable.

12.8 The exponential Brownian martingale

Fix $\theta \in \mathbb{R}$ and define $M_t = \exp\{\theta W_t - \frac{1}{2}\theta^2 t\}$. For $f(t,w) = e^{\theta w - \theta^2 t/2}$,

$$\begin{aligned}f_t &= -\tfrac{1}{2}\theta^2 f, \\ f_w &= \theta f, \\ f_{ww} &= \theta^2 f.\end{aligned}$$

The drift cancels under Itô's formula:

$$\begin{aligned} \mathrm{d}M_t &= \left(-\frac{1}{2}\theta^2 M_t + \frac{1}{2}\theta^2 M_t\right)\mathrm{d}t + \theta M_t\,\mathrm{d}W_t \\ &= \theta M_t\,\mathrm{d}W_t. \end{aligned}$$

Since Brownian increments are independent,

$$\begin{aligned} \mathbb{E}[M_t \mid \mathcal{F}_s] &= M_s \mathbb{E}\left[\exp\{\theta(W_t - W_s) - \tfrac{1}{2}\theta^2(t-s)\}\right] \\ &= M_s e^{-\theta^2(t-s)/2} e^{\theta^2(t-s)/2} = M_s. \end{aligned}$$

The conditional-expectation check proves the true martingale property without relying solely on a formal zero-drift calculation.

12.9 Multivariate product rule from quadratic covariation

For diffusions $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ and $\mathrm{d}Y_t = c_t\,\mathrm{d}t + d_t\,\mathrm{d}W_t$ driven by the same Brownian motion, the bilinear product $f(x,y) = xy$ has derivatives $f_x = y$, $f_y = x$, $f_{xy} = 1$, with $f_{xx} = f_{yy} = 0$. Consequently,

$$\begin{aligned} \mathrm{d}(X_t Y_t) &= X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}[X,Y]_t \\ &= (X_t c_t + Y_t a_t + b_t d_t)\,\mathrm{d}t \\ &\quad + (X_t d_t + Y_t b_t)\,\mathrm{d}W_t. \end{aligned}$$

For independent Brownian drivers the cross-variation term is zero; for correlated Brownian drivers with instantaneous correlation ρ, it becomes $\rho b_t d_t\,\mathrm{d}t$. This factor is fundamental to multidimensional SDEs and stochastic-volatility models.

12.10 Practice Set 2: evaluating $\int_0^T \sin(W_s)\,\mathrm{d}W_s$

Choose the antiderivative $f(x) = -\cos x$, whose derivatives are $f'(x) = \sin x$ and $f''(x) = \cos x$. Itô's formula for $f(W_t)$ gives

$$\begin{aligned} \mathrm{d}[-\cos(W_t)] &= f'(W_t)\,\mathrm{d}W_t + \frac{1}{2}f''(W_t)\,\mathrm{d}t \\ &= \sin(W_t)\,\mathrm{d}W_t + \frac{1}{2}\cos(W_t)\,\mathrm{d}t. \end{aligned}$$

Integrating and remembering $W_0 = 0$,

$$\begin{aligned} -\cos(W_T) + \cos(W_0) &= \int_0^T \sin(W_s)\,\mathrm{d}W_s + \frac{1}{2}\int_0^T \cos(W_s)\,\mathrm{d}s, \\ \int_0^T \sin(W_s)\,\mathrm{d}W_s &= 1 - \cos(W_T) - \frac{1}{2}\int_0^T \cos(W_s)\,\mathrm{d}s. \end{aligned}$$

The result eliminates the stochastic integral but necessarily retains an ordinary time integral. A classical chain rule would miss that term.

12.11 Practice Set 2: the SDE for a cubic transform

If $\mathrm{d}X_t = X_t^{-1}\,\mathrm{d}t + X_t^2\,\mathrm{d}W_t$ on a time interval where $X_t \neq 0$ and the SDE is well-defined, let $Y_t = X_t^3$. With $f(x) = x^3$, $f'(x) = 3x^2$, $f''(x) = 6x$. Thus

$$\begin{aligned}\mathrm{d}Y_t &= 3X_t^2\,\mathrm{d}X_t + \frac{1}{2}(6X_t)(X_t^2)^2\,\mathrm{d}t \\ &= 3X_t^2\left(X_t^{-1}\,\mathrm{d}t + X_t^2\,\mathrm{d}W_t\right) + 3X_t^5\,\mathrm{d}t \\ &= (3X_t + 3X_t^5)\,\mathrm{d}t + 3X_t^4\,\mathrm{d}W_t.\end{aligned}$$

If desired, $X_t = Y_t^{1/3}$ (real cube root) gives the resulting SDE entirely in Y:

$$\mathrm{d}Y_t = (3Y_t^{1/3} + 3Y_t^{5/3})\,\mathrm{d}t + 3Y_t^{4/3}\,\mathrm{d}W_t,$$

again interpreted up to the first hitting time of the singular state $X = 0$. The domain restriction matters because the original coefficient $1/X_t$ is undefined at zero.

12.12 Itô formula for W_t^3 from first principles

With $f(x) = x^3$ and $\mathrm{d}W_t$ as the differential,

$$\begin{aligned}\mathrm{d}(W_t^3) &= 3W_t^2\,\mathrm{d}W_t + \frac{1}{2}\cdot 6W_t\,\mathrm{d}t \\ &= 3W_t^2\,\mathrm{d}W_t + 3W_t\,\mathrm{d}t, \\ W_T^3 &= 3\int_0^T W_t^2\,\mathrm{d}W_t + 3\int_0^T W_t\,\mathrm{d}t.\end{aligned}$$

The integral correction is present even though $\mathbb{E}[W_t] = 0$ for every t. This demonstrates that a stochastic differential identity is pathwise and does not reduce to an identity of expectations.

12.13 Dynkin's formula and the generator from Itô calculus

Let X_t satisfy $\mathrm{d}X_t = a(X_t)\,\mathrm{d}t + b(X_t)\,\mathrm{d}W_t$ and take $f \in C^2$ with conditions making the expectations finite. Itô's formula gives

$$\begin{aligned}f(X_t) - f(X_0) = &\int_0^t \left[a(X_s)f'(X_s) + \frac{1}{2}b(X_s)^2 f''(X_s)\right]\,\mathrm{d}s \\ &+ \int_0^t b(X_s)f'(X_s)\,\mathrm{d}W_s.\end{aligned}$$

Define $Lf(x) = a(x)f'(x) + \frac{1}{2}b(x)^2 f''(x)$. Taking expectations and using the mean-zero property of the stochastic integral yields Dynkin's identity

$$\mathbb{E}[f(X_t)] - \mathbb{E}[f(X_0)] = \mathbb{E}\left[\int_0^t Lf(X_s)\,\mathrm{d}s\right].$$

Conditioning at time $s \le t$ yields the martingale statement

$$M_t^f = f(X_t) - f(X_0) - \int_0^t Lf(X_u)\,\mathrm{d}u,$$

which is a local martingale, and a true martingale under suitable integrability. This formula links infinitesimal generators, conditional expectations, and PDE methods without first assuming a closed-form pricing formula.

12.14 Deriving a backward equation from conditional expectations

Let $u(x,t) = \mathbb{E}_x[g(X_T) \mid X_t = x]$ for a time-homogeneous Markov diffusion with sufficient smoothness, and let $s < t < T$. The Markov/tower property implies $u(X_s, s) = \mathbb{E}[g(X_T) \mid \mathcal{F}_s]$, a martingale. Apply Itô's formula to the function $u(X_s, s)$:

$$\begin{aligned}\mathrm{d}u(X_s,s) = {} & [u_t(X_s,s) + (Lu)(X_s,s)]\,\mathrm{d}s \\ & + b(X_s)u_x(X_s,s)\,\mathrm{d}W_s.\end{aligned}$$

Since the process is a martingale, its predictable finite-variation drift must vanish (under regularity assumptions), leading to

$$u_t + Lu = 0, \qquad u(x,T) = g(x).$$

Discounting at constant rate r changes the martingale to $e^{-rs}u(X_s,s)$ and adds $-ru$ to the PDE. This is the probabilistic origin of the Feynman–Kac relationship used in derivative pricing.

12.15 An Itô calculus checklist for transformed processes

Given an SDE $\mathrm{d}X_t = a(t,X_t)\,\mathrm{d}t + b(t,X_t)\,\mathrm{d}W_t$ and a twice-space-differentiable once-time-differentiable function $f(t,x)$, perform the following mathematical steps:

1. Compute f_t, f_x, and f_{xx} explicitly, while keeping the arguments (t, X_t).
2. Substitute $\mathrm{d}X_t$ into $f_x\,\mathrm{d}X_t$; this supplies both a drift contribution and a Brownian contribution.
3. Compute the quadratic-variation correction $\frac{1}{2}f_{xx}b^2\,\mathrm{d}t$; do not replace it by a second ordinary-time derivative.
4. Collect all $\mathrm{d}t$ terms and all $\mathrm{d}W_t$ terms separately.
5. Verify dimensional consistency and, when possible, check the resulting expectation or distribution against an independent known identity.

For example, $f(t,x) = e^{-rt}x$ and $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ give

$$\begin{aligned}\mathrm{d}(e^{-rt}S_t) &= -re^{-rt}S_t\,\mathrm{d}t + e^{-rt}\,\mathrm{d}S_t \\ &= (-re^{-rt}S_t + re^{-rt}S_t)\,\mathrm{d}t + \sigma e^{-rt}S_t\,\mathrm{d}W_t \\ &= \sigma e^{-rt}S_t\,\mathrm{d}W_t.\end{aligned}$$

This verifies the local martingale structure of a discounted no-dividend stock under the risk-neutral measure, subject to the integrability needed for the true martingale property.

LECTURE 13

SDE Solutions and Basic Simulation Schemes

13.1 Stochastic differential equations as integral equations

Consider

$$\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t, \qquad X_0 = x_0.$$

A strong solution is an adapted process satisfying

$$X_t = x_0 + \int_0^t a(s, X_s)\,\mathrm{d}s + \int_0^t b(s, X_s)\,\mathrm{d}W_s$$

almost surely. Conditions such as global Lipschitz continuity and linear growth of a and b ensure existence and pathwise uniqueness in standard settings. Numerical discretizations approximate the integral equation, not an ordinary differentiable trajectory.

13.2 Exact solutions for benchmark processes

Example 13.1 (Generalized GBM)**.** For $\mathrm{d}X_t = \alpha X_t\,\mathrm{d}t + \beta X_t\,\mathrm{d}W_t$, Itô's formula applied to $\log X_t$ gives

$$\begin{aligned}\mathrm{d}\log X_t &= \alpha\,\mathrm{d}t + \beta\,\mathrm{d}W_t - \tfrac{1}{2}\beta^2\,\mathrm{d}t,\\ X_t &= X_0 \exp\left\{(\alpha - \beta^2/2)t + \beta W_t\right\}.\end{aligned}$$

Example 13.2 (Vasicek short-rate model)**.** Suppose $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \eta\,\mathrm{d}W_t$, where $\kappa > 0$. Multiply by $e^{\kappa t}$ and use the product rule:

$$\begin{aligned}\mathrm{d}(e^{\kappa t} r_t) &= \kappa e^{\kappa t} r_t\,\mathrm{d}t + e^{\kappa t}\,\mathrm{d}r_t\\ &= \kappa\theta e^{\kappa t}\,\mathrm{d}t + \eta e^{\kappa t}\,\mathrm{d}W_t.\end{aligned}$$

Therefore

$$r_t = \theta + (r_0 - \theta)e^{-\kappa t} + \eta \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s.$$

The stochastic integral is centered Gaussian with variance

$$\eta^2 \int_0^t e^{-2\kappa(t-s)}\,\mathrm{d}s = \frac{\eta^2}{2\kappa}(1 - e^{-2\kappa t}).$$

Remark 13.3 (CIR rate and variance process). The Cox–Ingersoll–Ross model takes $\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \eta\sqrt{X_t}\,\mathrm{d}W_t$. Its square-root diffusion and suitable parameter conditions support nonnegative trajectories; the strict Feller condition $2\kappa\theta \geq \eta^2$ prevents an attainable zero boundary when initialized positively. A naive Gaussian Euler step can become negative even though the exact continuous-time process is nonnegative.

13.3 Euler–Maruyama discretization

Let $t_n = nh$ and $\Delta W_n = \sqrt{h}Z_n$ with independent $Z_n \sim \mathcal{N}(0,1)$. Approximating the drift and volatility as constant over each step produces

$$\widehat{X}_{n+1} = \widehat{X}_n + a(t_n, \widehat{X}_n)h + b(t_n, \widehat{X}_n)\Delta W_n.$$

This is the Euler–Maruyama scheme. Its one-step innovations depend only on present information. For GBM, Euler yields $\widehat{S}_{n+1} = \widehat{S}_n(1 + rh + \sigma\sqrt{h}Z_n)$, which can become negative, unlike the exact exponential update. When an exact transition is known, it should generally be preferred to Euler for that component.

13.4 From Itô's formula to an integrating factor

An SDE may look nonlinear in its current coordinate yet become tractable after a suitable transformation. A linear mean-reverting diffusion $\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \eta\,\mathrm{d}W_t$ illustrates the general method. Rearrange as $\mathrm{d}X_t + \kappa X_t\,\mathrm{d}t = \kappa\theta\,\mathrm{d}t + \eta\,\mathrm{d}W_t$ and multiply by $e^{\kappa t}$. Since the deterministic factor has zero quadratic variation, the Itô product rule gives

$$\begin{aligned}\mathrm{d}(e^{\kappa t}X_t) &= e^{\kappa t}\,\mathrm{d}X_t + \kappa e^{\kappa t}X_t\,\mathrm{d}t \\ &= \kappa\theta e^{\kappa t}\,\mathrm{d}t + \eta e^{\kappa t}\,\mathrm{d}W_t.\end{aligned}$$

After integration, the exact conditional distribution over a step h is

$$X_{t+h} \mid X_t = x \sim \mathcal{N}\left(\theta + (x-\theta)e^{-\kappa h},\ \frac{\eta^2}{2\kappa}(1 - e^{-2\kappa h})\right).$$

This is a better simulator for Vasicek than Euler, because it avoids discretization bias at the grid points and reproduces the exact conditional moments.

13.5 Euler bias for GBM and the lognormal benchmark

Euler–Maruyama applied to $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ gives

$$\widehat{S}_{n+1} = \widehat{S}_n(1 + rh + \sigma\sqrt{h}Z_{n+1}).$$

Its conditional first moment is exact over one step to first order:

$$\mathbb{E}[\widehat{S}_{n+1} \mid \widehat{S}_n = s] = s(1 + rh),$$

whereas exact GBM has conditional mean $se^{rh} = s(1 + rh + r^2h^2/2 + \cdots)$. Thus the per-step weak error in the mean is $O(h^2)$ and the accumulated error over T/h steps is $O(h)$. Furthermore, the Euler multiplier may be negative when

$$Z_{n+1} < -\frac{1 + rh}{\sigma\sqrt{h}},$$

which has positive Gaussian probability for every fixed $h > 0$. A method that exactly preserves nonnegativity may therefore be preferable for pricing payoffs sensitive to extreme paths.

13.6 Solving the Vasicek SDE by an integrating factor

Consider the Ornstein–Uhlenbeck short-rate model $\mathrm{d}R_t = \kappa(\theta - R_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, with $\kappa > 0$, constant θ, and $\sigma > 0$. Multiplying by the deterministic integrating factor $e^{\kappa t}$ and using Itô's product rule,

$$\begin{aligned}\mathrm{d}(e^{\kappa t}R_t) &= e^{\kappa t}\,\mathrm{d}R_t + \kappa e^{\kappa t}R_t\,\mathrm{d}t \\ &= e^{\kappa t}[\kappa\theta - \kappa R_t]\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t + \kappa e^{\kappa t}R_t\,\mathrm{d}t \\ &= \kappa\theta e^{\kappa t}\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t.\end{aligned}$$

Integrating and simplifying yields

$$\begin{aligned}e^{\kappa t}R_t - R_0 &= \kappa\theta\int_0^t e^{\kappa s}\,\mathrm{d}s + \sigma\int_0^t e^{\kappa s}\,\mathrm{d}W_s \\ &= \theta(e^{\kappa t} - 1) + \sigma\int_0^t e^{\kappa s}\,\mathrm{d}W_s, \\ R_t &= \theta + (R_0 - \theta)e^{-\kappa t} + \sigma\int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s.\end{aligned}$$

The Itô integral of the deterministic function is normal with mean zero and variance from the Itô isometry:

$$\begin{aligned}\mathrm{Var}\left(\sigma\int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s\right) &= \sigma^2\int_0^t e^{-2\kappa(t-s)}\,\mathrm{d}s \\ &= \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}).\end{aligned}$$

Thus

$$R_t \mid R_0 = r_0 \sim \mathcal{N}\left(\theta + (r_0 - \theta)e^{-\kappa t}, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})\right).$$

The exact conditional transition allows an exact-in-law simulation at any sequence of observation times. It is an essential benchmark for measuring discretization error of Euler–Maruyama.

13.7 Euler–Maruyama from the SDE integral equation

For $\mathrm{d}X_t = a(X_t)\,\mathrm{d}t + b(X_t)\,\mathrm{d}W_t$, the integral equation on $[t_n, t_{n+1}]$ is

$$X_{t_{n+1}} - X_{t_n} = \int_{t_n}^{t_{n+1}} a(X_s)\,\mathrm{d}s + \int_{t_n}^{t_{n+1}} b(X_s)\,\mathrm{d}W_s.$$

Approximate each coefficient by its left-endpoint value. Then

$$\begin{aligned}\int_{t_n}^{t_{n+1}} a(X_s)\,\mathrm{d}s &\approx a(X_{t_n})h, \\ \int_{t_n}^{t_{n+1}} b(X_s)\,\mathrm{d}W_s &\approx b(X_{t_n})(W_{t_{n+1}} - W_{t_n}), \\ \widehat{X}_{n+1} &= \widehat{X}_n + a(\widehat{X}_n)h + b(\widehat{X}_n)\sqrt{h}Z_n,\end{aligned}$$

where the Z_n are independent standard normals. The approximation is adapted: the new state depends on the information available up to t_n and on the newly sampled noise, not on future coefficients.

13.8 Exact versus Euler simulation for GBM

For $\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W$, the Euler step is

$$\widehat{S}_{n+1} = \widehat{S}_n(1 + rh + \sigma\sqrt{h}Z_n).$$

The exact step on the same Brownian increment is

$$S_{t_{n+1}} = S_{t_n} e^{(r-\sigma^2/2)h+\sigma\sqrt{h}Z_n}.$$

Taylor-expanding the exact multiplier $e^A = 1 + A + A^2/2 + O(A^3)$ for $A = (r - \sigma^2/2)h + \sigma\sqrt{h}Z_n$ gives

$$\begin{aligned} e^A &= 1 + \sigma\sqrt{h}Z_n + (r - \tfrac{1}{2}\sigma^2)h + \tfrac{1}{2}\sigma^2 h Z_n^2 + O_{L^p}(h^{3/2}) \\ &= 1 + rh + \sigma\sqrt{h}Z_n + \tfrac{1}{2}\sigma^2 h(Z_n^2 - 1) + O_{L^p}(h^{3/2}). \end{aligned}$$

Euler omits the centered term $\frac{1}{2}\sigma^2 h(Z_n^2 - 1)$ that Milstein retains. Euler can also produce negative stock values if $Z_n < -(1 + rh)/(\sigma\sqrt{h})$, whereas exact GBM simulation and exponential-log schemes preserve positivity automatically.

13.9 Mean reversion and the numerical stability of Euler for Vasicek

For $R_{n+1} = R_n + \kappa(\theta - R_n)h + \sigma\sqrt{h}Z_n$,

$$R_{n+1} - \theta = (1 - \kappa h)(R_n - \theta) + \sigma\sqrt{h}Z_n.$$

Taking expectations produces $\mathbb{E}[R_n - \theta] = (1 - \kappa h)^n(R_0 - \theta)$. The numerical mean reverts to θ when $|1 - \kappa h| < 1$, or $0 < \kappa h < 2$. If $\kappa h > 2$, the numerical mean oscillates with increasing magnitude even though the continuous Vasicek model is stable for every $\kappa > 0$. This illustrates why convergence as $h \to 0$ and stability for a fixed finite step are distinct issues.

13.10 Numerical positivity of a square-root diffusion

Consider the CIR variance model $\mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \xi\sqrt{V_t}\,\mathrm{d}W_t$, with nonnegative initial variance and positive coefficients. Under standard CIR conditions it admits a nonnegative strong solution; if $2\kappa\theta \geq \xi^2$, the process started positive does not hit zero. A simple Euler step is

$$\widehat{V}_{n+1} = \widehat{V}_n + \kappa(\theta - \widehat{V}_n)h + \xi\sqrt{\widehat{V}_n}\sqrt{h}Z_n.$$

Conditional on $\widehat{V}_n = v > 0$, the new value is normal with mean $m(v) = v + \kappa(\theta - v)h$ and standard deviation $s(v) = \xi\sqrt{vh}$. Therefore

$$\begin{aligned} \mathbb{P}(\widehat{V}_{n+1} < 0 \mid \widehat{V}_n = v) &= \mathbb{P}\left(Z_n < -\frac{m(v)}{s(v)}\right) \\ &= \Phi\left(-\frac{v + \kappa(\theta - v)h}{\xi\sqrt{vh}}\right) > 0. \end{aligned}$$

Thus Euler can produce negative values even when the true variance process is nonnegative. Subsequent square-root updates are then undefined unless the numerical algorithm specifies a correction such as full truncation, positivity-preserving splitting, or an exact CIR transition sampler. Different fixes induce different biases and convergence properties.

13.11 Conditional moments of one Euler–Maruyama update

For $\widehat{X}_{n+1} = x + a(x)h + b(x)\sqrt{h}Z_n$, given $\widehat{X}_n = x$,

$$\begin{aligned}
\mathbb{E}[\widehat{X}_{n+1} - x \mid x] &= a(x)h + b(x)\sqrt{h}\mathbb{E}[Z_n] = a(x)h, \\
\mathrm{Var}(\widehat{X}_{n+1} - x \mid x) &= b(x)^2 h\,\mathrm{Var}(Z_n) = b(x)^2 h, \\
\mathbb{E}[(\widehat{X}_{n+1} - x)^2 \mid x] &= a(x)^2 h^2 + b(x)^2 h.
\end{aligned}$$

The first two conditional moments match the leading small-time moments of the diffusion. This explains why Euler is a natural first-order weak approximation even though its pathwise error is generally only of strong order $1/2$.

13.12 Exact step of the Vasicek process and a simulation benchmark

Starting from $R_{t_n} = r_n$, the exact Vasicek transition over step h follows from the integrating-factor solution:

$$R_{t_n+h} = \theta + (r_n - \theta)e^{-\kappa h} + \sigma \int_{t_n}^{t_n+h} e^{-\kappa(t_n+h-s)}\,\mathrm{d}W_s.$$

The stochastic integral is independent of $\mathcal{F}_{t_n}$ and normal with variance

$$\sigma^2 \int_{t_n}^{t_n+h} e^{-2\kappa(t_n+h-s)}\,\mathrm{d}s = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa h}).$$

Hence one can sample exactly on a discrete grid:

$$R_{n+1} = \theta + (R_n - \theta)e^{-\kappa h} + \sigma\sqrt{\frac{1 - e^{-2\kappa h}}{2\kappa}}Z_n.$$

This exact transition and Euler's $R_{n+1} = R_n + \kappa(\theta - R_n)h + \sigma\sqrt{h}Z_n$ agree to first order in conditional mean and variance as $h \to 0$ because $e^{-\kappa h} = 1 - \kappa h + O(h^2)$ and $(1 - e^{-2\kappa h})/(2\kappa) = h + O(h^2)$. For finite h, exact sampling avoids Euler's transition-distribution bias at essentially the same computational cost.

13.13 Exact CIR transitions in principle

For $\mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \xi\sqrt{V_t}\,\mathrm{d}W_t$, the conditional distribution over a step has a scaled noncentral chi-square form under the usual CIR parameter restrictions. Set

$$c_h = \frac{\xi^2(1-e^{-\kappa h})}{4\kappa},$$
$$\nu = \frac{4\kappa\theta}{\xi^2},$$
$$\lambda_h(v) = \frac{4\kappa e^{-\kappa h} v}{\xi^2(1-e^{-\kappa h})}.$$

Then

$$V_{t+h} \mid V_t = v \overset{d}{=} c_h\, \chi'^2_\nu(\lambda_h(v)).$$

This representation produces nonnegative transitions and is preferable to raw Euler when an accurate noncentral chi-square sampler is available. In stochastic-volatility models one must still correctly couple the variance and stock Brownian drivers; independently sampling their marginal transitions does not by itself preserve leverage correlation.

13.14 Numerical experiments for strong and weak convergence

To test strong convergence of a scheme against an exactly simulable benchmark, use a single fine Brownian path and aggregate fine increments to construct all coarser steps. For each replication i, obtain an exact terminal value $X_T^{(i)}$ and approximations $\widehat{X}_T^{(h,i)}$. Then estimate

$$E_{\text{strong}}(h) = \frac{1}{M}\sum_{i=1}^{M} |X_T^{(i)} - \widehat{X}_T^{(h,i)}|.$$

For weak convergence, use the same scheme to estimate

$$E_{\text{weak}}(h) = \left| \frac{1}{M}\sum_{i=1}^{M} f(\widehat{X}_T^{(h,i)}) - \mathbb{E}[f(X_T)] \right|.$$

Because the weak error estimate includes Monte Carlo noise, its apparent convergence may flatten out when discretization bias falls below its standard error. A reliable weak-order experiment either increases M as h decreases, uses control variates, or employs analytic conditional moments when available. Log–log plots of estimated error against h can suggest an order, but a fitted slope alone is not a proof of strong or weak convergence.

13.15 An exact-vs-Euler stock simulation test

Under GBM with constant r, σ, the exact one-step conditional mean is $\mathbb{E}[S_{t+h} \mid S_t = s] = se^{rh}$, while the Euler conditional mean is $s(1+rh)$. Their difference is

$$\begin{aligned} se^{rh} - s(1+rh) &= s\left(1 + rh + \frac{1}{2}r^2h^2 + O(h^3) - 1 - rh\right) \\ &= \frac{1}{2}sr^2h^2 + O(h^3). \end{aligned}$$

This is a one-step weak bias of order h^2 for the observable $f(s) = s$. Over T/h steps, accumulation leads to a first-order global bias, consistent with ordinary Euler theory.

LECTURE 14

Milstein, Higher-Order Weak Schemes, and Multivariate Itô Calculus

14.1 Itô–Taylor operators

For a time-homogeneous scalar SDE $\mathrm{d}X = a(X)\,\mathrm{d}t + b(X)\,\mathrm{d}W$, define the differential operators

$$L^0 f(x) = a(x)f'(x) + \tfrac{1}{2}b(x)^2 f''(x), \qquad L^1 f(x) = b(x)f'(x).$$

Then Itô's formula is concisely written $\mathrm{d}f(X_t) = (L^0 f)(X_t)\,\mathrm{d}t + (L^1 f)(X_t)\,\mathrm{d}W_t$. Applying this identity to the drift and diffusion coefficients inside the integral solution supplies successive correction terms.

14.2 Derivation of the Milstein update

Expand the diffusion coefficient over one short step:

$$b(X_s) \approx b(X_{t_n}) + b'(X_{t_n})b(X_{t_n})(W_s - W_{t_n}) + exthigher - orderterms.$$

Substituting into the stochastic integral gives

$$\begin{aligned}\int_{t_n}^{t_{n+1}} b(X_s)\,\mathrm{d}W_s &\approx b(X_{t_n})\Delta W_n + b'(X_{t_n})b(X_{t_n})\int_{t_n}^{t_{n+1}} (W_s - W_{t_n})\,\mathrm{d}W_s \\ &= b(X_{t_n})\Delta W_n + \tfrac{1}{2}b(X_{t_n})b'(X_{t_n})\left((\Delta W_n)^2 - h\right).\end{aligned}$$

Consequently, the Milstein approximation is

$$\widehat{X}_{n+1} = \widehat{X}_n + a_n h + b_n \Delta W_n + \frac{1}{2} b_n b_n' \left((\Delta W_n)^2 - h\right).$$

Under standard regularity assumptions, scalar Euler has strong order $1/2$ and Milstein strong order 1, reflecting the improved approximation of individual sample paths.

14.3 Strong versus weak error

Definition 14.1 (Convergence orders)**.** A scheme $\widehat{X}_T^{(h)}$ has strong order γ if $\mathbb{E}\left|X_T - \widehat{X}_T^{(h)}\right| = O(h^\gamma)$ (or the corresponding specified L^p norm), and weak order β if

$$\left|\mathbb{E}[f(X_T)] - \mathbb{E}[f(\widehat{X}_T^{(h)})]\right| = O(h^\beta)$$

for a suitable class of sufficiently smooth test functions f.

Strong convergence is important when approximating realized paths or path-dependent functionals; weak convergence is often sufficient for expected payoffs. Classical Euler–Maruyama generally has weak order 1 under smoothness conditions. A numerical method may have higher weak than strong order; the relevant criterion depends on the computational task and the payoff's regularity.

14.4 A simplified weak second-order expansion

For a scalar time-homogeneous SDE with sufficiently smooth coefficients, one useful simplified weak order-two scheme has increment

$$\begin{aligned}\widehat{X}_{n+1} - \widehat{X}_n = a_n h + b_n \Delta W_n + \tfrac{1}{2} b_n b_n' \big((\Delta W_n)^2 - h\big) \\ + \tfrac{1}{2}\big(a_n' b_n + a_n b_n' + \tfrac{1}{2} b_n^2 b_n''\big)\Delta W_n h \\ + \tfrac{1}{2}\big(a_n a_n' + \tfrac{1}{2} b_n^2 a_n''\big) h^2.\end{aligned}$$

This is an approximation to the relevant Itô–Taylor expansion and should not be asserted to have strong order two; replacing certain multiple stochastic integrals by simplified expressions targets weak expectation accuracy. The derivatives are evaluated at $\widehat{X}_n$.

14.5 Multivariate Itô formula

Let $X_t \in \mathbb{R}^d$ satisfy

$$\mathrm{d}X_t^i = a_i(t, X_t)\,\mathrm{d}t + \sum_{k=1}^{m} b_{ik}(t, X_t)\,\mathrm{d}W_t^k,$$

with independent driving Brownian motions. For $f \in C^{1,2}([0,T] \times \mathbb{R}^d)$,

$$\begin{aligned}\mathrm{d}f(t, X_t) = \left[f_t + \sum_{i=1}^{d} a_i f_{x_i} + \frac{1}{2}\sum_{i,j=1}^{d}\left(\sum_{k=1}^{m} b_{ik} b_{jk}\right) f_{x_i x_j}\right]\mathrm{d}t \\ + \sum_{k=1}^{m}\left(\sum_{i=1}^{d} b_{ik} f_{x_i}\right)\mathrm{d}W_t^k.\end{aligned}$$

Writing $B = (b_{ik})$, the second-order operator is $(1/2)\sum_{ij}(BB^\top)_{ij}\partial_{x_i x_j}$, which directly links the Hessian correction to the covariance matrix. High-order multivariate schemes require approximating iterated stochastic integrals, including cross integrals (Lévy areas); these cannot in general be reduced to independent scalar Milstein corrections.

14.6 A detailed Itô–Taylor expansion

For a time-homogeneous scalar SDE, the integral representation over $[t, t+h]$ is

$$X_{t+h} = X_t + \int_t^{t+h} a(X_s)\,\mathrm{d}s + \int_t^{t+h} b(X_s)\,\mathrm{d}W_s.$$

Apply Itô's formula separately to $a(X_s)$ and $b(X_s)$:

$$a(X_s) = a(X_t) + \int_t^s (L^0 a)(X_u)\,\mathrm{d}u + \int_t^s (L^1 a)(X_u)\,\mathrm{d}W_u,$$
$$b(X_s) = b(X_t) + \int_t^s (L^0 b)(X_u)\,\mathrm{d}u + \int_t^s (L^1 b)(X_u)\,\mathrm{d}W_u.$$

Substituting into the original equation creates iterated integrals. Introduce

$$I_{00} = \int_t^{t+h}\int_t^s \mathrm{d}u\,\mathrm{d}s = \frac{h^2}{2},$$
$$I_{11} = \int_t^{t+h}\int_t^s \mathrm{d}W_u\,\mathrm{d}W_s = \frac{(\Delta W)^2 - h}{2},$$
$$I_{10} = \int_t^{t+h}\int_t^s \mathrm{d}W_u\,\mathrm{d}s,$$
$$I_{01} = \int_t^{t+h}\int_t^s \mathrm{d}u\,\mathrm{d}W_s.$$

Since $I_{10} + I_{01} = h\Delta W$, replacing the two mixed integrals by $h\Delta W/2$ in the appropriate weak expansion leads to the simplified second-order scheme seen in the lecture. This substitution is not a pathwise identity. Indeed I_{10} and I_{01} are different stochastic integrals, so higher *strong* order requires accurate treatment of their joint distribution.

14.7 Differential operators in higher dimensions

For vector-valued $X_t \in \mathbb{R}^d$ driven by m Brownian motions, define

$$L^0 = \partial_t + \sum_{i=1}^d a_i \partial_{x_i} + \frac{1}{2}\sum_{i,j=1}^d (BB^\top)_{ij}\partial_{x_i x_j}, \qquad L^k = \sum_{i=1}^d b_{ik}\partial_{x_i}.$$

An iterated stochastic Taylor series applies these operators to coordinate functions, drift coefficients, and diffusion coefficients. Terms involving $L^k b_{ij}$ measure how the jth diffusion direction changes along the kth Brownian direction. When these derivatives do not commute, the resulting approximation contains mixed stochastic areas and cannot be reconstructed from Brownian increments alone without additional random variables or structural simplifications. This is an important conceptual distinction between scalar Milstein and genuinely multidimensional strong simulation.

14.8 The second-order multivariate approximation in the lecture

For the vector SDE introduced earlier, use the operators $L^0, L^1, \ldots, L^m$. A simplified weak-order-two update for coordinate i is expressed in the lecture in the form

$$\begin{aligned}\widehat{X}_i(t+h) = \widehat{X}_i(t) + a_i h + \sum_{k=1}^m b_{ik}\Delta W_k + \frac{1}{2}(L^0 a_i)h^2 \\ + \frac{1}{2}\sum_{k=1}^m \left(L^k a_i + L^0 b_{ik}\right)\Delta W_k h \\ + \frac{1}{2}\sum_{j=1}^m\sum_{k=1}^m (L^j b_{ik})\left(\Delta W_j \Delta W_k - V_{jk}\right).\end{aligned}$$

All coefficients and their operator derivatives are evaluated at the current approximation $(t, \widehat{X}(t))$. The random matrix V in a common weak Taylor construction has $V_{jj} = h$, and for $j < k$ one chooses independent Rademacher values $V_{jk} = \pm h$ with equal probability, setting $V_{kj} = -V_{jk}$. The supplied Lecture 14 slide instead writes $V_{kj} = V_{jk}$, a symmetry convention that differs from the familiar antisymmetric weak-Taylor surrogate. Because different stochastic Taylor schemes can require different auxiliary random matrices, a numerical implementation must verify the corresponding moment conditions and the intended convention rather than combining one convention with another schemeś coefficients. Importantly, these supplementary signs are not the actual Lévy areas. They are random surrogates selected so that relevant low-order moments of the numerical increment match those of the exact SDE to the required weak accuracy.

The scalar version with $m = 1$ has $V_{11} = h$ and recovers the familiar factor $(\Delta W)^2 - h$. The vector version exposes the main additional complexity: distinct noise directions contribute mixed products whose distributional treatment matters for convergence. In implementation, the Brownian increments must be sampled consistently with the covariance of the driving Brownian motions; correlated Brownian drivers should be whitened or represented through a suitable factor matrix before this formula is used.

14.9 Deriving Milstein from one repeated stochastic integral

Assume scalar $\mathrm{d}X_t = a(X_t)\,\mathrm{d}t + b(X_t)\,\mathrm{d}W_t$ and smooth coefficients. Over $[t, t+h]$, start with the integral representation

$$X_{t+h} = X_t + \int_t^{t+h} a(X_s)\,\mathrm{d}s + \int_t^{t+h} b(X_s)\,\mathrm{d}W_s.$$

The Euler approximation freezes both coefficients at X_t. To refine the stochastic term, Taylor-expand $b(X_s)$:

$$\begin{aligned} b(X_s) &\approx b(X_t) + b'(X_t)(X_s - X_t) \\ &\approx b(X_t) + b'(X_t)b(X_t)(W_s - W_t), \end{aligned}$$

where the leading stochastic variation of $X_s - X_t$ is $b(X_t)(W_s - W_t)$. Substitute into the stochastic integral:

$$\int_t^{t+h} b(X_s)\,\mathrm{d}W_s \approx b(X_t)\Delta W + b'(X_t)b(X_t)\int_t^{t+h} (W_s - W_t)\,\mathrm{d}W_s.$$

To evaluate the iterated integral, apply Itô's formula to $(W_s - W_t)^2$:

$$\begin{aligned} \mathrm{d}(W_s - W_t)^2 &= 2(W_s - W_t)\,\mathrm{d}W_s + \mathrm{d}s, \\ (W_{t+h} - W_t)^2 &= 2\int_t^{t+h} (W_s - W_t)\,\mathrm{d}W_s + h, \\ \int_t^{t+h} (W_s - W_t)\,\mathrm{d}W_s &= \frac{1}{2}\{(\Delta W)^2 - h\}. \end{aligned}$$

Hence the scalar Milstein update is

$$\begin{aligned} \widehat{X}_{n+1} = \widehat{X}_n &+ a(\widehat{X}_n)h + b(\widehat{X}_n)\Delta W_n \\ &+ \frac{1}{2}b(\widehat{X}_n)b'(\widehat{X}_n)\{(\Delta W_n)^2 - h\}. \end{aligned}$$

For $b(x) = \sigma x$ it becomes $\widehat{S}_{n+1} = \widehat{S}_n[1 + rh + \sigma\Delta W + \frac{1}{2}\sigma^2((\Delta W)^2 - h)]$, exactly matching the additional order-h Taylor term of the exponential stock solution.

14.10 Strong and weak errors are different statements

For an approximation $\widehat{X}_T^{(h)}$ computed with time step h, a strong order p statement means

$$\left(\mathbb{E}[|X_T - \widehat{X}_T^{(h)}|^q]\right)^{1/q} \leq Ch^p$$

for a specified $q \geq 1$ and appropriate assumptions, using the *same Brownian path* for the exact and numerical solutions. A weak order p statement instead means

$$\left|\mathbb{E}[f(X_T)] - \mathbb{E}[f(\widehat{X}_T^{(h)})]\right| \leq C_f h^p$$

for a suitable class of smooth test functions f. The first measures path accuracy; the second measures accuracy of expectations. The standard globally Lipschitz scalar Euler scheme generally has strong order $1/2$ and weak order 1 for sufficiently regular coefficients and observables; scalar Milstein has strong order 1 under stronger smoothness. A discontinuous digital payoff may reduce weak convergence order, so quoted rates should always be paired with assumptions on the payoff.

14.11 Operators and the beginning of Itô–Taylor expansions

Define operators on smooth functions f by

$$\begin{aligned} L^0 f(x) &= a(x)f'(x) + \frac{1}{2}b(x)^2 f''(x), \\ L^1 f(x) &= b(x)f'(x). \end{aligned}$$

Itô's formula implies, with f time-independent,

$$f(X_{t+h}) = f(X_t) + \int_t^{t+h} (L^0 f)(X_s)\,\mathrm{d}s + \int_t^{t+h} (L^1 f)(X_s)\,\mathrm{d}W_s.$$

Take $f(x) = x$. Then $L^0 f = a$ and $L^1 f = b$, recovering the original SDE. Apply the same identity to the coefficient functions a and b, and substitute the resulting integrals into the SDE:

$$\begin{aligned} X_{t+h} = X_t &+ a(X_t)h + b(X_t)\Delta W \\ &+ \int_t^{t+h}\int_t^s (L^0 a)(X_u)\,\mathrm{d}u\,\mathrm{d}s \\ &+ \int_t^{t+h}\int_t^s (L^1 a)(X_u)\,\mathrm{d}W_u\,\mathrm{d}s \\ &+ \int_t^{t+h}\int_t^s (L^0 b)(X_u)\,\mathrm{d}u\,\mathrm{d}W_s \\ &+ \int_t^{t+h}\int_t^s (L^1 b)(X_u)\,\mathrm{d}W_u\,\mathrm{d}W_s. \end{aligned}$$

This is the structural origin of a stochastic Taylor scheme. Higher-order algorithms approximate the nested integrals while respecting their different stochastic sizes; one must not treat them as ordinary deterministic double integrals.

14.12 Multidimensional Itô, covariance, and the Heston example

Let a vector process satisfy $\mathrm{d}X_t = a(X_t)\,\mathrm{d}t + B(X_t)\,\mathrm{d}W_t$, where W_t is m-dimensional standard Brownian motion and B is a $d \times m$ diffusion matrix. For $f \in C^2(\mathbb{R}^d)$,

$$\begin{aligned}
\mathrm{d}f(X_t) &= \sum_i \partial_i f(X_t)\,\mathrm{d}X_{i,t} + \frac{1}{2}\sum_{i,j} \partial_{ij} f(X_t)\,\mathrm{d}[X_i, X_j]_t, \\
\mathrm{d}[X_i, X_j]_t &= (BB^\top)_{ij}(X_t)\,\mathrm{d}t, \\
\mathrm{d}f(X_t) &= \left\{ a \cdot \nabla f + \frac{1}{2}\,\mathrm{tr}(BB^\top \nabla^2 f) \right\} \mathrm{d}t + (\nabla f)^\top B\,\mathrm{d}W_t.
\end{aligned}$$

For the Heston system

$$\begin{aligned}
\mathrm{d}S_t &= rS_t\,\mathrm{d}t + \sqrt{v_t} S_t\,\mathrm{d}W_t^{(1)}, \\
\mathrm{d}v_t &= \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}, \\
\mathrm{d}[W^{(1)}, W^{(2)}]_t &= \rho\,\mathrm{d}t,
\end{aligned}$$

the instantaneous covariance of the price and variance noises is $\rho\xi S_t v_t\,\mathrm{d}t$. This generates a mixed derivative term $\rho\xi s v \partial_{sv} f$ in the two-factor pricing generator. Numerical discretization must account for this covariance and the constraint $v_t \geq 0$.

14.13 Milstein's local moment correction has zero conditional mean

The Milstein correction is $M = \frac{1}{2} b(x) b'(x)[(\Delta W)^2 - h]$ at the current state x. If $\Delta W = \sqrt{h} Z$ with $Z \sim \mathcal{N}(0,1)$, then

$$\begin{aligned}
\mathbb{E}[M \mid X_t = x] &= \frac{1}{2} b(x) b'(x) h[\mathbb{E}(Z^2) - 1] \\
&= \frac{1}{2} b(x) b'(x) h(1 - 1) = 0.
\end{aligned}$$

Its conditional variance is

$$\begin{aligned}
\mathrm{Var}(M \mid X_t = x) &= \frac{1}{4} b(x)^2 b'(x)^2 h^2\,\mathrm{Var}(Z^2 - 1) \\
&= \frac{1}{4} b(x)^2 b'(x)^2 h^2 [\mathbb{E}(Z^4) - (\mathbb{E}Z^2)^2] \\
&= \frac{1}{2} b(x)^2 b'(x)^2 h^2.
\end{aligned}$$

The correction changes pathwise accuracy while not altering the leading first conditional moment. Because it has typical size h, it is one order smaller than the leading Brownian increment of size $h^{1/2}$ and one order larger than many neglected terms of size $h^{3/2}$.

14.14 Multivariate correlation implemented through independent normals

For correlated Brownian increments $\Delta W_1, \Delta W_2$ satisfying $\mathrm{Cov}(\Delta W_1, \Delta W_2) = \rho h$, generate independent standard normals Z_1, Z_2 and set

$$\Delta W_1 = \sqrt{h} Z_1,$$
$$\Delta W_2 = \sqrt{h}(\rho Z_1 + \sqrt{1-\rho^2} Z_2).$$

As already established for correlated Gaussian variables, $\mathrm{Var}(\Delta W_1) = \mathrm{Var}(\Delta W_2) = h$ and $\mathrm{Cov}(\Delta W_1, \Delta W_2) = \rho h$. Euler discretization of Heston is then

$$S_{n+1} = S_n + r S_n h + S_n \sqrt{v_n} \Delta W_1,$$
$$v_{n+1} = v_n + \kappa(\theta - v_n) h + \xi \sqrt{v_n} \Delta W_2.$$

This raw scheme requires a nonnegativity treatment when v_n becomes negative; the formulas are not a substitute for a fully specified positivity-preserving Heston simulator.

14.15 Stochastic iterated integrals and their order of magnitude

For fixed interval $[t, t+h]$, define the multiple integrals

$$I_{(0)} = \int_t^{t+h} \mathrm{d}s = h,$$
$$I_{(1)} = \int_t^{t+h} \mathrm{d}W_s = \Delta W,$$
$$I_{(1,1)} = \int_t^{t+h} \int_t^s \mathrm{d}W_u \, \mathrm{d}W_s = \frac{1}{2}[(\Delta W)^2 - h],$$
$$I_{(0,0)} = \int_t^{t+h} \int_t^s \mathrm{d}u \, \mathrm{d}s = \frac{1}{2} h^2.$$

The mixed integrals $I_{(1,0)} = \int_t^{t+h} \int_t^s \mathrm{d}W_u \, \mathrm{d}s$ and $I_{(0,1)} = \int_t^{t+h} \int_t^s \mathrm{d}u \, \mathrm{d}W_s$ both have mean zero. By Fubini and Itô isometry,

$$I_{(1,0)} = \int_t^{t+h} (t+h-u) \, \mathrm{d}W_u,$$
$$\mathrm{Var}(I_{(1,0)}) = \int_t^{t+h} (t+h-u)^2 \, \mathrm{d}u = \frac{h^3}{3},$$
$$I_{(0,1)} = \int_t^{t+h} (s-t) \, \mathrm{d}W_s,$$
$$\mathrm{Var}(I_{(0,1)}) = \int_t^{t+h} (s-t)^2 \, \mathrm{d}s = \frac{h^3}{3}.$$

Their covariance is

$$\mathrm{Cov}(I_{(1,0)}, I_{(0,1)}) = \int_t^{t+h} (t+h-s)(s-t) \, \mathrm{d}s = \frac{h^3}{6}.$$

Thus both are of stochastic order $h^{3/2}$, but they are correlated and cannot be replaced by independent Gaussian variables without further justification.

14.16 A bridge decomposition for higher-order stochastic integrals

Let $\Delta W = \sqrt{h}Z_1$ with $Z_1 \sim \mathcal{N}(0,1)$. Since $I_{(1,0)}$ and ΔW are jointly Gaussian,

$$\operatorname{Cov}(I_{(1,0)}, \Delta W) = \int_t^{t+h} (t+h-u)\,\mathrm{d}u = \frac{h^2}{2}.$$

Their conditional mean is therefore

$$\mathbb{E}[I_{(1,0)} \mid \Delta W] = \frac{h^2/2}{h}\Delta W = \frac{h}{2}\Delta W,$$

with conditional variance

$$\operatorname{Var}(I_{(1,0)} \mid \Delta W) = \frac{h^3}{3} - \frac{(h^2/2)^2}{h} = \frac{h^3}{12}.$$

Hence one exact joint construction is

$$I_{(1,0)} = \frac{h}{2}\Delta W + \sqrt{\frac{h^3}{12}}Z_2,$$

where Z_2 is independent of Z_1. This type of conditional Gaussian simulation underlies practical higher-order weak schemes and illustrates why matching only marginal variances can be insufficient.

14.17 Implementable Euler and Milstein schemes

Algorithm 14.2 (Scalar SDE path simulation)**.** For $\mathrm{d}X_t = a(X_t)\,\mathrm{d}t + b(X_t)\,\mathrm{d}W_t$ on $[0,T]$, choose N and $h = T/N$.

1. Initialize the two path approximations $E_0 = M_0 = X_0$ and a common random-number stream.
2. For $n = 0, \ldots, N-1$, draw $Z_n \sim \mathcal{N}(0,1)$ and set $\Delta W_n = \sqrt{h}Z_n$.
3. Compute Euler $E_{n+1} = E_n + a(E_n)h + b(E_n)\Delta W_n$.
4. Compute Milstein $M_{n+1} = M_n + a(M_n)h + b(M_n)\Delta W_n + \frac{1}{2}b(M_n)b'(M_n)[(\Delta W_n)^2 - h]$.
5. Store time values as required by the payoff; for terminal-only quantities, retaining the current states may suffice.

Both algorithms require $O(N)$ coefficient evaluations per path, and the scalar Milstein scheme additionally requires b'. Their relative advantages depend on the payoff, coefficient regularity, ability to simulate iterated stochastic integrals, and whether one needs pathwise accuracy or only an expectation. The scalar formula is not a general multidimensional Milstein method: noncommuting diffusion fields generate Lévy-area terms that cannot be replaced by independent scalar squares.

LECTURE 15

Extrapolation, Discretization Bias, and Quasi-Monte Carlo

15.1 Bias and statistical error are different

When an SDE must be discretized, the Monte Carlo estimator uses simulated terminal values $\widehat{X}_T^{(h)}$. Its total estimation error can be decomposed as

$$\widehat{\theta}_{n,h} - \theta = \left(\widehat{\theta}_{n,h} - \mathbb{E}[f(\widehat{X}_T^{(h)})]\right) + \left(\mathbb{E}[f(\widehat{X}_T^{(h)})] - \mathbb{E}[f(X_T)]\right).$$

The first term is statistical Monte Carlo noise, usually $O_{\mathbb{P}}(n^{-1/2})$. The second is discretization bias, often $O(h^p)$ for weak order p. Increasing n cannot remove a fixed step-size bias, and reducing h does not by itself reduce the sampling variance.

15.2 Richardson extrapolation

Suppose a weak approximation satisfies a more detailed expansion

$$\mathbb{E}[f(\widehat{X}_T^{(h)})] = \theta + ch^p + O(h^{p+1}).$$

For the same model, the half-step approximation satisfies $\mathbb{E}[f(\widehat{X}_T^{(h/2)})] = \theta + c(h/2)^p + O(h^{p+1})$. Eliminate the leading bias via

$$\begin{aligned}\theta_h^{\text{ext}} &= \frac{2^p \mathbb{E}[f(\widehat{X}_T^{(h/2)})] - \mathbb{E}[f(\widehat{X}_T^{(h)})]}{2^p - 1} \\ &= \theta + O(h^{p+1}).\end{aligned}$$

When implementing this with Monte Carlo, coupling the coarse and fine estimates using shared Brownian increments can reduce the variance of their difference. Strong errors and covariance structure matter even when the final objective is weak convergence.

15.3 Quasi-Monte Carlo integration

For $g : [0,1]^d \to \mathbb{R}$ and $\theta = \int_{[0,1]^d} g(u)\,\mathrm{d}u$, crude Monte Carlo samples independently, whereas quasi-Monte Carlo (QMC) uses carefully distributed deterministic points $v_1, \ldots, v_n$:

$$\widehat{\theta}_{\text{QMC}} = \frac{1}{n}\sum_{i=1}^{n} g(v_i).$$

Definition 15.1 (Star discrepancy)**.** For $P_n = \{v_1, \dots, v_n\} \subset [0,1]^d$, define

$$D_n^*(P_n) = \sup_{z \in [0,1]^d} \left| \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{[0,z)}(v_i) - \prod_{j=1}^{d} z_j \right|.$$

The star discrepancy measures how well the empirical point frequencies agree with volume in axis-aligned rectangles anchored at the origin. Low-discrepancy constructions such as Sobol' sequences spread points more evenly than a typical independent random sample in low and moderate effective dimension.

Theorem 15.2 (Koksma–Hlawka inequality)**.** *If g has finite variation $V_{\mathrm{HK}}(g)$ in the Hardy–Krause sense, then*

$$\left| \widehat{\theta}_{\mathrm{QMC}} - \theta \right| \le V_{\mathrm{HK}}(g) D_n^*(P_n).$$

For appropriate low-discrepancy sequences, a typical discrepancy bound is $O((\log n)^d / n)$, which can be better than $n^{-1/2}$ for fixed small d and large n. The bound can be uninformative for nonsmooth payoffs, singular transforms, or high dimension. Randomized QMC adds a random scrambling or shift while preserving favorable spatial distribution, allowing replication-based standard error estimates under an appropriate randomization scheme.

15.4 A transparent derivation of extrapolation weights

Suppose a numerical scheme has weak bias $ch + dh^2 + o(h^2)$. Let M_h and $M_{h/2}$ denote its expected payoff approximations. Then

$$M_h = \theta + ch + dh^2 + o(h^2),$$
$$M_{h/2} = \theta + \frac{c}{2}h + \frac{d}{4}h^2 + o(h^2).$$

Form $2M_{h/2} - M_h$:

$$\begin{aligned} 2M_{h/2} - M_h &= 2\theta + ch + \frac{d}{2}h^2 - \theta - ch - dh^2 + o(h^2) \\ &= \theta - \frac{d}{2}h^2 + o(h^2). \end{aligned}$$

Thus a first-order scheme can become second-order accurate for suitably smooth expectations by combining two mesh levels. If estimating M_h and $M_{h/2}$ by independent Monte Carlo runs, the extrapolated variance is $4\,\mathrm{Var}(\widehat{M}_{h/2}) + \mathrm{Var}(\widehat{M}_h)$, which can be costly. Instead, use a paired construction sharing the Brownian path, so

$$\mathrm{Var}(2\widehat{M}_{h/2} - \widehat{M}_h) = 4\,\mathrm{Var}(\widehat{M}_{h/2}) + \mathrm{Var}(\widehat{M}_h) - 4\,\mathrm{Cov}(\widehat{M}_{h/2}, \widehat{M}_h).$$

Positive coarse/fine covariance lowers the variance, an example of common random numbers in discretization error reduction.

15.5 QMC caveats in high-dimensional path simulation

A simulated Brownian trajectory observed at d time points can be expressed as a deterministic transform of a d-dimensional uniform vector. For example, the usual forward construction is

$$W_{t_j} = \sum_{k=1}^{j} \sqrt{t_k - t_{k-1}}\,\Phi^{-1}(u_k).$$

Although the input dimension equals the number of time steps, the payoff may depend principally on a few broad features of the path. Brownian bridge construction assigns early coordinates to coarse-scale path information, potentially reducing effective dimension for quasi-Monte Carlo. This can improve QMC performance even when the nominal dimension remains large. However, endpoint-heavy constructions are not universally optimal: a payoff controlled by an early barrier crossing may respond differently from a terminal-value payoff. Numerical experimentation with replicated randomized QMC is essential.

15.6 Richardson extrapolation from a weak-error expansion

Suppose a numerical expectation $P(h)$ has the asymptotic expansion

$$P(h) = P + ch^p + dh^{p+1} + O(h^{p+2}),$$

where P is the desired price, and $p > 0$. Compute the same approximation at steps h and $h/2$:

$$\begin{aligned} P(h/2) &= P + c\frac{h^p}{2^p} + d\frac{h^{p+1}}{2^{p+1}} + O(h^{p+2}), \\ 2^pP(h/2) - P(h) &= (2^p - 1)P \\ &\quad + dh^{p+1}\left(\frac{2^p}{2^{p+1}} - 1\right) + O(h^{p+2}) \\ &= (2^p - 1)P - \frac{1}{2}dh^{p+1} + O(h^{p+2}). \end{aligned}$$

Dividing by $2^p - 1$ yields

$$\begin{aligned} P_{\mathrm{RE}}(h) &= \frac{2^pP(h/2) - P(h)}{2^p - 1} \\ &= P - \frac{d}{2(2^p - 1)}h^{p+1} + O(h^{p+2}). \end{aligned}$$

The leading h^p bias is canceled. For the usual first-order weak Euler error, $p = 1$ and $P_{\mathrm{RE}}(h) = 2P(h/2) - P(h)$.

15.7 Why coupling coarse and fine trajectories matters

To compute a pathwise extrapolated payoff, use the same underlying Brownian motion at both resolutions. If the fine increments are ΔW_{2k} and ΔW_{2k+1}, set the coarse increment $\Delta W_k^{\mathrm{coarse}} =$

$\Delta W_{2k} + \Delta W_{2k+1}$. Its distribution is correct because

$$\begin{aligned}\mathbb{E}[\Delta W_k^{\text{coarse}}] &= 0,\\ \operatorname{Var}(\Delta W_k^{\text{coarse}}) &= \operatorname{Var}(\Delta W_{2k}) + \operatorname{Var}(\Delta W_{2k+1})\\ &= h/2 + h/2 = h.\end{aligned}$$

Consider $Y = 2f(\widehat{X}_T^{(h/2)}) - f(\widehat{X}_T^{(h)})$. Its variance is

$$\begin{aligned}\operatorname{Var}(Y) = 4\operatorname{Var}(f(\widehat{X}_T^{(h/2)})) + \operatorname{Var}(f(\widehat{X}_T^{(h)}))\\ - 4\operatorname{Cov}(f(\widehat{X}_T^{(h/2)}), f(\widehat{X}_T^{(h)})).\end{aligned}$$

Under coupling, the two payoffs are often highly positively correlated, making the negative covariance term helpful. Simulating two independent Brownian paths would leave the weak bias cancellation intact in expectation but could substantially inflate statistical noise.

15.8 Star discrepancy and the Koksma–Hlawka principle

Let $u_1, \dots, u_M \in [0,1]^d$ be a deterministic point set and $f : [0,1]^d \to \mathbb{R}$. The quasi-Monte Carlo estimator is $Q_M(f) = M^{-1}\sum_{i=1}^M f(u_i)$. Its star discrepancy is

$$D_M^* = \sup_{a \in [0,1]^d} \left| \frac{1}{M}\sum_{i=1}^M \mathbf{1}_{\{u_i \in [0,a)\}} - \prod_{j=1}^d a_j \right|,$$

where $[0,a) = \prod_{j=1}^d [0,a_j)$. For functions of bounded variation in the Hardy–Krause sense, the Koksma–Hlawka bound is

$$\left| Q_M(f) - \int_{[0,1]^d} f(u)\,\mathrm{d}u \right| \le V_{\text{HK}}(f) D_M^*.$$

The bound is deterministic. Standard Monte Carlo's $M^{-1/2}$ rate is stochastic and follows from variance calculations; one should not simply interpret low discrepancy as an unconditional improved error guarantee for every payoff. Discontinuities and the effective dimension of a high-dimensional Brownian-path functional matter greatly.

15.9 Randomized quasi-Monte Carlo and valid uncertainty estimates

Let $\mathcal{P}_M^{(1)}, \dots, \mathcal{P}_M^{(R)}$ be independent randomizations of a low-discrepancy point set, each preserving uniform marginals. Form estimates

$$\begin{aligned}Q_M^{(r)} &= \frac{1}{M}\sum_{u \in \mathcal{P}_M^{(r)}} f(u), \qquad r = 1, \dots, R,\\ \overline{Q}_{M,R} &= \frac{1}{R}\sum_{r=1}^R Q_M^{(r)},\\ s_Q^2 &= \frac{1}{R-1}\sum_{r=1}^R (Q_M^{(r)} - \overline{Q}_{M,R})^2.\end{aligned}$$

The standard error estimated from the R independent randomizations is $s_Q/\sqrt{R}$. The M points within one randomized low-discrepancy set are typically dependent, so treating all MR points as independent for confidence intervals is generally invalid.

15.10 Quantifying the total mean-square error of a discretized Monte Carlo price

Suppose $P = \mathbb{E}[f(X_T)]$, but we simulate $f(\widehat{X}_T^{(h)})$ instead. With M independent numerical paths, define $\widehat{P}_{M,h} = M^{-1}\sum_{i=1}^{M} f(\widehat{X}_T^{(h,i)})$. Let $P_h = \mathbb{E}[f(\widehat{X}_T^{(h)})]$. By adding and subtracting P_h,

$$\widehat{P}_{M,h} - P = (\widehat{P}_{M,h} - P_h) + (P_h - P).$$

The first part has mean zero and the second is deterministic. Expanding the squared error,

$$\begin{aligned}\mathbb{E}[(\widehat{P}_{M,h} - P)^2] &= \mathbb{E}[(\widehat{P}_{M,h} - P_h)^2] + (P_h - P)^2 \\ &\quad + 2(P_h - P)\mathbb{E}[\widehat{P}_{M,h} - P_h] \\ &= \frac{\mathrm{Var}(f(\widehat{X}_T^{(h)}))}{M} + (P_h - P)^2.\end{aligned}$$

Thus both M and h must be chosen. If weak bias is $O(h^p)$, squared bias is $O(h^{2p})$. Increasing M only shrinks statistical error and will not eliminate a persistent discretization bias.

15.11 Balancing time-step count against paths

If a trajectory requires $N = T/h$ Euler updates, the work per path is proportional to N. Suppose variance is approximately v and weak bias approximately ch^p, so the MSE is $v/M + c^2h^{2p}$. For a desired root mean squared error of order ε, a natural balance is

$$\begin{aligned}\frac{v}{M} &\asymp \varepsilon^2, \\ c^2h^{2p} &\asymp \varepsilon^2, \\ M &\asymp \varepsilon^{-2}, \\ h &\asymp \varepsilon^{1/p}, \\ N &\asymp \varepsilon^{-1/p}.\end{aligned}$$

Thus total work $MN \asymp \varepsilon^{-(2+1/p)}$. Richardson extrapolation can improve the weak bias order and thereby reduce the cost of attaining a fixed total error, provided the strong error, variance, and extra fine-path computation are also accounted for.

15.12 The multilevel Monte Carlo telescoping identity

Let $P_\ell = \mathbb{E}[f(\widehat{X}_T^{(h_\ell)})]$ with nested step sizes $h_\ell = T2^{-\ell}$. The finest-level expectation can be written exactly as

$$\begin{aligned} P_L &= P_0 + \sum_{\ell=1}^{L} (P_\ell - P_{\ell-1}) \\ &= \mathbb{E}[f(\widehat{X}_T^{(h_0)})] + \sum_{\ell=1}^{L} \mathbb{E}\left[f(\widehat{X}_T^{(h_\ell)}) - f(\widehat{X}_T^{(h_{\ell-1})}) \right]. \end{aligned}$$

Estimate each expectation with its own independent Monte Carlo sample size M_ℓ, but couple the coarse and fine paths *within* each difference. The estimator is

$$\begin{aligned} \widehat{P}_{\mathrm{ML}} = &\frac{1}{M_0} \sum_{i=1}^{M_0} f(X_0^{(i)}) \\ &+ \sum_{\ell=1}^{L} \frac{1}{M_\ell} \sum_{i=1}^{M_\ell} \left[f(X_\ell^{(i)}) - f(X_{\ell-1}^{(i)}) \right]. \end{aligned}$$

Independence across level estimators gives

$$\mathrm{Var}(\widehat{P}_{\mathrm{ML}}) = \frac{V_0}{M_0} + \sum_{\ell=1}^{L} \frac{V_\ell}{M_\ell},$$

where V_ℓ is the variance of the coupled level difference. Because fine and coarse approximations converge to the same limiting path, V_ℓ often decreases with level. This can substantially lower cost compared with simulating all paths at the finest resolution.

15.13 Cost-optimal distribution of multilevel samples

Let C_ℓ be the work per sample and fix a target sampling variance $\sum_\ell V_\ell / M_\ell = \varepsilon^2$. Minimize total work $\sum_\ell C_\ell M_\ell$ using a Lagrange multiplier λ:

$$\begin{aligned} \mathscr{L} &= \sum_\ell C_\ell M_\ell + \lambda \left(\sum_\ell \frac{V_\ell}{M_\ell} - \varepsilon^2 \right), \\ 0 &= \frac{\partial \mathscr{L}}{\partial M_\ell} = C_\ell - \lambda \frac{V_\ell}{M_\ell^2}, \\ M_\ell &= \sqrt{\lambda} \sqrt{\frac{V_\ell}{C_\ell}}. \end{aligned}$$

Substitute into the variance constraint:

$$\begin{aligned} \varepsilon^2 &= \frac{1}{\sqrt{\lambda}} \sum_\ell \sqrt{V_\ell C_\ell}, \\ \sqrt{\lambda} &= \varepsilon^{-2} \sum_\ell \sqrt{V_\ell C_\ell}, \\ M_\ell &= \varepsilon^{-2} \left(\sum_k \sqrt{V_k C_k} \right) \sqrt{\frac{V_\ell}{C_\ell}}. \end{aligned}$$

Thus expensive high-resolution levels receive fewer samples when their level-difference variance has become small. This is an optional extension to the lecture's extrapolation discussion, not an assumption needed for Richardson extrapolation itself.

15.14 Bias-canceling extrapolation with independent uncertainty accounting

Suppose $\widehat{P}_h = M^{-1}\sum_{i=1}^{M} Y_h^{(i)}$ and $\widehat{P}_{h/2} = M^{-1}\sum_{i=1}^{M} Y_{h/2}^{(i)}$ use Brownian-coupled coarse and fine samples. The first-order Richardson estimator is

$$\begin{aligned}\widehat{P}_{\mathrm{RE}} &= 2\widehat{P}_{h/2} - \widehat{P}_h \\ &= \frac{1}{M}\sum_{i=1}^{M}[2Y_{h/2}^{(i)} - Y_h^{(i)}].\end{aligned}$$

The correct standard error is computed from the *paired* values $R_i = 2Y_{h/2}^{(i)} - Y_h^{(i)}$. Because R_i are independent across paths but their coarse/fine components are dependent within each path,

$$\mathrm{Var}(\widehat{P}_{\mathrm{RE}}) = \frac{1}{M}\left[4\,\mathrm{Var}(Y_{h/2}) + \mathrm{Var}(Y_h) - 4\,\mathrm{Cov}(Y_{h/2}, Y_h)\right].$$

Ignoring the covariance can dramatically misstate estimator uncertainty. Extrapolation reduces discretization bias under an error expansion; it does not promise lower Monte Carlo variance without adequate path coupling.

LECTURE 16

Poisson Processes, Jump Models, and Introduction to PDE Pricing

16.1 A Poisson process and exponential waiting times

Definition 16.1 (Homogeneous Poisson process)**.** A counting process $(N_t)_{t\geq 0}$ with rate $\lambda > 0$ satisfies $N_0 = 0$, has independent stationary increments, and for $s < t$,

$$N_t - N_s \sim \text{Poisson}(\lambda(t-s)).$$

Hence $\mathbb{P}(N_t = k) = e^{-\lambda t}(\lambda t)^k/k!$, $\mathbb{E}[N_t] = \lambda t$, and $\text{Var}(N_t) = \lambda t$. Let T_1 denote the first arrival time. The event $\{T_1 > t\}$ is equivalent to having no arrivals in $(0, t]$, so

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = e^{-\lambda t},$$
$$F_{T_1}(t) = 1 - e^{-\lambda t}, \qquad f_{T_1}(t) = \lambda e^{-\lambda t}.$$

The inter-arrival durations are independent $\text{Exp}(\lambda)$ random variables. To simulate all jump times before T, generate exponentials $E_j = -\log U_j/\lambda$ and form the partial sums $T_k = \sum_{j=1}^{k} E_j$ until exceeding T. Alternatively, simulate the number of events as $\text{Poisson}(\lambda T)$ and conditional uniform jump times, then sort them.

16.2 Jump-diffusion asset models

A simple jump-diffusion can be written

$$\frac{\mathrm{d}S_t}{S_{t-}} = (\mu - \lambda\kappa)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t,$$

where S_{t-} is the left limit, $J > 0$ is a random gross jump multiplier, and $\kappa = \mathbb{E}[J-1]$. The compensator $-\lambda\kappa$ corrects for the average jump contribution. In a risk-neutral model with no dividends, replace μ by r, with appropriate assumptions that ensure the discounted stock is a martingale. Conditional on independently generated jump times, one can propagate the diffusion exactly between arrivals and multiply the stock by each jump factor at an event. This event-driven construction avoids missing discontinuities on a fixed time grid.

16.3 A first derivation of the Black–Scholes PDE

For a European payoff $H(S_T)$, let $V(t, s)$ denote its smooth model price in the risk-neutral GBM model. Itô's formula yields

$$\mathrm{d}V(t, S_t) = \left(V_t + rS_tV_s + \tfrac{1}{2}\sigma^2 S_t^2 V_{ss}\right)\,\mathrm{d}t + \sigma S_t V_s\,\mathrm{d}W_t.$$

Form the delta-hedged portfolio consisting of one option and short V_s shares. The Brownian fluctuation cancels, leaving instantaneous drift

$$\mathrm{d}V - V_s\,\mathrm{d}S = \left(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{ss}\right)\,\mathrm{d}t.$$

The portfolio value is $V - SV_s$. Absence of arbitrage requires its riskless drift to be $r(V - SV_s)\,\mathrm{d}t$. Equating terms and rearranging produces

$$V_t + \tfrac{1}{2}\sigma^2 s^2 V_{ss} + rsV_s - rV = 0, \qquad V(T,s) = H(s).$$

The PDE is a *backward terminal-value problem*: its terminal payoff is known, and the objective is to infer earlier values. Numerical methods solve this PDE when a convenient closed form does not exist.

16.4 The generator and its probabilistic meaning

For a diffusion with drift rs and variance rate $\sigma^2 s^2$, the infinitesimal generator acting on smooth functions is

$$(\mathcal{A}f)(s) = rsf'(s) + \tfrac{1}{2}\sigma^2 s^2 f''(s).$$

The pricing equation is $V_t + \mathcal{A}V - rV = 0$. It expresses the same risk-neutral discounting structure as $V(t,s) = \mathbb{E}^{\mathbb{Q}}_{t,s}[e^{-r(T-t)}H(S_T)]$ under conditions of Feynman–Kac. The relationship between stochastic processes, conditional expectations, and parabolic PDEs is one of the organizing ideas of numerical finance.

16.5 Conditional jump simulation and compound Poisson losses

Suppose a jump process has independent jump sizes $Y_1, Y_2, \ldots$ and rate-λ counting process N_t. Define the compound Poisson process

$$J_t = \sum_{k=1}^{N_t} Y_k.$$

If $\mathbb{E}[|Y_1|] < \infty$, conditioning on N_t gives

$$\begin{aligned}\mathbb{E}[J_t] &= \mathbb{E}\left[\mathbb{E}\left[\sum_{k=1}^{N_t} Y_k \mid N_t\right]\right] \\ &= \mathbb{E}[N_t]\mathbb{E}[Y_1] = \lambda t\mathbb{E}[Y_1].\end{aligned}$$

If $\mathbb{E}[Y_1^2] < \infty$, the law of total variance gives

$$\begin{aligned}\mathrm{Var}(J_t) &= \mathbb{E}[N_t\,\mathrm{Var}(Y_1)] + \mathrm{Var}(N_t\mathbb{E}[Y_1]) \\ &= \lambda t\,\mathrm{Var}(Y_1) + \lambda t(\mathbb{E}[Y_1])^2 \\ &= \lambda t\mathbb{E}[Y_1^2].\end{aligned}$$

These identities are direct tests for a jump simulator and explain why jumps create additional variability beyond the Brownian diffusion term.

16.6 Risk-neutral jump compensation

Consider a stock with proportional random jumps, interpreted through

$$\frac{\mathrm{d}S_t}{S_{t-}} = b\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (Y-1)\,\mathrm{d}N_t.$$

Over an infinitesimal period $\mathrm{d}t$, the expected jump contribution to $\mathrm{d}S_t/S_{t-}$ equals $\lambda\mathbb{E}[Y-1]\,\mathrm{d}t$. Therefore the total expected percentage drift is $b + \lambda\mathbb{E}[Y-1]$. To make the undiscounted risk-neutral stock have conditional expected return r, choose

$$b = r - \lambda\mathbb{E}[Y-1].$$

If dividends are paid continuously at rate q, replace r by $r-q$. Missing the jump compensation changes forward prices and invalidates the discounted-martingale property. Exact event-time simulation can combine this compensated deterministic drift with Brownian innovations and multiplicative jump factors.

16.7 PDE versus integro-differential equations

A pure diffusion has a local generator containing only derivatives. A jump-diffusion's generator also contains a nonlocal expectation over post-jump states. With jump multiplier Y and intensity λ, one schematic risk-neutral generator is

$$(\mathcal{A}f)(s) = (r-\lambda\kappa)sf'(s) + \frac{1}{2}\sigma^2 s^2 f''(s) + \lambda\mathbb{E}[f(sY) - f(s)].$$

Accordingly, an option value generally satisfies a partial integro-differential equation (PIDE)

$$V_t + \mathcal{A}V - rV = 0$$

instead of a purely differential Black–Scholes PDE. Numerically, the integral introduces nonlocal coupling between stock-grid nodes, illustrating why simulation is particularly attractive for certain jump and path-dependent models.

16.8 Poisson waiting times from the zero-event probability

For a Poisson process N_t of rate λ, $N_t \sim \text{Poisson}(\lambda t)$. In particular,

$$\mathbb{P}(N_t = 0) = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t}.$$

If T_1 is the first arrival time, the event $\{T_1 > t\}$ is exactly $\{N_t = 0\}$. Consequently,

$$\begin{aligned}\mathbb{P}(T_1 > t) &= e^{-\lambda t},\\ F_{T_1}(t) &= 1 - e^{-\lambda t},\\ f_{T_1}(t) &= \lambda e^{-\lambda t}, \qquad t > 0.\end{aligned}$$

The memoryless property follows:

$$\begin{aligned}\mathbb{P}(T_1 > s+t \mid T_1 > s) &= \frac{\mathbb{P}(T_1 > s+t)}{\mathbb{P}(T_1 > s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}.\end{aligned}$$

Independent exponentially distributed gaps therefore generate successive jump times exactly by cumulative addition.

16.9 Compensating the jump drift under the pricing measure

Consider a multiplicative jump diffusion with strictly positive jump multipliers J_k:

$$\frac{\mathrm{d}S_t}{S_{t-}} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t.$$

If independent jump multipliers are i.i.d. with mean $m_J = \mathbb{E}[J]$, then over a short interval h the expected relative jump contribution is approximately $\lambda(m_J - 1)h$. Hence under a no-dividend risk-neutral model the total expected instantaneous return must equal r, so the continuous drift must be

$$\mu_{\mathbb{Q}} = r - \lambda(m_J - 1).$$

Writing $\kappa_J = m_J - 1$, the risk-neutral dynamics become

$$\frac{\mathrm{d}S_t}{S_{t-}} = (r - \lambda\kappa_J)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t.$$

Between jumps the process evolves geometrically, while at each jump time it is multiplied by J_k. Therefore

$$S_T = S_0 \exp\left\{(r - \lambda\kappa_J - \tfrac{1}{2}\sigma^2)T + \sigma W_T\right\} \prod_{k=1}^{N_T} J_k.$$

Using the Poisson probability generating function $\mathbb{E}[a^{N_T}] = e^{\lambda T(a-1)}$ gives

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[S_T] &= S_0 e^{(r-\lambda\kappa_J)T}\mathbb{E}\left[\prod_{k=1}^{N_T} J_k\right] \\ &= S_0 e^{(r-\lambda\kappa_J)T}\mathbb{E}[m_J^{N_T}] \\ &= S_0 e^{(r-\lambda\kappa_J)T} e^{\lambda T(m_J-1)} \\ &= S_0 e^{rT},\end{aligned}$$

verifying the risk-neutral mean condition.

16.10 Black–Scholes PDE from a self-financing delta hedge

For a European option with $V = V(S,t)$ smooth and with stock dynamics $\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W$ under $\mathbb{Q}$, Itô's formula gives

$$\mathrm{d}V = \left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)\,\mathrm{d}t + \sigma S V_S\,\mathrm{d}W.$$

Form a self-financing position long one option and short $\Delta = V_S$ shares. Its risky diffusion term vanishes:

$$\begin{aligned}\mathrm{d}\Pi &= \mathrm{d}V - \Delta\,\mathrm{d}S \\ &= \left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)\,\mathrm{d}t + (V_S - \Delta)\sigma S\,\mathrm{d}W \\ &= \left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)\,\mathrm{d}t.\end{aligned}$$

Its value is $\Pi = V - SV_S$, and the riskless return condition requires $\mathrm{d}\Pi = r\Pi\,\mathrm{d}t$. Consequently,

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} = r(V - SV_S),$$
$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0.$$

The terminal condition is $V(S,T) = g(S)$. This derivation is schematic in that it suppresses the self-financing cash bookkeeping associated with continuous rebalancing; the correct portfolio strategy includes the cash account and uses a gain process, not the ordinary differential of $V - SV_S$ as if Δ were fixed.

16.11 The jump-diffusion pricing equation and its nonlocal term

For a smooth European option with the jump law above, the discounted price is a martingale only if its drift equals rV. Applying the jump Itô formula gives the integro-differential equation

$$\begin{aligned}0 = V_t + (r - \lambda\kappa_J)sV_s + \frac{1}{2}\sigma^2 s^2 V_{ss} - rV \\ + \lambda\mathbb{E}_J[V(js,t) - V(s,t)].\end{aligned}$$

The integral term is *nonlocal*: pricing at stock value s depends on values of the function at all possible jumped values js. A finite-difference implementation therefore generally needs both spatial derivatives and numerical quadrature (or a related transform) for the jump term.

16.12 Simulation of a compound Poisson sum

Let $N_T \sim \mathrm{Poisson}(\lambda T)$ and let independent jump sizes $Y_1, Y_2, \ldots$ have mean m_Y and variance s_Y^2, independent of N_T. Define $J_T = \sum_{k=1}^{N_T} Y_k$, where the sum is zero for $N_T = 0$. Conditioning on

$N_T = n$,

$$\mathbb{E}[J_T \mid N_T = n] = n m_Y,$$
$$\mathrm{Var}(J_T \mid N_T = n) = n s_Y^2.$$

Apply the law of total expectation and variance:

$$\begin{aligned}\mathbb{E}[J_T] &= \mathbb{E}[N_T m_Y] = \lambda T m_Y,\\ \mathrm{Var}(J_T) &= \mathbb{E}[\mathrm{Var}(J_T \mid N_T)] + \mathrm{Var}(\mathbb{E}[J_T \mid N_T])\\ &= s_Y^2 \mathbb{E}[N_T] + m_Y^2 \mathrm{Var}(N_T)\\ &= \lambda T(s_Y^2 + m_Y^2) = \lambda T \mathbb{E}[Y_1^2].\end{aligned}$$

An exact simulator samples N_T first, then samples that many jump sizes and sums them. If jump times are required for path-dependent contracts, instead simulate successive exponential waiting times until their cumulative sum first exceeds T.

16.13 Exact simulation of lognormal jump multipliers

If $J = e^{\mu_J + \sigma_J Z}$ with $Z \sim \mathcal{N}(0,1)$, its moments are

$$\begin{aligned}\mathbb{E}[J^k] &= e^{k\mu_J + k^2\sigma_J^2/2},\\ \mathbb{E}[J] &= e^{\mu_J + \sigma_J^2/2},\\ \kappa_J &= \mathbb{E}[J-1] = e^{\mu_J + \sigma_J^2/2} - 1.\end{aligned}$$

For the jump diffusion, the risk-neutral compensating drift becomes $r - \lambda[e^{\mu_J + \sigma_J^2/2} - 1]$. A simulation that uses drift r *and* positive-mean multiplicative jumps is not risk-neutral: its discounted stock price would systematically increase in expectation.

16.14 Exponential martingales and a risk-neutral jump model

For an independent Brownian motion and compound Poisson jump multipliers, consider $S_t = S_0 e^{at + \sigma W_t} \prod_{j=1}^{N_t} J_j$, where $J_j > 0$ are i.i.d. and $N_t \sim \mathrm{Poisson}(\lambda t)$. Independence gives

$$\begin{aligned}\mathbb{E}[S_t] &= S_0 e^{at} \mathbb{E}[e^{\sigma W_t}] \mathbb{E}\left[\prod_{j=1}^{N_t} J_j\right]\\ &= S_0 e^{at} e^{\sigma^2 t/2} e^{\lambda t(\mathbb{E}[J]-1)}.\end{aligned}$$

The discounted process $e^{-rt} S_t$ has constant expectation equal to S_0 precisely when

$$\begin{aligned}a + \frac{1}{2}\sigma^2 + \lambda(\mathbb{E}[J] - 1) &= r,\\ a &= r - \frac{1}{2}\sigma^2 - \lambda(\mathbb{E}[J] - 1).\end{aligned}$$

The same parameter produces the continuous drift $r - \lambda(\mathbb{E}[J] - 1)$ in the jump SDE because the Brownian exponential already contains the $-\sigma^2/2$ Itô adjustment. This double-check avoids the common error of compensating the Brownian variance term twice.

16.15 Conditional Monte Carlo over jump counts

For a European payoff $g(S_T)$ in a jump-diffusion model, apply the law of total expectation:

$$\begin{aligned} V_0 &= e^{-rT}\mathbb{E}[g(S_T)] \\ &= e^{-rT}\sum_{n=0}^{\infty}\mathbb{P}(N_T=n)\mathbb{E}[g(S_T)\mid N_T=n] \\ &= e^{-rT}\sum_{n=0}^{\infty}\frac{e^{-\lambda T}(\lambda T)^n}{n!}\mathbb{E}[g(S_T)\mid N_T=n]. \end{aligned}$$

Conditional on $N_T = n$ and lognormal multipliers $J_j = e^{\mu_J+\sigma_J Z_j}$, the terminal log-price is normal:

$$\log S_T \mid N_T = n \sim \mathcal{N}\left(\log S_0 + (r-\lambda\kappa_J-\tfrac{1}{2}\sigma^2)T + n\mu_J,\ \sigma^2 T + n\sigma_J^2\right).$$

Thus conditional European call prices can be evaluated using lognormal integrals, leaving only a Poisson-weighted sum. Truncating the sum after $n_{\max}$ events creates a tail error controlled by $\mathbb{P}(N_T > n_{\max})$ times appropriate payoff moment bounds; simply discarding large counts without checking their probability can be unsafe.

16.16 An event-driven jump-diffusion simulator

Algorithm 16.2 (Exact jump times and continuous evolution)**.** To simulate S_T for a multiplicative jump diffusion with independent jumps:

1. Initialize $t \leftarrow 0$ and $S \leftarrow S_0$.
2. Draw an exponential waiting time $E = -\log U/\lambda$ with $U \sim \text{Uniform}(0,1)$. Set $t_{\text{next}} = t + E$.
3. If $t_{\text{next}} > T$, evolve S geometrically over $T - t$ using the compensated continuous drift and a normal Brownian increment, and terminate.
4. Otherwise evolve S geometrically for duration E, sample an independent jump multiplier J, and apply $S \leftarrow SJ$.
5. Set $t \leftarrow t_{\text{next}}$ and return to step 2.

For an exact constant-coefficient model, Brownian increments over disjoint interjump intervals are independent normals with variances equal to the interval lengths. A grid-based Bernoulli approximation that takes $\Delta N \in \{0,1\}$ with probability λh is only an approximation and can fail if λh is not small; the exact Poisson process permits multiple jumps in a short interval. Event-driven simulation avoids this approximation while retaining the correct jump-time distribution.

LECTURE 17

Finite Differences for the Black–Scholes PDE

17.1 From option valuation to a parabolic PDE

For a non-dividend-paying stock under risk-neutral geometric Brownian motion, the value $V(s,t)$ of a smooth European contingent claim solves

$$V_t + \frac{1}{2}\sigma^2 s^2 V_{ss} + rsV_s - rV = 0, \qquad 0 \le t < T,$$

with terminal condition $V(s,T) = H(s)$. The PDE is backward in calendar time. To write it as a forward evolution equation in time-to-maturity, introduce $\tau = T - t$ and $v(s,\tau) = V(s, T-\tau)$. The chain rule gives $v_\tau = -V_t$, so

$$v_\tau = \frac{1}{2}\sigma^2 s^2 v_{ss} + rsv_s - rv, \qquad v(s,0) = H(s).$$

We work on a truncated spatial interval $0 = s_0 < s_1 < \cdots < s_M = s_{\max}$ and time grid $\tau_n = n\Delta\tau$, $n = 0, \dots, N$, with $N\Delta\tau = T$. Let $v_i^n \approx v(s_i, \tau_n)$. The design of a reliable finite-difference scheme requires two independent ingredients: consistent approximations of derivatives and appropriate boundary conditions.

17.2 Finite-difference formulas from Taylor expansions

For simplicity begin with an equally spaced spatial grid $s_{i+1} - s_i = h$. Expand a smooth function f about s_i:

$$f(s_i + h) = f_i + hf_i' + \frac{h^2}{2}f_i'' + \frac{h^3}{6}f_i''' + O(h^4),$$
$$f(s_i - h) = f_i - hf_i' + \frac{h^2}{2}f_i'' - \frac{h^3}{6}f_i''' + O(h^4).$$

Subtract the expansions, divide by $2h$, and obtain the central first derivative:

$$f'(s_i) = \frac{f_{i+1} - f_{i-1}}{2h} + O(h^2).$$

Alternatively, the one-sided first differences are

$$f'(s_i) = \frac{f_{i+1} - f_i}{h} + O(h), \qquad f'(s_i) = \frac{f_i - f_{i-1}}{h} + O(h).$$

Adding the Taylor expansions gives

$$f_{i+1} + f_{i-1} = 2f_i + h^2 f_i'' + O(h^4),$$
$$f''(s_i) = \frac{f_{i+1} - 2f_i + f_{i-1}}{h^2} + O(h^2).$$

For a nonuniform grid, write $h_- = s_i - s_{i-1}$ and $h_+ = s_{i+1} - s_i$. A useful local approximation is

$$v_{ss}(s_i) \approx \frac{2}{h_- + h_+} \left(\frac{v_{i+1} - v_i}{h_+} - \frac{v_i - v_{i-1}}{h_-} \right).$$

This formula is consistent for smooth functions as the grid is refined; the precise global order depends on mesh regularity.

17.3 Explicit time stepping

Define local coefficients for an interior node by

$$a_i = \frac{1}{2}\sigma^2 s_i^2, \qquad b_i = r s_i.$$

On a uniform grid with central differences, write the spatial operator in the form

$$(\mathcal{L}_h v)_i = \alpha_i (v_{i-1} - v_i) + \beta_i (v_{i+1} - v_i) - r v_i,$$

where

$$\alpha_i = \frac{a_i}{h^2} - \frac{b_i}{2h}, \qquad \beta_i = \frac{a_i}{h^2} + \frac{b_i}{2h}.$$

Indeed,

$$\alpha_i (v_{i-1} - v_i) + \beta_i (v_{i+1} - v_i) = \frac{a_i}{h^2}(v_{i-1} - 2v_i + v_{i+1}) + \frac{b_i}{2h}(v_{i+1} - v_{i-1}).$$

A forward difference in τ gives $v_i^{n+1} = v_i^n + \Delta\tau (\mathcal{L}_h v^n)_i$, or

$$v_i^{n+1} = \Delta\tau \alpha_i v_{i-1}^n + \left[1 - \Delta\tau(\alpha_i + \beta_i + r)\right] v_i^n + \Delta\tau \beta_i v_{i+1}^n.$$

When all three coefficients are nonnegative and sum to $1 - r\Delta\tau$ with $r \geq 0$, the update has a positivity-preserving structure. A sufficient local step restriction is

$$\Delta\tau \leq \frac{1}{\max_i (\alpha_i + \beta_i + r)}$$

provided $\alpha_i, \beta_i \geq 0$ for all interior nodes.

17.4 Initial and boundary data

The terminal payoff becomes the numerical initial condition $v_i^0 = H(s_i)$. For the Black–Scholes European call, the natural asymptotic boundaries are

$$v(0,\tau) = 0, \qquad v(s,\tau) \sim s - Ke^{-r\tau} \quad (s \to \infty).$$

For the European put,

$$v(0,\tau) = Ke^{-r\tau}, \qquad v(s,\tau) \to 0 \quad (s \to \infty).$$

These conditions follow from the probabilistic payoff and no-arbitrage bounds and are more accurate at a finite upper boundary than arbitrarily holding the upper-node value fixed. At $s = 0$, the PDE reduces to $v_\tau = -rv$ for payoffs consistent with an absorbing zero stock price. An explicit lower-boundary update is $v_0^{n+1} = (1 - r\Delta\tau)v_0^n$; exact discounting or a higher-order discretization is also possible.

17.5 Consistency obtained by inserting the exact solution

Consider the differential operator

$$(\mathcal{L}v)(s,\tau) = a(s)v_{ss}(s,\tau) + b(s)v_s(s,\tau) - rv(s,\tau).$$

Let v be a sufficiently smooth exact solution of $v_\tau = \mathcal{L}v$. Evaluate the numerical explicit residual at a grid point:

$$R_i^n = \frac{v(s_i,\tau_{n+1}) - v(s_i,\tau_n)}{\Delta\tau} - (\mathcal{L}_h v(\cdot,\tau_n))_i.$$

The time Taylor expansion yields

$$\frac{v(s_i,\tau_{n+1}) - v(s_i,\tau_n)}{\Delta\tau} = v_\tau(s_i,\tau_n) + \frac{\Delta\tau}{2}v_{\tau\tau}(s_i,\tau_n) + O(\Delta\tau^2).$$

With centered spatial derivatives on a uniform grid,

$$\mathcal{L}_h v = \mathcal{L}v + O(h^2).$$

Substituting $v_\tau = \mathcal{L}v$ gives

$$R_i^n = O(\Delta\tau) + O(h^2).$$

This is the consistency error in the differential equation. It does not prove the numerical solution converges, because errors can amplify when the scheme is iterated. A separate stability argument is necessary.

17.6 Why the high-stock boundary must be handled carefully

For a call with terminal payoff $(s-K)^+$, the asymptotic form $v(s,\tau) \sim s - Ke^{-r\tau}$ is determined by the event that a sufficiently large stock price almost surely leaves the call in the money at maturity. More precisely, the Black–Scholes formula has $d_1, d_2 \to +\infty$ as $s \to \infty$, so $\Phi(d_1), \Phi(d_2) \to 1$ and

$$v(s,\tau) = s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2) \sim s - Ke^{-r\tau}.$$

Imposing the constant upper-node condition $v(s_{\max}, \tau) = v(s_{\max}, 0) = s_{\max} - K$ ignores the discounting of the strike and can introduce a bias even when the spatial mesh is extremely fine. Choosing a more accurate asymptotic condition is therefore a distinct numerical issue from refining the derivative stencil.

17.7 Nonuniform-grid first derivatives

For $h_- = s_i - s_{i-1}$ and $h_+ = s_{i+1} - s_i$, a three-point nonuniform derivative approximation can be obtained by eliminating the second derivative from asymmetric Taylor expansions. The result is

$$f'(s_i) \approx -\frac{h_+}{h_-(h_- + h_+)} f_{i-1} + \frac{h_+ - h_-}{h_- h_+} f_i + \frac{h_-}{h_+(h_- + h_+)} f_{i+1}.$$

If $h_- = h_+ = h$, it reduces to $(f_{i+1} - f_{i-1})/(2h)$. In practice, clustering points near a strike can increase local pricing accuracy without multiplying the total number of nodes as aggressively as a globally fine uniform grid.

17.8 Finite-difference derivatives from Taylor's theorem

Let f possess four continuous derivatives near x, and let $h > 0$. Taylor-expanding on the right and left,

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{6} f'''(x) + \frac{h^4}{24} f^{(4)}(x) + O(h^5),$$
$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(x) - \frac{h^3}{6} f'''(x) + \frac{h^4}{24} f^{(4)}(x) + O(h^5).$$

Subtract and divide by $2h$ to derive the centered first derivative:

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3} f'''(x) + O(h^5),$$
$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{6} f'''(x) + O(h^4).$$

Add and subtract $2f(x)$ to derive the centered second derivative:

$$f(x+h) - 2f(x) + f(x-h) = h^2 f''(x) + \frac{h^4}{12} f^{(4)}(x) + O(h^6),$$
$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{h^2}{12} f^{(4)}(x) + O(h^4).$$

The leading error is second order in h for both centered formulas. In contrast, a forward first derivative satisfies $(f(x+h) - f(x))/h = f'(x) + (h/2)f''(x) + O(h^2)$, only first-order accurate.

17.9 Turning a terminal-value PDE into an initial-value problem

Under Black–Scholes, $V_t + \frac{1}{2}\sigma^2 s^2 V_{ss} + rsV_s - rV = 0$, with maturity condition $V(s,T) = g(s)$. Define time-to-maturity $\tau = T - t$ and $v(s,\tau) = V(s, T-\tau)$. By the chain rule,

$$\begin{aligned}\frac{\partial v}{\partial \tau}(s,\tau) &= \frac{\partial V}{\partial t}(s, T-\tau)\frac{\partial (T-\tau)}{\partial \tau} \\ &= -V_t(s, T-\tau).\end{aligned}$$

Thus the equation is a forward evolution in τ,

$$\begin{aligned}v_\tau &= \frac{1}{2}\sigma^2 s^2 v_{ss} + rsv_s - rv, \\ v(s,0) &= g(s).\end{aligned}$$

This transformation is not optional bookkeeping: an incorrect sign on v_τ leads to evolving a diffusion equation backward in its stable numerical time direction.

17.10 Deriving the complete explicit three-point stencil

Choose $s_i = i\Delta s$, $\tau_n = n\Delta\tau$, and denote $v_i^n \approx v(s_i, \tau_n)$. Replace the time derivative by a forward difference and the spatial derivatives by centered differences:

$$\begin{aligned}\frac{v_i^{n+1} - v_i^n}{\Delta\tau} = \frac{1}{2}\sigma^2 s_i^2 \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{(\Delta s)^2} \\ + rs_i \frac{v_{i+1}^n - v_{i-1}^n}{2\Delta s} - rv_i^n.\end{aligned}$$

Solve for the new value and collect the three neighboring coefficients:

$$\begin{aligned}v_i^{n+1} = v_i^n + \Delta\tau \left[\left(\frac{\sigma^2 s_i^2}{2(\Delta s)^2} - \frac{rs_i}{2\Delta s} \right) v_{i-1}^n \right. \\ \left. + \left(-\frac{\sigma^2 s_i^2}{(\Delta s)^2} - r \right) v_i^n + \left(\frac{\sigma^2 s_i^2}{2(\Delta s)^2} + \frac{rs_i}{2\Delta s} \right) v_{i+1}^n \right].\end{aligned}$$

Since $s_i = i\Delta s$, the weights simplify to

$$\begin{aligned}a_i &= \frac{\Delta\tau}{2}(\sigma^2 i^2 - ri), \\ b_i &= 1 - \Delta\tau(\sigma^2 i^2 + r), \\ c_i &= \frac{\Delta\tau}{2}(\sigma^2 i^2 + ri), \\ v_i^{n+1} &= a_i v_{i-1}^n + b_i v_i^n + c_i v_{i+1}^n.\end{aligned}$$

The sum $a_i + b_i + c_i = 1 - r\Delta\tau$ reflects the discounting term. At $i = 0$ and $i = M$ the update references boundary conditions rather than an interior three-point stencil.

17.11 Deriving the no-dividend call boundary conditions

At zero stock price, a geometric Brownian motion remains zero, so a call is worthless: $v(0,\tau) = 0$. At very large s, the probability of expiring out-of-the-money becomes negligible and the call approximates a forward position:

$$v(s_{\max},\tau) \approx s_{\max} - Ke^{-r\tau}.$$

For a put the corresponding far-right boundary is near zero, while at zero $v(0,\tau) = Ke^{-r\tau}$. The boundaries are asymptotic approximations on a finite spatial grid; $s_{\max}$ should be sufficiently far into the relevant distribution tail that boundary truncation error is small relative to discretization and sampling errors.

17.12 The local truncation error of the explicit Black–Scholes scheme

For a sufficiently smooth exact solution $v(s,\tau)$, Taylor expansion in time gives

$$\frac{v(s,\tau+h_t) - v(s,\tau)}{h_t} = v_\tau + \frac{h_t}{2}v_{\tau\tau} + O(h_t^2).$$

With spatial grid h_s,

$$\begin{aligned}\frac{v(s+h_s,\tau) - v(s-h_s,\tau)}{2h_s} &= v_s + \frac{h_s^2}{6}v_{sss} + O(h_s^4),\\ \frac{v(s+h_s,\tau) - 2v(s,\tau) + v(s-h_s,\tau)}{h_s^2} &= v_{ss} + \frac{h_s^2}{12}v_{ssss} + O(h_s^4).\end{aligned}$$

Insert these into the difference equation, subtract the exact PDE $v_\tau = \frac{1}{2}\sigma^2 s^2 v_{ss} + rsv_s - rv$, and collect the residual:

$$\begin{aligned}\mathcal{R} = \frac{h_t}{2}v_{\tau\tau} - \frac{\sigma^2 s^2 h_s^2}{24}v_{ssss} - \frac{rsh_s^2}{6}v_{sss}\\ + O(h_t^2 + h_s^4).\end{aligned}$$

Under smoothness and bounded-domain assumptions, this is $O(h_t + h_s^2)$. It establishes local consistency but does not by itself control error growth as time steps are repeated.

17.13 The log-price transform to a heat equation

Let $x = \log(s/K)$ and $\tau = T - t$. Write $v(s,\tau) = Ke^{ax+b\tau}u(x,\tau)$, where a, b are chosen to remove the first-derivative and zero-order terms. Because $s\partial_s = \partial_x$ and $s^2\partial_{ss} = \partial_{xx} - \partial_x$, the Black–Scholes PDE in x becomes

$$v_\tau = \frac{1}{2}\sigma^2 v_{xx} + \left(r - \frac{1}{2}\sigma^2\right)v_x - rv.$$

Let $D = \sigma^2/2$ and $c = r - \sigma^2/2$. Substituting $v = e^{ax+b\tau}u$ gives

$$u_\tau + bu = D(u_{xx} + 2au_x + a^2 u) + c(u_x + au) - ru.$$

Choose $2Da + c = 0$, so $a = -c/(2D)$, and choose $b = Da^2 + ca - r = -c^2/(4D) - r$. The equation simplifies to $u_\tau = Du_{xx}$. This transformation connects option pricing directly to classical heat-equation numerics and explains why Fourier stability calculations for diffusion equations are relevant to Black–Scholes grids.

17.14 Nonuniform-grid derivatives from three neighboring points

Let $x_{i-1} < x_i < x_{i+1}$ and define $h_- = x_i - x_{i-1} > 0$, $h_+ = x_{i+1} - x_i > 0$. Taylor expansion about x_i gives

$$f_{i+1} = f_i + h_+ f_i' + \frac{h_+^2}{2} f_i'' + \frac{h_+^3}{6} f_i''' + O(h_+^4),$$
$$f_{i-1} = f_i - h_- f_i' + \frac{h_-^2}{2} f_i'' - \frac{h_-^3}{6} f_i''' + O(h_-^4).$$

Multiply the first equation by h_-^2, the second by $-h_+^2$, and divide by $h_- h_+ (h_- + h_+)$. The f'' terms cancel and the approximation to the first derivative is

$$f'(x_i) \approx -\frac{h_+}{h_-(h_- + h_+)} f_{i-1} + \frac{h_+ - h_-}{h_- h_+} f_i + \frac{h_-}{h_+(h_- + h_+)} f_{i+1}.$$

For the second derivative, subtract the left and right secant slopes and divide by their average separation:

$$f''(x_i) \approx \frac{2}{h_- + h_+} \left(\frac{f_{i+1} - f_i}{h_+} - \frac{f_i - f_{i-1}}{h_-} \right)$$
$$= \frac{2}{h_-(h_- + h_+)} f_{i-1} - \frac{2}{h_- h_+} f_i + \frac{2}{h_+(h_- + h_+)} f_{i+1}.$$

When $h_- = h_+ = h$, these reduce to the familiar centered $[f_{i+1} - f_{i-1}]/(2h)$ and $[f_{i+1} - 2f_i + f_{i-1}]/h^2$. On an arbitrary nonsmooth nonuniform mesh, the second-derivative approximation can have a leading truncation term proportional to $h_+ - h_-$; one should not automatically claim uniform second-order accuracy without mesh regularity assumptions.

17.15 Localizing the grid near a strike

At expiry, the European call payoff $g(s) = (s - K)^+$ is only Lipschitz continuous and is not differentiable at $s = K$. A uniform grid may waste spatial nodes in regions where the solution is smooth while undersampling the kink. A nonuniform mesh can place smaller h_- and h_+ around the strike, reducing local spatial error there. However, clustering nodes changes the finite-difference weights, matrix condition numbers, and stability restrictions. A sensible mesh is therefore selected jointly with the time-stepping method and boundary conditions rather than solely by the terminal payoff geometry.

17.16 A complete explicit Black–Scholes solver

Algorithm 17.1 (Forward time-to-maturity finite differences)**.** Choose stock nodes $0 = s_0 < \cdots < s_M = s_{\max}$ and time-to-maturity levels $\tau_n = n\Delta\tau$. For a European call:

1. Initialize $v_i^0 = (s_i - K)^+$ for $i = 0, \ldots, M$.
2. At every time level n, set the boundary values $v_0^{n+1} = 0$ and $v_M^{n+1} \approx s_{\max} - Ke^{-r\tau_{n+1}}$.
3. For each interior $i = 1, \ldots, M-1$, evaluate the spatial finite-difference weights and set $v_i^{n+1} = a_i v_{i-1}^n + b_i v_i^n + c_i v_{i+1}^n$.
4. Continue until the grid reaches $\tau = T$, then interpolate the grid values around $s = S_0$ to estimate the current price.

Each level takes $O(M)$ arithmetic operations, giving $O(MN)$ total cost and $O(M)$ memory if only two time layers are retained. An explicit scheme needs a stable time increment: a nonnegative-coefficient condition is often sufficient for a monotone maximum-norm estimate but can be restrictive near large stock nodes. The interpolation error at S_0 is separate from the underlying PDE discretization error.

LECTURE 18

Stability, Upwinding, Implicit Schemes, and Crank–Nicolson

18.1 Monotonicity and upwind differencing

Central differencing of the drift term is second-order in space but can produce $\alpha_i < 0$ where diffusion is weak relative to the drift. Such negative coefficients may cause oscillations, loss of positivity, or poor stability even when a formally higher-order truncation error is advertised. For $b_i > 0$, a forward first difference for v_s gives

$$\alpha_i^{\text{fwd}} = \frac{a_i}{h^2}, \qquad \beta_i^{\text{fwd}} = \frac{a_i}{h^2} + \frac{b_i}{h},$$

whereas a backward first difference gives

$$\alpha_i^{\text{bwd}} = \frac{a_i}{h^2} - \frac{b_i}{h}, \qquad \beta_i^{\text{bwd}} = \frac{a_i}{h^2}.$$

For the *forward-in-τ* parabolic pricing PDE, select a first-derivative stencil that gives nonnegative off-diagonal coefficients; which one is stable depends on the PDE sign convention and whether one views the convection term in conservative form. On the above grid with $b_i \geq 0$, the displayed forward formula preserves nonnegative off-diagonal weights. One may use central differences whenever $\alpha_i, \beta_i \geq 0$ and switch to a one-sided formula only where the condition fails. This is the scheme-selection idea discussed under upstream weighting in the lectures.

18.2 Matrix formulation

For interior nodes, collect the spatial discretization into a matrix A_h such that $(A_h v)_i = (\mathcal{L}_h v)_i$. Define $M = -\Delta\tau A_h$, with boundary rows adjusted to impose the chosen conditions. The explicit scheme becomes

$$v^{n+1} = (I - M)v^n.$$

At an interior row the nonzero coefficients of M are

$$M_{i,i-1} = -\Delta\tau\alpha_i, \quad M_{i,i} = \Delta\tau(\alpha_i + \beta_i + r), \quad M_{i,i+1} = -\Delta\tau\beta_i.$$

The matrix is tridiagonal, a computational advantage: linear systems can be solved in $O(M)$ work using the Thomas algorithm when suitable pivoting conditions hold.

18.3 Fully implicit Euler method

Evaluate the spatial right-hand side at the new time level:

$$\frac{v_i^{n+1} - v_i^n}{\Delta\tau} = \alpha_i(v_{i-1}^{n+1} - v_i^{n+1}) + \beta_i(v_{i+1}^{n+1} - v_i^{n+1}) - rv_i^{n+1}.$$

Multiplying by $\Delta\tau$ and moving new unknowns to the left produces

$$(I + M)v^{n+1} = v^n.$$

Under standard monotonicity conditions, the implicit method is stable for all positive time steps even though excessively large steps degrade accuracy. In the constant-coefficient scalar test equation $y' = -\lambda y$, $\lambda \geq 0$, implicit Euler has amplification factor $(1 + \lambda\Delta\tau)^{-1}$, whose magnitude never exceeds one.

18.4 Crank–Nicolson as trapezoidal time integration

Average the spatial operator at the old and new times:

$$\frac{v^{n+1} - v^n}{\Delta\tau} = \frac{1}{2}A_h(v^{n+1} + v^n).$$

Rearrangement gives

$$\left(I - \frac{\Delta\tau}{2}A_h\right)v^{n+1} = \left(I + \frac{\Delta\tau}{2}A_h\right)v^n,$$
$$\left(I + \frac{1}{2}M\right)v^{n+1} = \left(I - \frac{1}{2}M\right)v^n.$$

For a smooth solution on a uniform spatial mesh, Crank–Nicolson has local discretization errors of order $O(\Delta\tau^2 + h^2)$ in the standard global consistency sense, whereas forward and backward Euler are first-order in time. For nonsmooth option payoffs, startup damping (for example, two implicit half-steps) can reduce the initial spurious oscillations associated with Crank–Nicolson.

18.5 A lower-boundary Crank–Nicolson equation

At $s = 0$ the PDE reduces to $\partial_\tau v = -rv$. Applying the trapezoidal method gives

$$\frac{v_0^{n+1} - v_0^n}{\Delta\tau} = -\frac{r}{2}(v_0^{n+1} + v_0^n),$$
$$\left(1 + \frac{r\Delta\tau}{2}\right)v_0^{n+1} = \left(1 - \frac{r\Delta\tau}{2}\right)v_0^n.$$

Accordingly, the first row of $M/2$ has diagonal entry $r\Delta\tau/2$ and no off-diagonal entries. This is equivalent to setting $\alpha_0 = \beta_0 = 0$ while retaining the discount term.

18.6 Positivity and a discrete maximum principle

Suppose the explicit coefficients are nonnegative:

$$A_i = \Delta\tau\alpha_i \geq 0, \quad B_i = 1 - \Delta\tau(\alpha_i + \beta_i + r) \geq 0, \quad C_i = \Delta\tau\beta_i \geq 0.$$

For $r \geq 0$, they satisfy $A_i + B_i + C_i = 1 - r\Delta\tau \leq 1$. If all values at time n are nonnegative, then $v_i^{n+1} = A_i v_{i-1}^n + B_i v_i^n + C_i v_{i+1}^n \geq 0$. Hence, under suitable nonnegative boundary conditions, the scheme preserves nonnegative option values.

Similarly, if the two solutions v and w have ordered initial and boundary values, and use the same coefficients, then $v^n \leq w^n$ componentwise implies $v^{n+1} \leq w^{n+1}$ componentwise. This discrete comparison principle mirrors the continuous parabolic PDE's comparison principle. Negative coefficients destroy this simple reasoning; an apparently small numerical truncation error may coexist with negative computed option prices or oscillatory Greeks.

18.7 Fourier stability for a constant-coefficient heat equation

A standard model problem is $u_\tau = \kappa u_{xx}$ with $\kappa > 0$. The forward-time, centered-space scheme on a uniform grid is

$$u_i^{n+1} = u_i^n + \nu(u_{i-1}^n - 2u_i^n + u_{i+1}^n), \qquad \nu = \frac{\kappa\Delta\tau}{h^2}.$$

For a Fourier error mode $e_i^n = G^n e^{\mathrm{i}i\xi}$, substitute into the scheme:

$$\begin{aligned} G &= 1 + \nu(e^{-\mathrm{i}\xi} - 2 + e^{\mathrm{i}\xi}) \\ &= 1 - 2\nu(1 - \cos\xi) = 1 - 4\nu\sin^2(\xi/2). \end{aligned}$$

For stability it is necessary and sufficient on the infinite/periodic uniform grid that $|G| \leq 1$ for every ξ. Since $0 \leq \sin^2(\xi/2) \leq 1$, this holds exactly when $0 \leq \nu \leq 1/2$. Consequently,

$$\Delta\tau \leq \frac{h^2}{2\kappa}.$$

An explicit diffusion scheme therefore requires a time step of order h^2 as the spatial mesh is refined, which can be costly. Fully implicit and Crank–Nicolson schemes avoid this particular stability restriction, although they require solving linear systems.

18.8 Why a centered first difference can give negative option prices

For a uniform stock grid, the explicit centered scheme uses $a_i = \frac{1}{2}\Delta\tau(\sigma^2 i^2 - ri)$, $b_i = 1 - \Delta\tau(\sigma^2 i^2 + r)$ and $c_i = \frac{1}{2}\Delta\tau(\sigma^2 i^2 + ri)$. A sufficient condition for preserving nonnegativity when all previous values and boundary data are nonnegative is $a_i, b_i, c_i \geq 0$. The condition $a_i \geq 0$ requires

$$\begin{aligned} \sigma^2 i^2 - ri &\geq 0, \\ i(\sigma^2 i - r) &\geq 0. \end{aligned}$$

For $i > 0$, this fails near the lower stock boundary whenever $i < r/\sigma^2$. Thus even as $\Delta\tau \to 0$, the centered drift coefficient can have the wrong sign on a coarse spatial grid. Reducing the time step alone cannot repair this issue.

18.9 Using a monotone drift discretization

When $r > 0$, the forward spatial approximation $v_s(s_i, \tau) \approx (v_{i+1} - v_i)/\Delta s$ leads to

$$\begin{aligned} v_i^{n+1} &= v_i^n + \Delta\tau \left[\frac{1}{2}\sigma^2 i^2 (v_{i+1}^n - 2v_i^n + v_{i-1}^n) \right. \\ &\qquad \left. + ri(v_{i+1}^n - v_i^n) - rv_i^n \right] \\ &= \frac{1}{2}\Delta\tau\sigma^2 i^2 v_{i-1}^n \\ &\qquad + \left[1 - \Delta\tau(\sigma^2 i^2 + ri + r)\right] v_i^n \\ &\qquad + \Delta\tau \left(\frac{1}{2}\sigma^2 i^2 + ri \right) v_{i+1}^n. \end{aligned}$$

The off-diagonal weights are nonnegative; it remains to choose $\Delta\tau$ so the diagonal weight is also nonnegative. This forward spatial difference is sometimes called upstream/upwind weighting relative to the backward pricing PDE convention. The choice of direction must be justified by the sign convention and the PDE's evolution form, not selected mechanically from the word "upwind."

18.10 Deriving implicit Euler as a tridiagonal linear system

Write the spatial operator at interior nodes as $(Lv)_i = \alpha_i v_{i-1} + \beta_i v_i + \gamma_i v_{i+1}$, including the negative discount term in β_i. The fully implicit method evaluates L at the new time level:

$$\begin{aligned} \frac{v_i^{n+1} - v_i^n}{\Delta\tau} &= \alpha_i v_{i-1}^{n+1} + \beta_i v_i^{n+1} + \gamma_i v_{i+1}^{n+1}, \\ -\Delta\tau\alpha_i v_{i-1}^{n+1} + (1 - \Delta\tau\beta_i) v_i^{n+1} - \Delta\tau\gamma_i v_{i+1}^{n+1} &= v_i^n. \end{aligned}$$

Collecting the unknown interior values gives $(I - \Delta\tau L_h)v^{n+1} = v^n + b_{\text{boundary}}^{n+1}$, where L_h is tridiagonal and b carries known boundary contributions. In a monotone scheme, $I - \Delta\tau L_h$ often has a positive diagonal, nonpositive off-diagonals, and an inverse that preserves componentwise nonnegativity (an M-matrix structure). Implicit stability does not imply the resulting grid values automatically have high spatial accuracy.

18.11 Crank–Nicolson derived by averaging two slopes

The trapezoidal rule for the spatial operator gives

$$\begin{aligned} \frac{v^{n+1} - v^n}{\Delta\tau} &= \frac{1}{2}L_h v^{n+1} + \frac{1}{2}L_h v^n, \\ v^{n+1} - \frac{\Delta\tau}{2}L_h v^{n+1} &= v^n + \frac{\Delta\tau}{2}L_h v^n, \\ \left(I - \frac{\Delta\tau}{2}L_h\right) v^{n+1} &= \left(I + \frac{\Delta\tau}{2}L_h\right) v^n + b_{\text{boundary}}. \end{aligned}$$

For a smooth sufficiently regular solution this improves temporal truncation error to second order, $O((\Delta\tau)^2)$ globally, compared with first-order Euler time stepping. Nonsmooth maturity payoffs may cause oscillations or order degradation. Rannacher smoothing (two initial half steps of implicit Euler) is a common practical remedy before switching to Crank–Nicolson.

18.12 A von Neumann stability calculation for the heat equation

For $u_\tau = \kappa u_{xx}$ on a uniform grid, the explicit scheme is $u_i^{n+1} = u_i^n + \nu(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$, where $\nu = \kappa\Delta\tau/(\Delta x)^2$. Insert a Fourier mode $u_i^n = G^n e^{\mathrm{i}\theta i}$:

$$\begin{aligned} G^{n+1}e^{\mathrm{i}\theta i} &= G^n e^{\mathrm{i}\theta i}\{1 + \nu(e^{\mathrm{i}\theta} - 2 + e^{-\mathrm{i}\theta})\}, \\ G &= 1 + \nu(2\cos\theta - 2) \\ &= 1 - 4\nu\sin^2(\theta/2). \end{aligned}$$

Stability requires $|G| \le 1$ for every θ. Since $0 \le \sin^2(\theta/2) \le 1$, the strongest condition is $1 - 4\nu \ge -1$, or $0 \le \nu \le 1/2$. Thus the forward Euler heat-equation method requires

$$\Delta\tau \le \frac{(\Delta x)^2}{2\kappa}.$$

For implicit Euler, the amplification factor is $G = (1 + 4\nu\sin^2(\theta/2))^{-1}$, with $|G| \le 1$ for every $\nu \ge 0$. For Crank–Nicolson,

$$G_{\mathrm{CN}} = \frac{1 - 2\nu\sin^2(\theta/2)}{1 + 2\nu\sin^2(\theta/2)},$$

which satisfies $|G_{\mathrm{CN}}| \le 1$ for $\nu \ge 0$ but can be negative for large ν, creating damped temporal oscillations of high-frequency modes. This explains the distinction between absolute stability and non-oscillatory monotonicity.

18.13 Deriving the Thomas algorithm for tridiagonal matrices

Consider the system $a_i x_{i-1} + b_i x_i + c_i x_{i+1} = d_i$ for $i = 1, \dots, M-1$, with $a_1 = 0$ and $c_{M-1} = 0$ after incorporating boundary conditions. Eliminate the lower diagonal recursively. For the first row define $\widetilde{c}_1 = c_1/b_1$ and $\widetilde{d}_1 = d_1/b_1$. The first equation becomes $x_1 + \widetilde{c}_1 x_2 = \widetilde{d}_1$. At row i, substitute the previous expression $x_{i-1} = \widetilde{d}_{i-1} - \widetilde{c}_{i-1}x_i$:

$$\begin{aligned} a_i(\widetilde{d}_{i-1} - \widetilde{c}_{i-1}x_i) + b_i x_i + c_i x_{i+1} &= d_i, \\ (b_i - a_i\widetilde{c}_{i-1})x_i + c_i x_{i+1} &= d_i - a_i\widetilde{d}_{i-1}. \end{aligned}$$

Define the effective pivot $m_i = b_i - a_i\widetilde{c}_{i-1}$. If $m_i \neq 0$,

$$\begin{aligned} \widetilde{c}_i &= \frac{c_i}{m_i}, \\ \widetilde{d}_i &= \frac{d_i - a_i\widetilde{d}_{i-1}}{m_i}. \end{aligned}$$

At the last row, $x_{M-1} = \widetilde{d}_{M-1}$. Back substitution yields

$$x_i = \widetilde{d}_i - \widetilde{c}_i x_{i+1}, \qquad i = M-2, \dots, 1.$$

Only constant work is required per grid node, giving an $O(M)$ tridiagonal solve. Implicit Euler and Crank–Nicolson use one such solve per time layer; explicitly computing A^{-1} is slower, less stable, and unnecessary.

18.14 Fully implicit temporal consistency

If the exact smooth solution satisfies $v_\tau = Lv$, then the backward Euler time difference is

$$\frac{v(\tau+h)-v(\tau)}{h} = v_\tau(\tau+h) - \frac{h}{2}v_{\tau\tau}(\tau+h) + O(h^2).$$

Thus the residual of the implicit Euler PDE equation is $O(h)$ in time plus the chosen spatial truncation error. Crank–Nicolson averages Lv at both endpoints, approximating the time integral $\int_\tau^{\tau+h} Lv(s)\,\mathrm{d}s$ by the trapezoidal rule. For smooth Lv, this has error $O(h^3)$ in the integrated equation, and therefore local derivative residual $O(h^2)$. The distinction between local derivative residual and integrated local error matters when relating Taylor expansions to global convergence order.

18.15 An implementable Crank–Nicolson solver

Algorithm 18.1 (Tridiagonal Crank–Nicolson pricing)**.** Assume a finite stock grid and let L_h denote a tridiagonal discretization of the spatial Black–Scholes operator.

1. Construct $A = I - (\Delta\tau/2)L_h$ and $B = I + (\Delta\tau/2)L_h$ for interior nodes, together with correctly discretized boundary conditions.
2. Initialize v^0 from the terminal payoff and define the first boundary vector.
3. At time layer n, form $d = Bv^n + b^{n,n+1}_{\text{boundary}}$.
4. Solve $Av^{n+1} = d$ by the Thomas algorithm or a stable tridiagonal solver; do not explicitly form A^{-1}.
5. Repeat for all time layers, then interpolate the final stock grid to S_0.

After one $O(M)$ factorization, each time step is $O(M)$. When coefficients and the grid are unchanged across time, a single factorization may be reused. For American options the solve is replaced by an obstacle-constrained LCP, not by the same unconstrained linear algebra with a final unconstrained price.

LECTURE 19

Convergence, Dividends, and Market Calibration

19.1 Stability, consistency, and convergence

A numerical approximation v_i^n is *convergent* if v_i^n approaches the true solution evaluated at the corresponding mesh points as $\Delta\tau$ and the maximum spatial mesh size tend to zero. Consistency controls the local error made by inserting a smooth exact solution into the scheme. Stability controls the propagation of perturbations and rounding errors from one time step to the next. For well-posed linear initial-value problems, the Lax equivalence framework relates a consistent scheme's stability to its convergence, under the relevant assumptions.

For a one-step linear method $v^{n+1} = B_h v^n$, repeated application yields $v^n = B_h^n v^0$. Large powers of B_h magnify disturbances. In a positivity-preserving explicit update, nonnegative rows and an appropriate norm bound can help control these powers. For a stable implicit discretization, the matrix inverse $(I + M)^{-1}$ damps high-frequency components. Stability alone is not accuracy: truncating the stock domain too narrowly or using an inappropriate upper boundary can dominate the PDE discretization error.

19.2 Choice of computational domain and mesh

The stock interval $[0, s_{\max}]$ approximates an unbounded state space. If $s_{\max}$ is too small, boundary conditions contaminate prices near the current stock level. An effective procedure is to enlarge $s_{\max}$ until the value of the target option changes by less than a specified tolerance. Grid refinement in both space and time can then distinguish boundary truncation error from finite-difference error.

Use finer spatial nodes near strikes, barriers, or other payoff kinks. Time steps can be shortened around sudden changes in exercise constraints or around dividend dates. Numerical Greeks are especially sensitive to grid spacing and payoff nonsmoothness, so convergence checks should include sensitivities as well as the option price.

19.3 Continuous dividend yields

If the stock pays a constant continuous dividend yield q, its risk-neutral ex-dividend dynamics are

$$\mathrm{d}S_t = (r - q)S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

A self-financing position in the stock receives the dividend stream $qS_t\,\mathrm{d}t$. Delta-hedging the option and accounting for that income replaces rS by $(r - q)S$ in the PDE:

$$V_t + \frac{1}{2}\sigma^2 s^2 V_{ss} + (r - q)sV_s - rV = 0.$$

For European calls and puts, the put–call parity identity becomes

$$C_0 - P_0 = S_0 e^{-qT} - Ke^{-rT}.$$

Ignoring dividends produces systematically wrong forward levels, option values, and hedge deltas.

19.4 Black–Scholes closed forms and implied volatility

Let $\tau = T - t > 0$, with $s > 0$, $K > 0$, and $\sigma > 0$. For a European call under a constant dividend yield q, the Black–Scholes formula is

$$\begin{aligned} C(t, s; K, T, \sigma) &= se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \\ d_1 &= \frac{\log(s/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \\ d_2 &= d_1 - \sigma\sqrt{\tau}. \end{aligned}$$

The corresponding put is $P = Ke^{-r\tau}\Phi(-d_2) - se^{-q\tau}\Phi(-d_1)$. For an observed option market price C^{mkt}, its Black–Scholes implied volatility σ_{imp} solves

$$C(t, s; K, T, \sigma_{\text{imp}}) = C^{\text{mkt}}.$$

This is a numerical root-finding problem. Since call vega satisfies

$$\frac{\partial C}{\partial \sigma} = se^{-q\tau}\varphi(d_1)\sqrt{\tau} > 0,$$

the theoretical call price is strictly increasing in volatility for positive maturity and a nondegenerate underlying. Newton's method is

$$\sigma_{k+1} = \sigma_k - \frac{C(\sigma_k) - C^{\text{mkt}}}{\partial_\sigma C(\sigma_k)}.$$

Bracketing and careful handling of low-vega deep in/out-of-the-money contracts improve numerical robustness.

19.5 The volatility smile and fitting model parameters

If the Black–Scholes model with constant σ were exactly correct across all strikes and maturities, every liquid European option on the same underlying would imply the same volatility. In practice $\sigma_{\text{imp}}(K, T)$ varies with both strike and maturity, producing smiles, skews, and a volatility surface. This motivates more expressive models with local volatility, stochastic volatility, jumps, or mixtures.

Given market quotes C_l^{mkt} and a parameterized model price $C_l(\theta)$, calibration estimates

$$\widehat{\theta} \in \arg\min_{\theta \in \Theta} \sum_{l=1}^{m} w_l \big(C_l(\theta) - C_l^{\text{mkt}}\big)^2,$$

where $w_l > 0$ are chosen error weights. The objective is typically nonlinear, and reliable calibration requires constraints, sensible initial values, sufficient data quality, and a consistent numerical pricing algorithm. Models must fit market instruments well enough to be useful for marking-to-market and hedging, but a fit at observed strikes does not guarantee accurate extrapolation to unquoted payoffs.

19.6 The theta family of time-stepping schemes

Forward Euler, backward Euler, and Crank–Nicolson can be unified using a weight $\vartheta \in [0, 1]$:

$$\frac{v^{n+1} - v^n}{\Delta\tau} = (1 - \vartheta)A_h v^n + \vartheta A_h v^{n+1}.$$

Equivalently,

$$(I - \vartheta\Delta\tau A_h)v^{n+1} = (I + (1 - \vartheta)\Delta\tau A_h)v^n.$$

The choices $\vartheta = 0$, 1, and $1/2$ produce explicit Euler, implicit Euler, and Crank–Nicolson. For the scalar test equation $y' = -\lambda y$, the amplification factor is

$$G(z) = \frac{1 - (1 - \vartheta)z}{1 + \vartheta z}, \qquad z = \lambda\Delta\tau \geq 0.$$

For $\vartheta \geq 1/2$, $|G(z)| \leq 1$ for every $z \geq 0$. Crank–Nicolson is therefore stable on the negative real axis but not strongly damping for $z \to \infty$: $G(z) \to -1$ when $\vartheta = 1/2$. Implicit Euler has $G(z) \to 0$ and therefore damps stiff, rapidly oscillating modes, explaining why initial implicit steps can improve a Crank–Nicolson option solver near a nonsmooth payoff.

19.7 Black–Scholes parity under continuous dividends

The put–call parity relation with dividend yield q is derived by comparing a call-minus-put portfolio to a prepaid forward. At maturity, the long call and short put pay $S_T - K$. A prepaid forward delivering one share at T costs $S_0 e^{-qT}$ under the constant-yield model; a zero-coupon bond paying K at T costs Ke^{-rT}. By no arbitrage,

$$C_0 - P_0 = S_0 e^{-qT} - Ke^{-rT}.$$

This also identifies the maturity-T forward price

$$F_{0,T} = S_0 e^{(r-q)T}.$$

The forward relationship is useful when formulating option pricing and implied volatility in terms of log-moneyness $\log(K/F_{0,T})$ rather than raw strikes.

19.8 Consistency, stability, and convergence are not synonyms

Suppose a PDE evolution $v_\tau = Lv$ is approximated by $v^{n+1} = A_h v^n$, with spatial step Δs and temporal step $\Delta\tau$. Consistency is a statement about the residual obtained by inserting the *exact smooth* solution into the numerical equation. Stability controls how perturbations to initial data, roundoff, and local errors grow under repeated application of A_h. Convergence asserts the numerical values approach the exact solution as the grid is refined. In a suitable well-posed linear setting, consistency and stability together imply convergence (the Lax equivalence principle), but a local Taylor expansion by itself cannot guarantee global accuracy.

19.9 An explicit maximum-norm stability estimate

Consider a three-point update with nonnegative coefficients a_i, b_i, c_i satisfying $a_i + b_i + c_i \le 1$. For two grid solutions u^n and v^n, define the difference $e_i^n = u_i^n - v_i^n$. Then

$$\begin{aligned} |e_i^{n+1}| &= |a_i e_{i-1}^n + b_i e_i^n + c_i e_{i+1}^n| \\ &\le a_i |e_{i-1}^n| + b_i |e_i^n| + c_i |e_{i+1}^n| \\ &\le (a_i + b_i + c_i) \, \|e^n\|_\infty \\ &\le \|e^n\|_\infty . \end{aligned}$$

Taking the maximum over interior nodes and incorporating compatible boundary perturbations gives $\|e^{n+1}\|_\infty \le \|e^n\|_\infty$. Iteration yields $\|e^n\|_\infty \le \|e^0\|_\infty$. The argument exposes the role of coefficient positivity; a stencil with negative weights does not admit this simple monotonicity proof.

19.10 Dividend yield and the modified pricing PDE

With continuous proportional dividend yield q, the risk-neutral ex-dividend stock follows $\mathrm{d}S_t = (r-q)S_t \, \mathrm{d}t + \sigma S_t \, \mathrm{d}W_t$. Apply Itô's formula to an option $V(S,t)$:

$$\mathrm{d}V = \left(V_t + (r-q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) \mathrm{d}t + \sigma S V_S \, \mathrm{d}W.$$

Risk-neutral discounted valuation therefore leads to

$$\begin{aligned} V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + (r-q)SV_S - rV = 0, \\ V(S,T) = g(S). \end{aligned}$$

Put–call parity now reflects dividends and financing:

$$C_0 - P_0 = S_0 e^{-qT} - K e^{-rT}.$$

The stock term is multiplied by e^{-qT} because holding one unit of the current ex-dividend stock and receiving dividends differs from receiving exactly one stock at maturity without intermediate cash flows.

19.11 Vega and a justified Newton step for implied volatility

For a no-dividend Black–Scholes call $C(\sigma) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$, the vega is

$$\frac{\partial C}{\partial \sigma} = S_0 \varphi(d_1) \sqrt{T} > 0.$$

To see the cancellation of the d-derivative terms, note the Gaussian density identity $S_0\varphi(d_1) = Ke^{-rT}\varphi(d_2)$. Differentiating,

$$\begin{aligned} C_\sigma &= S_0\varphi(d_1)\frac{\partial d_1}{\partial \sigma} - Ke^{-rT}\varphi(d_2)\frac{\partial d_2}{\partial \sigma} \\ &= S_0\varphi(d_1)\left(\frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma}\right) \\ &= S_0\varphi(d_1)\frac{\partial}{\partial \sigma}(d_1 - d_2) \\ &= S_0\varphi(d_1)\sqrt{T}. \end{aligned}$$

Given an observed call price C^{mkt}, solving $C(\sigma) = C^{\text{mkt}}$ by Newton iteration yields

$$\sigma_{k+1} = \sigma_k - \frac{C(\sigma_k) - C^{\text{mkt}}}{S_0\varphi(d_1(\sigma_k))\sqrt{T}}.$$

Vega becomes small for deep in/out-of-the-money options or extremely short maturities, making an unguarded Newton step unstable. Bisection or bracketed root-finding is safer when a valid positive-volatility interval can be established.

19.12 Implied-volatility existence from price bounds

Fix $S_0, K, T, r > 0$ and consider the Black–Scholes European call $C^{\text{BS}}(\sigma)$ for $\sigma > 0$. Vega is strictly positive, so the price is strictly increasing in σ. Its zero-volatility limit is

$$\lim_{\sigma \downarrow 0} C^{\text{BS}}(\sigma) = (S_0 - Ke^{-rT})^+.$$

As $\sigma \to \infty$, $d_1 \to \infty$ and $d_2 \to -\infty$, giving

$$\lim_{\sigma \to \infty} C^{\text{BS}}(\sigma) = S_0.$$

Therefore any market call price strictly between the lower no-arbitrage bound and S_0 has a unique positive Black–Scholes implied volatility under this setup. If an observed quote violates these bounds, a solver may fail because no valid implied volatility exists, not because the root-finding algorithm is defective.

19.13 Least-squares calibration and overfitting

Suppose a parametric model $V^{\text{model}}(K_j, T_j; \theta)$ is fitted to observed prices V_j^{mkt} for $j = 1, \ldots, m$. A typical weighted calibration objective is

$$J(\theta) = \sum_{j=1}^{m} w_j \left[V^{\text{model}}(K_j, T_j; \theta) - V_j^{\text{mkt}}\right]^2.$$

Differentiating formally,

$$\frac{\partial J}{\partial \theta_k} = 2\sum_{j=1}^{m} w_j \left[V_j^{\text{model}}(\theta) - V_j^{\text{mkt}}\right] \frac{\partial V_j^{\text{model}}(\theta)}{\partial \theta_k}.$$

A numerical optimizer seeks a feasible θ that makes the gradient small, but local minima, parameter nonidentifiability, stale quotes, and the nonlinear sensitivity of options can all undermine the result. Weights based on bid–ask spreads or option-vega scaling are often preferable to treating every price residual as equally reliable. Exact fitting of all observed options is not a substitute for an arbitrage-consistent calibrated surface.

19.14 Crank–Nicolson boundary rows from the zero-stock ODE

At $s = 0$, the Black–Scholes PDE for a put reduces to $v_\tau = -rv$, because both sv_s and $s^2 v_{ss}$ vanish. Denote the boundary value by v_0^n. Apply Crank–Nicolson at the boundary:

$$\begin{aligned} \frac{v_0^{n+1} - v_0^n}{\Delta\tau} &= -\frac{r}{2}(v_0^{n+1} + v_0^n), \\ v_0^{n+1} - v_0^n &= -\frac{r\Delta\tau}{2} v_0^{n+1} - \frac{r\Delta\tau}{2} v_0^n, \\ \left(1 + \frac{r\Delta\tau}{2}\right) v_0^{n+1} &= \left(1 - \frac{r\Delta\tau}{2}\right) v_0^n. \end{aligned}$$

This is exactly the first-row equation obtained from the general CN matrix stencil by setting spatial neighbor weights $\alpha_0 = \beta_0 = 0$ and retaining the discounting term. It is a useful internal consistency check for the matrix assembly, and it also directly addresses the boundary-row calculation in Practice Set 2.

19.15 Choosing a financially meaningful upper stock boundary

In a geometric Brownian model over remaining time τ, $\log(S_T/s) \sim \mathcal{N}((r - \sigma^2/2)\tau, \sigma^2\tau)$. A tail-based provisional upper bound can be chosen so that the lognormal probability of reaching beyond $s_{\max}$ at expiry is small:

$$\mathbb{P}(S_T > s_{\max} \mid S_t = s) = 1 - \Phi\left(\frac{\log(s_{\max}/s) - (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}}\right).$$

To make this probability less than ε, it suffices to select

$$s_{\max} \geq s \exp\{(r - \sigma^2/2)\tau + \sigma\sqrt{\tau}\Phi^{-1}(1 - \varepsilon)\}.$$

For PDE errors, a terminal-tail bound is only a heuristic because paths can hit the spatial boundary at intermediate times and the payoff may be large on rare paths. More robust validation increases $s_{\max}$ until prices and sensitivities on the region of interest stabilize.

19.16 A robust implied-volatility root-finding algorithm

Algorithm 19.1 (Bracketed implied-volatility calibration)**.** Given a quoted European call price C^{mkt}, current stock S_0, strike K, maturity T, and constant interest rate r:

1. Verify the quote lies within the model's no-arbitrage call-price interval $[(S_0 - Ke^{-rT})^+, S_0]$, subject to numerical tolerances and dividend adjustments when needed.

2. Choose lower/upper volatility bounds $0 < \sigma_L < \sigma_U$ such that $C^{\mathrm{BS}}(\sigma_L) \le C^{\mathrm{mkt}} \le C^{\mathrm{BS}}(\sigma_U)$.
3. Repeatedly set $\sigma_M = (\sigma_L + \sigma_U)/2$ and evaluate the pricing residual $F(\sigma_M) = C^{\mathrm{BS}}(\sigma_M) - C^{\mathrm{mkt}}$.
4. If $F(\sigma_M) > 0$, replace $\sigma_U \leftarrow \sigma_M$; otherwise replace $\sigma_L \leftarrow \sigma_M$.
5. Stop when both the price residual and volatility interval satisfy chosen tolerances.

Because Black–Scholes call vega is positive, the residual is monotone, and bisection preserves the root bracket. Newton updates may converge faster when vega is not too small, but a bracketed method is robust for short maturities or extreme strikes where the derivative-based step can be unreliable. Always distinguish a numerical root tolerance from market quote uncertainty.

LECTURE 20

Calibration and Least-Squares Monte Carlo for American Options

20.1 The optimal stopping formulation

A European claim must be exercised at its prescribed maturity. An American claim permits exercise at a stopping time τ chosen using only information available up to that time. For a discounted nonnegative exercise payoff $g(S_t, t)$, its time-zero value is

$$V_0^{\mathrm{A}} = \sup_{\tau \in \mathcal{T}_{[0,T]}} \mathbb{E}^{\mathbb{Q}} \left[e^{-r\tau} g(S_\tau, \tau) \right],$$

where $\mathcal{T}_{[0,T]}$ is the set of admissible stopping times taking values in $[0, T]$. Restricting exercise to dates $0 = t_0 < \cdots < t_N = T$ gives a Bermudan approximation; its price is no greater than the fully American value because fewer exercise decisions are permitted.

20.2 Why naive pathwise exercise is biased

Suppose one first simulates an entire future trajectory and then retrospectively exercises at the most favorable date on that path. The rule

$$\max_{0 \le j \le N} e^{-rt_j} g(S_{t_j}, t_j)$$

uses future prices not available to the holder at earlier times. It is therefore an *anticipating* or clairvoyant maximum, rather than an admissible stopping strategy, and generally yields an upper-biased value. The critical numerical task is to estimate the *conditional continuation value* using information available at the decision date.

20.3 The Snell-envelope recursion

Set $G_j = g(S_{t_j}, t_j)$ and define the value process V_j through backward induction:

$$\begin{aligned} V_N &= G_N, \\ V_j &= \max\left\{ G_j, e^{-r(t_{j+1}-t_j)} \mathbb{E}^{\mathbb{Q}}[V_{j+1} \mid \mathcal{F}_{t_j}] \right\}, \qquad j = N-1, \ldots, 0. \end{aligned}$$

The conditional expectation is the continuation value of not exercising at t_j. The holder exercises when the immediate payoff is at least the continuation value. This is the discrete-time Snell envelope, the smallest supermartingale dominating the discounted payoff process.

20.4 Longstaff–Schwartz conditional regression

Simulate independent stock trajectories $S_{i,j}$ for paths $i = 1, \ldots, n$ and times $j = 0, \ldots, N$. For each exercise date, approximate continuation values using a finite set of basis functions $\psi_0, \ldots, \psi_M$:

$$C_j(s) \approx \sum_{k=0}^{M} \beta_{j,k} \psi_k(s).$$

For a plain American put, one may use monomials, Laguerre polynomials, or other basis functions in s. The regression at time t_j fits suitably discounted realized future cash flows on current state features:

$$\widehat{\beta}_j \in \arg\min_{\beta} \sum_{i \in \mathcal{I}_j} \left(Y_{i,j} - \sum_{k=0}^{M} \beta_k \psi_k(S_{i,j}) \right)^2 .$$

Here $\mathcal{I}_j$ is often the subset of in-the-money simulated paths, and $Y_{i,j}$ is the cash flow generated by the existing future exercise policy, discounted back to t_j. The estimated continuation function is $\widehat{C}_j(s) = \sum_k \widehat{\beta}_{j,k} \psi_k(s)$.

Algorithm 20.1 (Least-squares Monte Carlo)**.** Simulate n independent complete paths and initialize every path's exercise time at T with its terminal payoff. For $j = N - 1, \ldots, 1$, discount each path's currently designated future cash flow to t_j. Regress these discounted cash flows on basis functions of the current state, preferably using in-the-money paths. For each eligible path, compare immediate exercise $g(S_{i,j}, t_j)$ to estimated continuation $\widehat{C}_j(S_{i,j})$. If immediate exercise is larger, replace that path's future exercise time by t_j. After processing all dates, average the realized discounted cash flows from the resulting stopping policy. For evaluation, preferably apply the fitted exercise policy to an independent set of test paths.

20.5 Bias, basis selection, and state dependence

Using the same paths for estimating regression coefficients and evaluating the resulting stopping rule can introduce in-sample bias. An independent training/test split removes the most direct look-ahead from in-sample policy evaluation; such an out-of-sample stopping-policy estimate is a lower bound on the true optimal stopping value in expectation, when the policy is admissible. Regression with too few basis functions underfits continuation values; too many relative to the number of in-the-money paths causes unstable estimates. Path-dependent payoffs require additional state variables such as running averages or maxima. Regression provides a flexible method when the option cannot be represented by a tractable low-dimensional PDE.

20.6 Regression as a projection of future cash flows

The continuation value at a decision date t_j is the conditional expectation

$$C_j(X_j) = \mathbb{E}^{\mathbb{Q}}[e^{-r\Delta t} V_{j+1} \mid X_j],$$

where X_j represents the state sufficient for future pricing, possibly containing stock price, volatility, and path statistics. Least-squares Monte Carlo selects a finite-dimensional function space

$$\mathcal{H}_M = \operatorname{span}\{\psi_0, \ldots, \psi_M\}$$

and approximates C_j by its L^2 projection onto that space. If $\psi(X)$ is the vector of basis values, the population normal equations are

$$\mathbb{E}[\psi(X_j)\psi(X_j)^\top]\beta_j = \mathbb{E}[\psi(X_j)Y_j],$$

where Y_j is the discounted future cash flow under the current stopping policy. Empirical least squares replaces the expectations by sample averages. This reveals that the method estimates *conditional expectations*, not exercise probabilities directly.

20.7 An explicit two-date stopping example

Consider exercise dates t_1 and $t_2 = T$ and payoff $g(s) = (K - s)^+$. At maturity, $V_2 = g(S_T)$. At t_1, the true continuation value is

$$C_1(s) = e^{-r(T-t_1)}\mathbb{E}^{\mathbb{Q}}[(K - S_T)^+ \mid S_{t_1} = s].$$

Under GBM, this conditional expectation is the European put price with current stock s and remaining maturity $T - t_1$. The optimal Bermudan decision is exercise at t_1 when $(K - s)^+ \geq C_1(s)$. A least-squares approximation instead uses simulated pairs

$$\left(S_{t_1}^{(i)}, e^{-r(T-t_1)}(K - S_T^{(i)})^+\right).$$

The regression response is not the current intrinsic value: it is the realized, suitably discounted *future* cash flow. The current exercise payoff enters only when the regression estimate is compared with the immediate payoff to choose a stopping decision.

20.8 Conditional-policy lower bounds

An estimated exercise strategy can always be evaluated by simulating fresh paths and executing its stopping decisions using current and past states only. Each resulting exercise date $\widehat{\tau}$ is a valid stopping time. Thus

$$\mathbb{E}^{\mathbb{Q}}[e^{-r\widehat{\tau}}g(S_{\widehat{\tau}}, \widehat{\tau})] \leq V_0^{\mathrm{A}}.$$

The fresh-path Monte Carlo estimate of this policy's value is subject to random sampling error, but its population expectation is a lower bound on the optimal American price. This distinction is important: an in-sample training value may not have the same useful lower-bound interpretation if the evaluation procedure selects decisions using information leaked from the training paths.

20.9 Deriving the Snell envelope through optimal stopping

Let $0 = t_0 < \cdots < t_N = T$ and let $G_n = g(S_{t_n})$ be an adapted, integrable exercise-reward process. The time-n American value is

$$V_n = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_n} \mathbb{E}^{\mathbb{Q}}[e^{-r(t_\tau - t_n)}G_\tau \mid \mathcal{F}_{t_n}],$$

where $\mathcal{T}_n$ contains stopping times taking values in $\{n, \dots, N\}$. At N, one must exercise or expire: $V_N = G_N$. For $n < N$ every admissible decision is either to stop now or to wait one period and then behave optimally, giving

$$V_n = \max\left\{G_n, e^{-r(t_{n+1}-t_n)}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_{t_n}]\right\}.$$

Define discounted variables $Y_n = e^{-rt_n}V_n$ and $Z_n = e^{-rt_n}G_n$. Then

$$\begin{aligned} Y_N &= Z_N, \\ Y_n &= \max\{Z_n, \mathbb{E}^{\mathbb{Q}}[Y_{n+1} \mid \mathcal{F}_{t_n}]\}. \end{aligned}$$

Therefore $Y_n \geq \mathbb{E}^{\mathbb{Q}}[Y_{n+1} \mid \mathcal{F}_{t_n}]$, making Y a supermartingale that dominates Z. The backward recursion proves Y is the smallest such supermartingale. The first contact time $\tau^* = \inf\{n : Y_n = Z_n\}$ is optimal under standard finite-horizon integrability assumptions.

20.10 Conditional expectation as an L^2 regression target

At date t_n, define the future discounted cash flow from following a given continuation policy by $Y = e^{-r(t_{n+1}-t_n)}H_{n+1}$. The continuation value is

$$C_n(s) = \mathbb{E}^{\mathbb{Q}}[Y \mid S_{t_n} = s].$$

Choose basis functions $\phi_1, \dots, \phi_K$ and approximate $C_n(s) \approx \sum_{k=1}^{K} \beta_k \phi_k(s)$. In population least squares, minimize

$$J(\beta) = \mathbb{E}\left[\left(Y - \sum_{k=1}^{K} \beta_k \phi_k(S_{t_n})\right)^2\right].$$

Differentiate with respect to β_j:

$$\begin{aligned} \frac{\partial J}{\partial \beta_j} &= -2\mathbb{E}\left[\phi_j(S_{t_n})\left(Y - \sum_{k=1}^{K} \beta_k \phi_k(S_{t_n})\right)\right], \\ 0 &= \mathbb{E}[\phi_j(S_{t_n})Y] - \sum_{k=1}^{K} \beta_k \mathbb{E}[\phi_j(S_{t_n})\phi_k(S_{t_n})]. \end{aligned}$$

Writing $G_{jk} = \mathbb{E}[\phi_j \phi_k]$ and $b_j = \mathbb{E}[\phi_j Y]$, the normal equations are $G\beta = b$. In finite-sample regression with design matrix $\mathbf{X}$ and response $\mathbf{y}$, the ordinary least squares estimate satisfies

$$\widehat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{y}$$

when $\mathbf{X}^\top \mathbf{X}$ is invertible. A QR or SVD solver is numerically preferable to explicitly forming the inverse, especially with correlated polynomial basis functions.

20.11 Why the algorithm must run backward

At the maturity date set realized cash flow $H_N^{(i)} = g(S_N^{(i)})$ for every simulated path i. Moving from $n = N-1$ toward $n = 1$, use only the future cash flows already determined on each path to fit the continuation regression $\widehat{C}_n$. Set the exercise decision

$$D_n^{(i)} = \mathbf{1}_{\{g(S_n^{(i)}) \geq \widehat{C}_n(S_n^{(i)})\}},$$

subject to any in-the-money exercise restriction appropriate for the contract. If $D_n^{(i)} = 1$, replace the path's future cash flow and stopping date by the immediate exercise payoff and t_n; otherwise keep its previously determined future stopping decision. Working forward without knowing the continuation policy at later dates would leave the regression target undefined.

20.12 Look-ahead bias and independent policy evaluation

A naive estimator that maximizes discounted realized payoffs path by path,

$$\frac{1}{M}\sum_{i=1}^{M} \max_{0\leq n\leq N} e^{-rt_n} g(S_n^{(i)}),$$

allows the holder to observe the whole future path before choosing an exercise date. It is therefore generally *not* the value of an admissible stopping policy and can severely overestimate the American option value. In contrast, a legitimate trained stopping rule $\widehat{\tau}$ evaluated on independent test paths satisfies

$$\mathbb{E}^{\mathbb{Q}}[e^{-r\widehat{\tau}} g(S_{\widehat{\tau}})] \leq \sup_{\tau\in\mathcal{T}} \mathbb{E}^{\mathbb{Q}}[e^{-r\tau} g(S_\tau)]$$

when $\widehat{\tau}$ is a valid stopping time on test paths and the expectation is taken conditional on the training set. Thus out-of-sample policy valuation provides a lower bound in expectation to the optimally exercisable value. Monte Carlo sampling error may make an observed estimate exceed the true value, so a realized finite-sample number is not guaranteed to be a deterministic lower bound.

20.13 A two-date exercise example showing why conditional expectation matters

Let exercise be permitted at $t_0 = 0$ or $t_1 = T$. Suppose immediate exercise pays a known g_0, while the maturity payoff H_T is random. The true value is

$$V_0 = \max\{g_0, e^{-rT}\mathbb{E}^{\mathbb{Q}}[H_T]\}.$$

The incorrect clairvoyant rule produces the value $\mathbb{E}^{\mathbb{Q}}[\max\{g_0, e^{-rT}H_T\}]$. Because the maximum is convex,

$$\mathbb{E}^{\mathbb{Q}}[\max\{g_0, e^{-rT}H_T\}] \geq \max\{g_0, e^{-rT}\mathbb{E}^{\mathbb{Q}}[H_T]\},$$

and the inequality can be strict. For example let $g_0 = 1$, $r = 0$, and $H_T = 0$ or 4 with equal probabilities. The legitimate value is $\max\{1, 2\} = 2$; the clairvoyant value is $\frac{1}{2}\max\{1, 0\} + \frac{1}{2}\max\{1, 4\} = 2.5$. The half-unit difference is precisely the value of forbidden foresight, not an attainable trading gain.

20.14 Regression error changes the optimal exercise boundary

Let $C_n(s)$ be the true continuation value and $\widehat{C}_n(s) = C_n(s) + \varepsilon_n(s)$ an estimated regression. The optimal decision compares $g(s)$ with $C_n(s)$, whereas the learned decision compares $g(s)$ with $C_n(s) + \varepsilon_n(s)$. Their decisions can differ only when

$$|g(s) - C_n(s)| \leq |\varepsilon_n(s)|.$$

Indeed, if $g(s) - C_n(s) > |\varepsilon_n(s)|$, both comparisons exercise; if $C_n(s) - g(s) > |\varepsilon_n(s)|$, both continue. Therefore the principal decision error occurs near the exercise boundary, where immediate exercise and continuation are close. This motivates allocating regression accuracy to boundary regions rather than blindly increasing polynomial degree everywhere.

20.15 Why least-squares continuation is a projection

Let Y be the random discounted payoff from continuing, and let $\mathcal{G} = \sigma(S_t)$ represent the observable state at the decision date. The conditional expectation $m = \mathbb{E}[Y \mid \mathcal{G}]$ is the unique (up to almost-sure equivalence) minimizer of $\mathbb{E}[(Y - Z)^2]$ among square-integrable $\mathcal{G}$-measurable variables Z. To see this, write $Y - Z = (Y - m) + (m - Z)$ and expand:

$$\begin{aligned}\mathbb{E}[(Y - Z)^2] = \mathbb{E}[(Y - m)^2] + \mathbb{E}[(m - Z)^2] \\ + 2\mathbb{E}[(Y - m)(m - Z)].\end{aligned}$$

Since $m - Z$ is $\mathcal{G}$-measurable,

$$\begin{aligned}\mathbb{E}[(Y - m)(m - Z)] &= \mathbb{E}[\mathbb{E}[(Y - m)(m - Z) \mid \mathcal{G}]] \\ &= \mathbb{E}[(m - Z)\mathbb{E}[Y - m \mid \mathcal{G}]] = 0.\end{aligned}$$

Thus

$$\mathbb{E}[(Y - Z)^2] = \mathbb{E}[(Y - m)^2] + \mathbb{E}[(m - Z)^2] \geq \mathbb{E}[(Y - m)^2],$$

with equality only if $Z = m$ almost surely. Ordinary least squares onto a finite basis is the projection of that conditional expectation onto the smaller linear span of the chosen regressors.

20.16 Hedging a learned exercise policy versus pricing its value

Once a stopping policy $\widehat{\tau}$ has been trained, its expected discounted payoff $L(\widehat{\tau}) = \mathbb{E}[e^{-r\widehat{\tau}} g(S_{\widehat{\tau}})]$ is no greater than the supremal American value. It is important to distinguish estimating $L(\widehat{\tau})$ from estimating the continuation functions themselves. A training algorithm can appear to fit realized future outcomes extremely well while having poor out-of-sample stopping decisions. Independent pricing paths therefore assess the performance of the learned policy without reusing the same future payoffs that determined it. For more reliable uncertainty quantification, repeat complete training/evaluation cycles or use suitably designed sample splitting rather than treating highly correlated fitted-path payoffs as independent observations.

20.17 A reproducible least-squares Monte Carlo algorithm

Algorithm 20.2 (Longstaff–Schwartz with independent policy validation)**.** Given exercise times $t_0 < \cdots < t_N$, payoff $g(s)$, basis functions $\phi_1, \ldots, \phi_K$, and independent training/test path sets:

1. Simulate complete training stock paths and initialize their selected cash-flow value at maturity to $g(S_N)$, with stopping date t_N.
2. For $n = N-1, \ldots, 1$, form a regression sample from the training paths eligible to exercise at t_n; use their currently assigned future cash flows discounted back to t_n as response variables.
3. Regress those responses on $\phi_k(S_n)$ and store the coefficient vector $\widehat{\beta}_n$.
4. On each training path, if $g(S_n) \geq \sum_k \widehat{\beta}_{n,k}\phi_k(S_n)$ and early exercise is permitted, replace its later cash flow with $g(S_n)$ and its stopping date with t_n.
5. Once all coefficient vectors have been trained, apply their exercise decisions forward on *independent* test paths.
6. Discount the realized test-path exercise cash flows to time zero, average them, and compute uncertainty from independent test-path payoffs. Compare with a European lower bound and an American upper bound where available.

The basis, exercise-state selection, and training sample size are material modeling choices. For an arithmetic-average or barrier option, regression on S_n alone may be insufficient because the continuation value depends on additional state variables.

LECTURE 21

American Exercise Properties and the Complementarity PDE

21.1 General ordering properties

Let $V^{\mathrm{A}}(s,t)$ and $V^{\mathrm{E}}(s,t)$ denote American and European values for identical terminal payoffs. Since the holder may exercise immediately,

$$V^{\mathrm{A}}(s,t) \geq g(s,t).$$

Since waiting until expiry is an admissible American exercise strategy,

$$V^{\mathrm{A}}(s,t) \geq V^{\mathrm{E}}(s,t).$$

The difference is the early-exercise premium, which depends on rates, dividends, payoff shape, and the stock dynamics.

Proposition 21.1 (No early exercise for a non-dividend-paying American call)**.** *Assume $r \geq 0$, the stock pays no dividends, and the option is a standard call with strike $K > 0$. Early exercise is never strictly optimal before maturity.*

Proof. Put–call parity and nonnegativity of the European put imply

$$\begin{aligned} C^{\mathrm{E}}(s,t) &= P^{\mathrm{E}}(s,t) + s - Ke^{-r(T-t)} \\ &\geq s - Ke^{-r(T-t)} \geq s - K. \end{aligned}$$

The call price is also nonnegative, hence $C^{\mathrm{E}}(s,t) \geq (s-K)^+$. Holding the European call is at least as valuable as exercising the American call immediately. Because exercise at maturity remains available, the American call has no early-exercise advantage over its European counterpart. □

An American put can benefit from early exercise when deeply in the money, because receiving the strike cash early may be valuable. A dividend-paying call can also favor early exercise when foregone dividends outweigh the benefit of delaying strike payment.

21.2 Continuation and exercise regions

Write the differential operator, under a continuous dividend yield q, as

$$\mathcal{L}V = V_t + \tfrac{1}{2}\sigma^2 s^2 V_{ss} + (r-q)sV_s - rV.$$

Define the continuation region $\mathcal{C} = \{(s,t) : V^{\mathrm{A}}(s,t) > g(s,t)\}$ and exercise region $\mathcal{E} = \{(s,t) : V^{\mathrm{A}}(s,t) = g(s,t)\}$. Inside $\mathcal{C}$, the option is held and behaves locally like a European option, so $\mathcal{L}V^{\mathrm{A}} = 0$. In the exercise region, immediate exercise determines its value, while the superhedging condition gives $\mathcal{L}V^{\mathrm{A}} \leq 0$ in the relevant variational sense.

Theorem 21.2 (American option complementarity problem)**.** *Under sufficient regularity and standard no-arbitrage assumptions, the American value satisfies the obstacle system*

$$\begin{cases} -\mathcal{L}V \geq 0, \\ V - g \geq 0, \\ (V-g)(-\mathcal{L}V) = 0, \\ V(s,T) = g(s,T). \end{cases}$$

Equivalently, $\max\{\mathcal{L}V, g - V\} = 0$.

The first two inequalities require the discounted option value to respect its exercise opportunity and supermartingale property. The product condition enforces that at every point at least one constraint binds: either the holder exercises, or the PDE holds with equality. A classical smooth solution may not exist across the exercise boundary; variational inequalities and viscosity/weak-solution interpretations handle limited regularity.

21.3 Free boundaries and smooth pasting

For an American put on a non-dividend-paying underlying, exercise typically occurs for $s \leq s^*(t)$, where $s^*(t)$ is an unknown exercise boundary. In the stopping region, $V^{\mathrm{A}}(s,t) = K - s$. Under standard regularity for $t < T$, the free boundary satisfies value matching and smooth pasting:

$$V^{\mathrm{A}}(s^*(t),t) = K - s^*(t), \qquad V_s^{\mathrm{A}}(s^*(t),t) = -1.$$

The boundary is part of the solution, which distinguishes the American PDE from a standard linear terminal-boundary problem.

21.4 The American option obstacle as a variational inequality

For a standard European claim the value solves $\mathcal{L}V = 0$ in the full interior of its domain. For an American claim, the holder can terminate the contract, and the exercise payoff acts as an obstacle that must never exceed the option value. Formally, the system

$$V \geq g, \qquad -\mathcal{L}V \geq 0, \qquad (V-g)(-\mathcal{L}V) = 0$$

expresses the complementarity of the two alternatives. In the continuation set where $V > g$, the product condition forces $\mathcal{L}V = 0$. In the exercise set where $V = g$, the differential inequality need not be an equality. This latter region cannot generally be specified in advance, so the problem is a free-boundary problem rather than a conventional linear PDE.

21.5 A lower bound for an American put's early-exercise region

Consider a non-dividend-paying stock, positive interest $r > 0$, and an American put with strike K. A very small stock price makes the immediate exercise reward $K - s$ close to K. Waiting until expiration gives no more than a payment bounded by K, but receiving K in the future has

discounted value at most $Ke^{-r(T-t)} < K$. For sufficiently small $s > 0$ and sufficient time remaining, immediate exercise can be superior to waiting. More precisely, the continuation value must be compared to $K - s$, and the optimal exercise boundary $s^*(t)$ delineates this region. This explains economically why the no-early-exercise proof for a non-dividend-paying call cannot be reused for a put: the cash benefit from immediate exercise goes in the opposite direction.

21.6 Penalty approximations and obstacle enforcement

For a discretized operator A and right-hand side b, a penalty method seeks a vector satisfying

$$Av - b - \rho(g - v)^+ = 0,$$

where $\rho > 0$ is large. If a component v_i falls below the exercise payoff g_i, the positive-part term becomes $\rho(g_i - v_i)$ and pushes the equation toward a higher value. If $v_i \geq g_i$, the penalty vanishes and the equation reduces to the continuation equation. The limit $\rho \to \infty$ encourages the hard obstacle constraint, but in finite precision a very large ρ may cause poor conditioning and make the numerical linear systems difficult to solve. A converged computed price must be checked for both complementarity residual and discretization error.

21.7 Why dividends can make early call exercise worthwhile

A call owner receives no dividends before exercise, while a stock owner does. With constant yield $q > 0$, a deeply in-the-money European call has approximate value

$$C^{\mathrm{E}}(s,t) \approx se^{-q\tau} - Ke^{-r\tau}.$$

Immediate exercise yields approximately $s - K$ when $s \gg K$. Their difference is

$$(s - K) - \left(se^{-q\tau} - Ke^{-r\tau}\right) = s(1 - e^{-q\tau}) - K(1 - e^{-r\tau}).$$

For sufficiently large s and positive q, the foregone dividend benefit $s(1 - e^{-q\tau})$ can dominate the financing benefit $K(1 - e^{-r\tau})$ of postponing strike payment. Hence the no-early-exercise argument for a non-dividend-paying call no longer applies. In a continuous-yield setting, the exercise boundary for an American call is therefore nontrivial, as with an American put.

21.8 A rigorous non-exercise argument for a no-dividend American call

For a non-dividend-paying stock with $r \geq 0$, a European call with maturity T satisfies put–call parity:

$$\begin{aligned} C^E(s,t) &= P^E(s,t) + s - Ke^{-r(T-t)} \\ &\geq s - Ke^{-r(T-t)} \\ &\geq s - K. \end{aligned}$$

Also $C^E \geq 0$, so $C^E(s,t) \geq (s-K)^+$. If $r > 0$, $t < T$, and $s > K$, the bound is strict because $Ke^{-r(T-t)} < K$. The holder is never better off exercising immediately for $(s-K)^+$ than retaining a

European call; an American call can be held at least as long, so its value is the European price and there is no early exercise advantage. For $r = 0$ the strict inequality argument disappears, but early exercise still cannot *improve* value in the standard frictionless no-dividend model; under special degeneracies it may be value-equivalent.

21.9 Why puts and dividend-paying calls are different

For a put, immediate exercise returns $K - s$, allowing immediate receipt of strike cash that may earn interest. Consequently there can be a region where exercising is optimal even before maturity. For a stock paying continuous dividend yield $q > 0$, delaying exercise of a deep-in-the-money call forgoes ownership of dividends. A dividend-paying call can therefore have an early-exercise region. These are economic differences in cash-flow timing, not just differences in the curvature of payoff graphs.

21.10 The complementarity conditions from continuation and exercise

Let $g(s)$ be an American reward, and let $\mathcal{L}$ be the risk-neutral spatial generator, including discounting through $\mathcal{A}V = V_t + \mathcal{L}V - rV$. In the continuation region the usual pricing PDE holds, $\mathcal{A}V = 0$. Everywhere the holder can exercise, so $V \geq g$. In the exercise region $V = g$, and the supermartingale property implies $\mathcal{A}V \leq 0$ in the appropriate weak/viscosity interpretation. Thus the obstacle problem is

$$\begin{aligned} V - g &\geq 0, \\ -\mathcal{A}V &\geq 0, \\ (V - g)(-\mathcal{A}V) &= 0. \end{aligned}$$

The third condition says that at each point at least one of the first two constraints is active. In the continuation region $V > g$, so necessarily $\mathcal{A}V = 0$; in an exercise region $V = g$ and $-\mathcal{A}V$ can be strictly positive. It is generally wrong to impose the Black–Scholes equality on the entire exercise region.

21.11 Free boundaries and value matching

For an American put with strike K and typical parameters, an exercise boundary $s^*(t)$ separates exercise for $0 \leq s \leq s^*(t)$ from continuation for $s > s^*(t)$. The boundary is unknown in advance; it is part of the solution. Under regularity conditions, the value matches the payoff and satisfies smooth pasting:

$$\begin{aligned} V(s^*(t), t) &= K - s^*(t), \\ V_s(s^*(t), t) &= -1. \end{aligned}$$

The first equation ensures the holder is indifferent between exercising and continuing at the boundary. The second expresses smoothness of the value function across the optimal boundary; it should not be asserted without hypotheses in models with discontinuous jumps, discrete exercise opportunities, or irregular payoff structures.

21.12 A useful lower bound for the American put

Consider an American put with strike K, current stock price $s \geq 0$, constant interest $r > 0$, and maturity T. Immediate exercise is always admissible, giving $P^A(s,t) \geq (K-s)^+$. Holding the right until T yields the European put value, so

$$P^A(s,t) \geq \max\{(K-s)^+, P^E(s,t)\}.$$

There is also the upper bound $P^A(s,t) \leq K$ when $r \geq 0$: the exercise payoff is at most K and any delayed reward is discounted by at most one. More explicitly, for any exercise time $\tau \geq t$,

$$e^{-r(\tau-t)}(K-S_\tau)^+ \leq Ke^{-r(\tau-t)} \leq K,$$

so taking conditional expectations and a supremum preserves the inequality. These bounds are independent of the free-boundary numerical scheme and serve as validation checks for an American pricing implementation.

21.13 How to check the complementarity residual numerically

Given a computed grid value v and obstacle g, define slack variables

$$\begin{aligned} s_i &= v_i - g_i, \\ r_i &= (Av-b)_i. \end{aligned}$$

A correct solution to the discrete obstacle problem should have $s_i \geq -\varepsilon$, $r_i \geq -\varepsilon$, and $|s_i r_i| \leq \varepsilon$ for a tolerance ε scaled appropriately to the problem. Checking only $v_i \geq g_i$ is insufficient because a scheme that always outputs the payoff would satisfy that inequality while usually failing the PDE residual conditions in the continuation region.

21.14 American exercise as the smallest dominating supermartingale

Let $Z_n = e^{-rt_n}g(S_n)$ be the discounted exercise reward and $Y_N = Z_N$, $Y_n = \max\{Z_n, \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n]\}$. Induction shows $Y_n \geq Z_n$ and $Y_n \geq \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n]$, so Y is a supermartingale dominating Z. Now let U be any other supermartingale with $U_n \geq Z_n$. At maturity $U_N \geq Z_N = Y_N$. If $U_{n+1} \geq Y_{n+1}$, then

$$\begin{aligned} U_n &\geq \mathbb{E}[U_{n+1} \mid \mathcal{F}_n] \\ &\geq \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n], \end{aligned}$$

while also $U_n \geq Z_n$. Hence

$$U_n \geq \max\{Z_n, \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n]\} = Y_n.$$

Backward induction proves minimality. The obstacle PDE is a continuous-time analogue of this same supermartingale domination property: a pricing function must dominate exercise payoffs and evolve according to the martingale equation wherever immediate exercise is suboptimal.

21.15 A put–call comparison in the presence of dividends

For proportional dividend yield q, European put–call parity is $C^E - P^E = se^{-q\tau} - Ke^{-r\tau}$. It yields

$$\begin{aligned} C^E(s,t) &= P^E(s,t) + se^{-q\tau} - Ke^{-r\tau} \\ &\geq se^{-q\tau} - Ke^{-r\tau}. \end{aligned}$$

Because $se^{-q\tau} < s$ when $q > 0$, this lower bound may fail to exceed intrinsic value $s - K$. Thus the simple no-dividend inequality that ruled out early call exercise no longer applies. Near expiry, the benefit of immediately acquiring the dividend-paying stock can outweigh the interest lost by paying the strike early. The exact boundary is determined by the American obstacle problem rather than by parity alone.

21.16 The exercise-region diagnostic for an American put

In a numerical American put grid, define the exercise residual $E_i^n = v_i^n - (K - s_i)^+$. The exercise region consists approximately of nodes where E_i^n is near zero; the continuation region is where E_i^n is positive. To check whether the numerical boundary is financially reasonable, verify that immediate exercise is typically favored at low stock prices for positive interest rates, that option values dominate intrinsic value at every stock/time node, and that the computed boundary changes smoothly with time under a sufficiently resolved and regular model. A jagged boundary may indicate inadequate mesh resolution, oscillations in a Crank–Nicolson update, or an LCP convergence problem. A numerical plot alone does not prove free-boundary convergence: residual checks and grid refinements are needed.

LECTURE 22

Numerical American Options and Portfolio Risk Optimization

22.1 Discrete linear complementarity systems

Let $g_i = g(s_i)$ be the discretized exercise payoff and v_i^n the option value at time-to-expiry grid level τ_n. For European options, an explicit step has the form $\widetilde{v}^{n+1} = (I - M)v^n$. For American exercise, the node value must not fall below the payoff. The resulting componentwise complementarity problem is

$$\begin{cases} v^{n+1} - (I - M)v^n \geq 0, \\ v^{n+1} - g \geq 0, \\ \left(v^{n+1} - g\right) \odot \left(v^{n+1} - (I - M)v^n\right) = 0, \end{cases}$$

where $\odot$ denotes coordinatewise multiplication. The explicit solution, when the step is well-posed, is the simple projection

$$\widetilde{v}^{n+1} = (I - M)v^n, \qquad v^{n+1} = \max\{\widetilde{v}^{n+1}, g\},$$

with the maximum taken componentwise. The projection is the discrete analogue of comparing continuation value with immediate exercise.

22.2 Implicit and Crank–Nicolson complementarity

Fully implicit and Crank–Nicolson schemes yield linear complementarity problems (LCPs) of the generic form

$$\begin{cases} Av^{n+1} - b^n \geq 0, \\ v^{n+1} - g \geq 0, \\ (v^{n+1} - g) \odot (Av^{n+1} - b^n) = 0. \end{cases}$$

For implicit Euler, $A = I + M$ and $b^n = v^n$ (subject to boundary-row adjustments). For Crank–Nicolson, $A = I + M/2$ and $b^n = (I - M/2)v^n$. Unlike European pricing, one cannot simply solve $Av^{n+1} = b^n$ everywhere because the exercise obstacle may bind at an unknown subset of nodes.

22.3 Projected iterations and penalty approximation

A projected solver alternates a linear-system update with a projection onto $v \geq g$, or uses a projected successive over-relaxation method. Another approach introduces a penalty parameter $\rho > 0$ and approximately solves

$$Av - b - \rho(g - v)^+ = 0$$

with the positive part applied componentwise. As ρ increases, violating the intrinsic-value constraint becomes increasingly expensive. Excessively large penalties may make the nonlinear system ill-conditioned, so both penalty error and iterative tolerance must be controlled. Newton-type active-set solvers exploit the fact that the unknown exercise region can be represented by a set of binding inequality constraints.

22.4 Markowitz mean–variance optimization

Let $R = (R_1, \dots, R_d)^\top$ be the vector of asset returns, $m = \mathbb{E}[R]$, $\Sigma = \mathrm{Cov}(R)$, and $x \in \mathbb{R}^d$ the portfolio weights. The return is $R_p = x^\top R$, with

$$\mathbb{E}[R_p] = x^\top m, \qquad \mathrm{Var}(R_p) = x^\top \Sigma x.$$

The classical minimum-variance problem under a target return m_0 is

$$\min_x \ \frac{1}{2} x^\top \Sigma x, \qquad \mathbf{1}^\top x = 1, \quad m^\top x = m_0.$$

If short selling is prohibited, add componentwise constraints $x \geq 0$. Assuming Σ is positive definite and moment constraints feasible, the unconstrained solution can be derived from Lagrange multipliers.

Proposition 22.1 (Minimum-variance weights without inequality constraints)**.** *Let $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} m$, $C = m^\top \Sigma^{-1} m$, and $D = AC - B^2 > 0$. Then the minimizing portfolio is*

$$x^* = \Sigma^{-1} \left(\frac{C - B m_0}{D} \mathbf{1} + \frac{A m_0 - B}{D} m \right).$$

Proof. Set $\mathscr{L}(x, \lambda, \gamma) = \frac{1}{2} x^\top \Sigma x - \lambda(\mathbf{1}^\top x - 1) - \gamma(m^\top x - m_0)$. Differentiating gives $\Sigma x - \lambda \mathbf{1} - \gamma m = 0$, so

$$x = \Sigma^{-1}(\lambda \mathbf{1} + \gamma m).$$

Enforce the two constraints:

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} 1 \\ m_0 \end{pmatrix}.$$

Since the determinant is $D > 0$, matrix inversion yields $\lambda = (C - B m_0)/D$ and $\gamma = (A m_0 - B)/D$. Substituting proves the formula. □

22.5 VaR and Expected Shortfall optimization

When asset returns are non-Gaussian or portfolios contain nonlinear derivatives, variance may not adequately characterize downside risk. Let $L \in \mathbb{R}^d$ denote a vector of losses and $L_p = x^\top L$ the portfolio loss. One may minimize $\mathrm{VaR}_\alpha(L_p)$ or $\mathrm{ES}_\alpha(L_p)$ subject to budget, return, and exposure constraints. The VaR objective can be nonconvex, while Expected Shortfall admits a useful convex representation for suitable linear portfolio losses:

Theorem 22.2 (Expected Shortfall optimization identity)**.** *For an integrable loss Y and $\alpha \in (0, 1)$,*

$$\mathrm{ES}_\alpha(Y) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1 - \alpha} \mathbb{E}[(Y - \eta)^+] \right\}.$$

For simulated loss vectors $L^{(1)}, \dots, L^{(n)}$, this suggests the sample optimization

$$\min_{x,\eta} \left\{ \eta + \frac{1}{(1-\alpha)n} \sum_{i=1}^{n} (x^\top L^{(i)} - \eta)^+ \right\},$$

subject to the desired portfolio constraints. Introducing slack variables $z_i \geq x^\top L^{(i)} - \eta$ and $z_i \geq 0$ converts the objective to a linear program when the portfolio constraints are linear. This approach connects the course's simulation methods to computational risk management.

22.6 Complementarity as a discrete stopping decision

The explicit American-option update

$$v_i^{n+1} = \max\{g_i, ((I - M)v^n)_i\}$$

can be read as a one-step dynamic programming operation. The term $((I - M)v^n)_i$ approximates the discounted continuation value, and the intrinsic payoff g_i is the immediate exercise reward. For the update to be financially meaningful, the discretized continuation operator should preserve the relevant positivity/monotonicity properties. A negative finite-difference weight can cause a mathematically inconsistent continuation value and hence misclassify the exercise region. This is why monotone discretizations are particularly important for obstacle problems.

22.7 The convexity of sample Expected Shortfall

For a fixed simulated loss vector $L^{(i)}$, the function of portfolio weights x and tail threshold η given by

$$f_i(x, \eta) = (x^\top L^{(i)} - \eta)^+ = \max\{x^\top L^{(i)} - \eta, 0\}$$

is the maximum of two affine functions and is therefore convex. A nonnegative sum of convex functions is convex, so the sample ES objective

$$J(x, \eta) = \eta + \frac{1}{(1-\alpha)n} \sum_{i=1}^{n} f_i(x, \eta)$$

is jointly convex in (x, η). With affine budget/return constraints and convex position constraints, optimizing J is a convex program. The corresponding sample VaR objective is an order statistic of affine functions and generally lacks this convexity property. Convexity is a major computational advantage of ES for portfolio optimization.

Example 22.3 (Linear-program formulation)**.** Introduce auxiliary variables z_i for the positive deviations above η. Then sample ES minimization is equivalent to

$$\begin{aligned} \text{minimize} \quad & \eta + \frac{1}{(1-\alpha)n} \sum_{i=1}^{n} z_i \\ \text{subject to} \quad & z_i \geq x^\top L^{(i)} - \eta, \quad i = 1, \dots, n, \\ & z_i \geq 0, \quad i = 1, \dots, n, \\ & \mathbf{1}^\top x = 1, \qquad x \geq 0, \end{aligned}$$

plus any specified linear expected-return or exposure constraints. Every displayed objective and constraint is linear. Large simulation samples produce large but sparse linear programs, which standard optimization algorithms can handle without relying on a fragile numerical derivative of an empirical quantile.

22.8 Gaussian portfolio VaR and ES versus variance minimization

If $R \sim \mathcal{N}_d(m, \Sigma)$, the portfolio loss over the modeled horizon can be written $L_p = -x^\top R$ and is Gaussian with mean $-x^\top m$ and standard deviation $\sqrt{x^\top \Sigma x}$. Hence

$$\begin{aligned} \mathrm{VaR}_\alpha(L_p) &= -x^\top m + z_\alpha \sqrt{x^\top \Sigma x}, \\ \mathrm{ES}_\alpha(L_p) &= -x^\top m + \frac{\varphi(z_\alpha)}{1-\alpha} \sqrt{x^\top \Sigma x}. \end{aligned}$$

If the expected return is constrained to equal m_0, then $-x^\top m = -m_0$ is fixed. At conventional high confidence levels (so the positive normal coefficients apply), minimizing Gaussian VaR or ES is equivalent to minimizing the portfolio standard deviation. Both therefore select the same optimum as the Markowitz minimum-variance portfolio under identical linear constraints. This equivalence does not generally survive asymmetric, heavy-tailed, or option-induced return distributions. The goal of simulation-based risk optimization is precisely to accommodate such nonnormal behavior without imposing a multivariate Gaussian model.

.1 The obstacle inequality as a linear complementarity problem

Discretize $v_\tau = L_h v$ in time by implicit Euler, with $A = I - \Delta\tau L_h$. Let g_i be the exercise payoff at each stock node and let b^n include the previous-time solution and boundary contributions. The discrete American value must satisfy

$$\begin{aligned} v^{n+1} - g &\geq 0, \\ Av^{n+1} - b^n &\geq 0, \\ (v_i^{n+1} - g_i)(Av^{n+1} - b^n)_i &= 0 \quad \text{for each } i. \end{aligned}$$

At an interior continuation node, $v_i^{n+1} > g_i$ and the discrete PDE equation is binding: $(Av^{n+1})_i = b_i^n$. At an exercise node, $v_i^{n+1} = g_i$ and the PDE residual need only be nonnegative. Enforcing only $v^{n+1} \geq g$ after an unconstrained European solve can be inconsistent with the implicit coupling of neighboring nodes; a numerical LCP method resolves both conditions simultaneously.

.2 Projected Gauss–Seidel / SOR iteration derived coordinatewise

Suppose A has positive diagonal entries. For a linear system $Av = b$, the Gauss–Seidel update at coordinate i is

$$\widetilde{v}_i^{(k+1)} = \frac{b_i - \sum_{j<i} A_{ij} v_j^{(k+1)} - \sum_{j>i} A_{ij} v_j^{(k)}}{A_{ii}}.$$

Successive over-relaxation with $\omega > 0$ changes it to

$$y_i^{(k+1)} = (1-\omega)v_i^{(k)} + \omega\widetilde{v}_i^{(k+1)}.$$

To enforce the immediate-exercise constraint, project onto the half-line $[g_i, \infty)$:

$$v_i^{(k+1)} = \max\{g_i, y_i^{(k+1)}\}.$$

The resulting PSOR algorithm is widely used for American-option LCPs. Its convergence depends on properties of A and the choice of ω; the projection does not make every arbitrary linear system well posed.

.3 Deriving Markowitz minimum-variance weights

For a vector of asset returns R with mean μ and positive-definite covariance Σ, portfolio return is $R_p = w^\top R$ and its variance is

$$\begin{aligned}\operatorname{Var}(R_p) &= \operatorname{Var}\left(\sum_i w_i R_i\right) \\ &= \sum_{i,j} w_i w_j \operatorname{Cov}(R_i, R_j) = w^\top \Sigma w.\end{aligned}$$

Minimize the variance subject to full investment $\mathbf{1}^\top w = 1$. The Lagrangian is $\mathscr{L}(w, \lambda) = w^\top \Sigma w - \lambda(\mathbf{1}^\top w - 1)$. Differentiation yields

$$\begin{aligned}\nabla_w \mathscr{L} &= 2\Sigma w - \lambda \mathbf{1} = 0, \\ w &= \frac{\lambda}{2}\Sigma^{-1}\mathbf{1}, \\ 1 &= \mathbf{1}^\top w = \frac{\lambda}{2}\mathbf{1}^\top \Sigma^{-1}\mathbf{1}, \\ \lambda &= \frac{2}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}, \\ w^* &= \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.\end{aligned}$$

This is the unconstrained global minimum-variance portfolio; constraints $w_i \geq 0$, target-return equality, leverage limits, or nonquadratic transaction costs change the solution and usually require a constrained optimizer.

.4 Expected Shortfall as a convex optimization objective

For an integrable loss random variable L, Expected Shortfall at confidence level $\alpha \in (0,1)$ admits the variational representation

$$\mathrm{ES}_\alpha(L) = \min_{z \in \mathbb{R}} \left\{ z + \frac{1}{1-\alpha}\mathbb{E}[(L-z)^+] \right\}$$

under the usual conventions. Given M sampled asset-loss vectors $\ell^{(1)}, \dots, \ell^{(M)}$, portfolio losses are $L_i(w) = w^\top \ell^{(i)}$. The sample approximation is

$$\widehat{\mathrm{ES}}_\alpha(w) = \min_z \left\{ z + \frac{1}{(1-\alpha)M} \sum_{i=1}^{M} (w^\top \ell^{(i)} - z)^+ \right\}.$$

Introduce auxiliary variables $u_i \geq 0$ with $u_i \geq w^\top \ell^{(i)} - z$. Then minimizing sample ES subject to portfolio constraints becomes the linear program

$$\begin{aligned} \min_{w,z,u_1,\dots,u_M} \quad & z + \frac{1}{(1-\alpha)M} \sum_{i=1}^{M} u_i, \\ \text{subject to} \quad & u_i \geq w^\top \ell^{(i)} - z, \qquad i = 1, \dots, M, \\ & u_i \geq 0, \qquad i = 1, \dots, M, \\ & \mathbf{1}^\top w = 1, \end{aligned}$$

plus any desired linear return or position constraints. The objective is convex because each positive-part function is convex. Unlike the quantile-based VaR objective, this representation is computationally tractable for scenario portfolios and does not require normally distributed returns.

.5 Deriving the normal-case Markowitz frontier with two constraints

Minimize $w^\top \Sigma w$ subject to $\mathbf{1}^\top w = 1$ and $\mu^\top w = m_0$, where Σ is positive definite. With Lagrange multipliers λ, γ,

$$\begin{aligned} \mathscr{L} &= w^\top \Sigma w - \lambda(\mathbf{1}^\top w - 1) - \gamma(\mu^\top w - m_0), \\ 2\Sigma w &= \lambda \mathbf{1} + \gamma \mu, \\ w &= \frac{1}{2} \Sigma^{-1}(\lambda \mathbf{1} + \gamma \mu). \end{aligned}$$

Define $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \mu$, and $C = \mu^\top \Sigma^{-1} \mu$. The constraints imply

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = 2 \begin{pmatrix} 1 \\ m_0 \end{pmatrix}.$$

If $D = AC - B^2 > 0$, solve the 2×2 system:

$$\begin{aligned} \lambda &= \frac{2(C - B m_0)}{D}, \\ \gamma &= \frac{2(A m_0 - B)}{D}, \\ w^* &= \Sigma^{-1} \left[\frac{C - B m_0}{D} \mathbf{1} + \frac{A m_0 - B}{D} \mu \right]. \end{aligned}$$

Substituting w^* into $w^\top \Sigma w$ yields the frontier variance

$$\sigma_p^2(m_0) = \frac{A m_0^2 - 2 B m_0 + C}{D}.$$

This closed form depends on unrestricted weights and known Gaussian second moments. Sample-based ES portfolios can be more flexible because their objective uses the full scenario loss distribution rather than covariance alone.

.6 Deriving the Rockafellar–Uryasev quantile condition

Consider $F(z) = z + (1-\alpha)^{-1}\mathbb{E}[(L-z)^+]$. If the loss distribution is continuous at z, differentiate under the expectation:

$$\frac{\mathrm{d}}{\mathrm{d}z}(L-z)^+ = -\mathbf{1}_{\{L>z\}} \quad \text{for } L \neq z,$$
$$F'(z) = 1 - \frac{\mathbb{P}(L>z)}{1-\alpha}.$$

Setting $F'(z) = 0$ gives $\mathbb{P}(L > z) = 1-\alpha$, equivalently $\mathbb{P}(L \leq z) = \alpha$. Thus the minimizing threshold is an α-quantile, making the ES variational formula a direct generalization of tail-average risk. For distributions with atoms, subgradient conditions replace ordinary derivatives and still identify an appropriate quantile interval.

.7 Convexity of sample expected shortfall under portfolio mixing

For $0 \leq \lambda \leq 1$ and portfolios $w^{(1)}, w^{(2)}$, the positive-part map $x \mapsto x^+$ is convex. For each scenario $\ell^{(i)}$, and any thresholds z_1, z_2,

$$\Big(\lambda w^{(1)\top}\ell^{(i)} + (1-\lambda)w^{(2)\top}\ell^{(i)} - [\lambda z_1 + (1-\lambda)z_2]\Big)^+$$
$$\leq \lambda(w^{(1)\top}\ell^{(i)} - z_1)^+ + (1-\lambda)(w^{(2)\top}\ell^{(i)} - z_2)^+.$$

Sum over scenarios, multiply by the nonnegative factor $[M(1-\alpha)]^{-1}$, and add the convex combination of thresholds. Taking the infimum over thresholds on both sides establishes convexity of the sample ES portfolio objective (or directly use the epigraph linear-program representation). Consequently, if the feasible portfolio region is convex and a solution exists, every local minimizer is global. Value-at-Risk generally lacks this convexity property, which is one reason ES is attractive in numerical portfolio optimization.

.8 A complete two-scenario ES optimization illustration

Take two equally weighted loss scenarios $\ell^{(1)} = (3,0)^\top$, $\ell^{(2)} = (0,2)^\top$ and a fully invested two-asset portfolio $w = (w, 1-w)^\top$ with $0 \leq w \leq 1$. Losses are

$$L_1(w) = 3w,$$
$$L_2(w) = 2(1-w).$$

At confidence level $\alpha = 1/2$, the upper-half Expected Shortfall for two equally weighted scenarios is simply the larger loss:

$$\mathrm{ES}_{1/2}(w) = \max\{3w, 2(1-w)\}.$$

The first term increases with w; the second decreases. The convex maximum is minimized when the two scenario losses coincide:

$$\begin{aligned} 3w &= 2(1-w), \\ 5w &= 2, \\ w^* &= \frac{2}{5}, \\ 1-w^* &= \frac{3}{5}, \\ \mathrm{ES}_{1/2}(w^*) &= 3 \cdot \frac{2}{5} = \frac{6}{5}. \end{aligned}$$

Thus the optimal portfolio for this simple scenario ES problem equalizes the worst losses, not necessarily portfolio variance or expected loss. The same principle extends to many scenarios through the auxiliary-variable linear program.

.9 An implementable American-option LCP solver

Algorithm .4 (Projected SOR on each time layer)**.** Assume a tridiagonal obstacle system $Av \geq b$, $v \geq g$, $(Av-b)_i(v_i-g_i)=0$ with an admissible initial iterate $v^{(0)} \geq g$.

1. For iteration $k = 0, 1, \ldots$, sweep through the interior stock nodes in ascending order.
2. Compute the Gauss–Seidel candidate $\widetilde{v}_i = [b_i - \sum_{j<i} A_{ij} v_j^{(k+1)} - \sum_{j>i} A_{ij} v_j^{(k)}]/A_{ii}$.
3. Over-relax and project: $v_i^{(k+1)} = \max\{g_i, (1-\omega)v_i^{(k)} + \omega\widetilde{v}_i\}$.
4. Evaluate the change $\left\| v^{(k+1)} - v^{(k)} \right\|_\infty$, the feasibility slacks $v-g$, $Av-b$, and the complementarity products.
5. Stop only when both the iterate change and the complementarity/feasibility residuals meet suitable tolerances.

After solving each time layer, propagate the new American values into the following layer and repeat. For a well-behaved M-matrix discretization, projected iterations are effective; poor parameter choices or loss of monotonicity can prevent rapid convergence. One should report both a spatial/time-grid refinement study and an inner LCP convergence tolerance, because otherwise an iterative-solver error can be mistaken for a PDE discretization error.

LECTURE A

Numerical and Probability Reference

A.1 Common distributions

The following identities are repeatedly useful in Monte Carlo implementations and stochastic models. For $Z \sim \mathcal{N}(0,1)$, $\mathbb{E}[Z] = 0$, $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$, and $\mathbb{E}[e^{tZ}] = e^{t^2/2}$. For $N \sim \text{Poisson}(\lambda)$, $\mathbb{P}(N = k) = e^{-\lambda}\lambda^k/k!$, $\mathbb{E}[N] = \text{Var}(N) = \lambda$. For $X \sim \text{Exp}(\lambda)$, $\mathbb{P}(X > t) = e^{-\lambda t}$, $\mathbb{E}[X] = 1/\lambda$, and $\text{Var}(X) = 1/\lambda^2$. If $X \sim \mathcal{N}(m, v)$, then $\mathbb{E}[e^X] = e^{m+v/2}$ and e^X is lognormal.

A.2 Black–Scholes Greeks for a non-dividend-paying call

With $\tau = T - t$, $d_1 = (\log(s/K) + (r + \sigma^2/2)\tau)/(\sigma\sqrt{\tau})$ and $d_2 = d_1 - \sigma\sqrt{\tau}$,

$$\begin{aligned} C &= s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \\ \Delta &= C_s = \Phi(d_1), \\ \Gamma &= C_{ss} = \frac{\varphi(d_1)}{s\sigma\sqrt{\tau}}, \\ \text{Vega} &= C_\sigma = s\varphi(d_1)\sqrt{\tau}, \\ \Theta &= C_t = -\frac{s\sigma\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2). \end{aligned}$$

The derivations use $s\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$ and the chain rule. These Greek formulas provide independent test cases for finite-difference derivative calculations.

A.3 Geometric-average Asian option as a control

Suppose $0 < t_1 < \cdots < t_m = T$ and risk-neutral GBM. The logarithm of the discrete geometric average $G = (\prod_{j=1}^m S_{t_j})^{1/m}$ is

$$\log G = \log S_0 + (r - \tfrac{1}{2}\sigma^2)\bar{t} + \frac{\sigma}{m}\sum_{j=1}^m W_{t_j},$$

$$\bar{t} = \frac{1}{m}\sum_{j=1}^m t_j.$$

It is normally distributed, since linear combinations of jointly Gaussian Brownian values are Gaussian. Its variance is

$$\sigma_G^2 = \frac{\sigma^2}{m^2}\sum_{j=1}^m\sum_{k=1}^m \min(t_j, t_k).$$

Let $m_G = \log S_0 + (r - \sigma^2/2)\bar{t}$ and $v_G = \sigma_G^2$. If $v_G > 0$, its geometric Asian call value is

$$\begin{aligned} C_G &= e^{-rT}\mathbb{E}[(G-K)^+] \\ &= e^{-rT}\left[e^{m_G + v_G/2}\Phi(\widetilde{d}_1) - K\Phi(\widetilde{d}_2)\right], \\ \widetilde{d}_1 &= \frac{m_G - \log K + v_G}{\sqrt{v_G}}, \qquad \widetilde{d}_2 = \widetilde{d}_1 - \sqrt{v_G}. \end{aligned}$$

This explicitly known quantity is a strong candidate for a control variate when estimating arithmetic Asian calls.

LECTURE B

Additional Simulation Techniques from the Instructor Notes

B.1 Common random numbers

Suppose two policies have output random variables $Y_A = g_A(U)$ and $Y_B = g_B(U)$ and the goal is to estimate $\theta_A - \theta_B$. Independent simulation gives variance $(\mathrm{Var}(Y_A) + \mathrm{Var}(Y_B))/n$ for a difference of two sample means. If the same uniforms U_i drive both alternatives, the paired estimate $n^{-1}\sum_i (Y_{A,i} - Y_{B,i})$ has variance

$$\frac{1}{n}\left(\mathrm{Var}(Y_A) + \mathrm{Var}(Y_B) - 2\,\mathrm{Cov}(Y_A, Y_B)\right).$$

Thus common random numbers help when the two outputs are positively correlated. This is especially valuable when comparing close option prices, hedge strategies, or alternative parameter values rather than estimating each independently.

B.2 Brownian bridge sampling

A Brownian bridge generates intermediate values conditional on already sampled endpoints. Given $0 \le s < t < u$ and Brownian values $W_s = x$ and $W_u = y$, the conditional distribution of W_t is normal with

$$\mathbb{E}[W_t \mid W_s = x, W_u = y] = \frac{u-t}{u-s}x + \frac{t-s}{u-s}y,$$
$$\mathrm{Var}(W_t \mid W_s, W_u) = \frac{(t-s)(u-t)}{u-s}.$$

Therefore, after simulating $W_T = \sqrt{T}Z_1$, one can sample $W_{T/2}$ conditional on (W_0, W_T), then fill in remaining midpoints recursively. This is an exact method for constructing a discretely sampled Brownian path. In quasi-Monte Carlo, allocating early low-discrepancy coordinates to the large-scale bridge components may reduce effective dimension and improve the quality of path-dependent option integration.

B.3 A source coverage and ordering note

The supplied archive contains lecture slide files numbered 1–8 and 10–22, with multiple annotated/iPad variants for some of them. No file numbered Lecture 9 was supplied, so the main manuscript preserves that gap rather than inventing its contents. The file named `cs476_w25_lec20.pdf` repeats a Lecture 19 label on its first slide; its American-options content is

treated in Lecture 20, matching the filename and subsequent sequence. The two extra appendices cover reference formulas and supplemental material available in the instructor notes, practice sets, and formula sheet; they do not purport to replace an absent Lecture 9.